

The Second Monopoly

The Patent Expired. The Exclusion Didn't and It Doubled the Cost of American Living.

(formerly titled *The Forgotten Half of the Patent Clause*)

How a Constitutional Bargain Was Cancelled — Restoring It Frees Buyers from Surveillance, Stops Robots Replacing America's Drivers & Operators, and Returns the Joy of Operating Real Machines

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Foreword by Gord Magill, author of End of the Road: Inside the War on Truckers

This is the complete back-matter apparatus to the book above — its Endnotes, Endnote Commentary, Supreme Court case notes, Glossary, and Key Terms, published online so that every source link is live and the whole section is searchable.

Endnotes, Glossary & Key Terms

The complete back-matter apparatus, online

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Apparatus Convention

The book uses no superscript reference numbers in the running text. Citations of sources, court cases, and statutes appear in the Endnotes section at the back of the book. The Endnotes, Endnote Commentary, Glossary, and Key Terms follow the Chicago Manual of Style, 17th edition (author-date system), with the house exceptions noted below.

The order of entries is set out in full in the Endnotes Ordering section, which opens the Endnotes. Two departures from Chicago apply throughout the apparatus. Every entry ends with a source URL. And same-year works by one author take no a, b, or c suffix; an in-text citation such as “Martin (2025)” is disambiguated by context, the running text making the specific work plain.

The Endnote Commentary section, which collects in alphabetical order — keyed to the same author surname or case name as the Endnotes section — substantive notes on those sources: full passages a chapter quotes only in part, additional context the running text omits for the sake of pace, related material the curious reader will want and the impatient reader can skip. Not every Endnote has a corresponding Commentary entry; only those that

genuinely reward a longer note. An Endnote that does have a Commentary entry ends with the italicized notice *"See Endnote Commentary."*

The two sections are designed to be read together by a reader who wants depth, and skipped together by a reader who does not.

The Endnote Commentary is followed by a Glossary section, which collects in alphabetical order by abbreviation the acronyms used throughout the book — primarily federal agencies, statutes, and regulatory frameworks defined parenthetically at first use in the running text. It also defines the five kinds of intellectual property the book distinguishes: copyright, patent, trade dress, trade secret, and trademark. The Glossary is followed by a Key Terms section, which collects in alphabetical order by term the book's specific terms of art (such as bootleggers-and-Baptists, regulatory ratchet, anti-backsliding provision, three-tier breakdown, public domain in the patent-law sense, the Expired Patent Commons, remanufacturing in the book's specific sense, grandfather exemption, expired-patent commerce, constitutional forgetting, and two suspenders), each with a brief definition.

Each Endnote entry concludes with a URL — Justia for U.S. Supreme Court cases, Cornell University Legal Information Institute for federal statutes, Founders Online for Founding-era documents, the publisher's canonical URL for periodicals, books, and reports, with Amazon as a last resort for older works without stable publisher URLs — so that readers using a digital edition of the book can follow the source directly.

The reader who wants to verify a claim in the running text consults the Endnote entry alphabetized by the author or case name as they appear in the text. There are no chapter-specific endnote groupings; the Endnotes section is a single book-wide alphabetical list. The reader who wants additional context on a source — the full passage a chapter quotes only in part, or material the running text omits for the sake of pace — consults the parallel entry in the Endnote Commentary section, which is keyed to the same anchor and ends with the italicized notice *"See Endnote for full citation."*

The reader who wants the meaning of an acronym, or the difference between a patent and a trademark, consults the Glossary; the reader who wants to anchor a recurring concept con-

sults the Key Terms section. Each section does one job.

In-text citations follow Chicago author-date style: integrated into the sentence as “Yandle (1983) named the pattern,” with full names and titles reserved for the Endnote entry. Acronyms are spelled out in full at first appearance in the book with the abbreviation following in parentheses; thereafter only the abbreviation is used.

Endnotes Ordering

Sort order. Entries run by headword — the first significant element of the entry. The ten rules below govern placement throughout this section. Rules 2 through 5 also govern the Glossary and the Key Terms sections, each alphabetized by its own headwords. One exception: the Endnote Commentary opens with its entry on The Chicago Manual of Style*, ahead of alphabetical order, because that entry explains the conventions the whole apparatus follows.*

1. Headword. The headword is the author’s surname, the organization’s name, the case name, the statute’s name, or — for an entry with none of these — the entry’s own leading words.

2. Leading articles. A leading “The,” “A,” or “An” is disregarded for placement. An article anywhere else in the headword counts as an ordinary word.

3. Word by word, not letter by letter. Comparison proceeds one word at a time, and a word boundary sorts ahead of any letter. New Civil Liberties Alliance therefore precedes Newbold, Charles, though a letter-by-letter sort would reverse them. The abbreviation “v.” counts as a word, so United States Code precedes United States v. Ballin.

4. Case and punctuation. Comparison ignores capitalization, commas, periods, apostrophes, ampersands, and hyphens. Sears, Roebuck & Co. v. Stiffel Co. files under Sears; D’Angelo files as DAngelo.

5. Numerals. A numeral inside a headword sorts by value and not as a string, so U.S. Patent No. 39,285 precedes 139,668, which precedes 5,357,751, which precedes 12,196,640. A headword beginning with a numeral sorts ahead of every headword beginning with a letter, so United States v. 50 Acres of Land precedes United States v. Ballin.

6. Court cases. A case files under the first party named in its style. United States v. Moghadam files under United States, never under Moghadam. In re and Ex parte file under In and Ex.

7. Statutes and code sections. A statute files under its own name — National Appliance Energy Conservation Act of 1987. A codified statutory section files under United States Code, and the sections there run by title number, then by section number. A codified regulation files under Code of Federal Regulations, and the entries there run by title number, then by part number, then by section number.

8. Names as written. A name files as the source writes it, not as it would read expanded. United States Code files under United; U.S. Department of Energy files under US. Spelled-out names therefore precede their abbreviated counterparts, and the two form separate runs.

9. Several entries under one headword. Entries sharing a headword run by date of publication, earliest first. An entry carrying no publication date — a continuing series such as the Consumer Price Index, or an organization's standing web page — goes last among them, and those entries run alphabetically by title.

10. Topical entries. An entry standing for a body of material rather than a single document — Self-certification and certification by simulation; Nichia Corporation and the LED-to-laser lighting succession — files under its own leading word like any other entry.

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pered vehicle or engine, and each defeat-device part, counts as a separate violation. Source for the per-violation civil-penalty figures cited in the trucking chapter. https://www.ecfr.gov/current/title-40/chapter-I/subchapter-A/part-19/section-19.4.

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https://scholarship.law.wm.edu/british-colonial-antecedents/32/.

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https://www.freep.com/videos/money/cars/ford/2026/06/08/ford-ceo-jim-farley-right-to-repair/90465881007/. This footnote is also for the Foreword.
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Compco Corp. v. Day-Brite Lighting, Inc., 376 U.S. 234 (1964). Companion case to Sears, Roebuck & Co. v. Stiffel Co., decided the same day. Together the two cases established that the federal patent system's policy of leaving the unpatented public domain free for all to copy is supreme over state unfair-competition laws that would prohibit such copying. Cited in the chapter on objections considered as part of the Supreme Court doctrinal line that supports the book's structural Patent Clause argument.
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Congressional Research Service. *Diesel Fuel and Engines: An Analysis of EPA's New Regulations.* CRS Report RL30737. May 1, 2001. Neutral congressional analytical document reviewing the EPA heavy-duty diesel rule signed December 21, 2000 by Administrator Carol Browner and promulgated as final rule at 66 Fed. Reg. 5002 (January 18, 2001). The report explicitly characterizes industry reactions to the rule as "mixed," documenting concerns expressed by engine manufacturers and refiners about technological feasibility, cost, and the diesel-fuel sulfur-reduction component, all addressed by EPA through its formal Response to Comments document published with the final rule under the notice-and-comment rulemaking process of the Administrative Procedure Act, 5 U.S.C. § 553. Source for the trucking chapter's hedged characterization that the affected manufacturers are reported as having opposed the 2007 rule when it was proposed.
https://www.everycrsreport.com/reports/RL30737.html.

Connecticut Humanities. "Albert Augustus Pope, Transportation Pioneer." ConnecticutHistory.org. Pope began by importing English bicycles in the 1870s and contracting the Weed Sewing Machine Company of Hartford to build them under the Columbia name. By 1892 he was selling a safety bicycle that carried the Starley frame, tangential

spoke lacing, the Renold chain, and pneumatic tires — British designs, every one, built in Connecticut. That same year he took majority control of the Hartford Rubber Works, then employing roughly one hundred seventy people and running around the clock, along with a steel company and a nickel-plating works. The strategy was vertical integration, and it worked because the underlying designs cost him nothing. A firm that had to license the frame, the chain, and the tire could not have absorbed its suppliers; it would have been paying rent on the machine itself.
https://connecticuthistory.org/albert-augustus-pope-1843-1909/.

Conrad, Kylie, and John D. Graham. “The Benefits and Costs of Automotive Regulations for Low-Income Americans.” *Journal of Benefit-Cost Analysis* 12, no. 3 (Fall 2021): 518–549. Aggregates per-vehicle regulatory-cost data from agency Regulatory Impact Assessments compiled in the Office of Management and Budget’s annual reports to Congress, covering the period 1967–2015. Table 5 (p. 535) itemizes the per-vehicle cumulative cost addition in 2016 dollars: $2,708.23 for safety regulations, $1,049.19 for emissions regulations, and $2,439.14 for fuel-economy regulations, summing to $6,196.56. The authors describe the total burden as “in the vicinity of $6,000 to $7,000 per vehicle” and note that, against an average new-vehicle transaction price “approaching $40,000,” cumulative federal regulation accounts for 15 to 20 percent of the price the consumer pays. The cars-and-trucks chapter uses the Conrad and Graham (2021) figure as the post-1970 cost magnitude and Sperling et al. (2004) as the proportional-attribution evidence that essentially all of that cumulative burden was added after 1970.
https://doi.org/10.1017/bca.2021.12.

Consumer Product Safety Act of 1972, Pub. L. No. 92-573, 86 Stat. 1207. Codified as amended at 15 U.S.C. §§ 2051 et seq. Establishes the Consumer Product Safety Commission and authorizes the issuance of mandatory product-safety standards and recalls under section 15.
https://www.law.cornell.edu/uscode/text/15/chapter-47.

Consumer Product Safety Commission. “Requirements for Bicycles.” 16 C.F.R. Part 1512. First promulgated at 43 Fed. Reg. 60034 (December 22, 1978), with amendments in 1980,

1981, 1995, and 2003. Bicycles were among the first product categories the Commission regulated after Congress created it in 1972. The date carries the weight here. Every American bicycle discussed in this chapter's historical section — the Columbia safeties of the 1890s, the Indianapolis chain that went to Kitty Hawk, the balloon-tire machines of the 1930s, the middleweights and lightweights of the 1950s — was designed, built, and sold before any federal design standard for bicycles existed. Secondary accounts disagree about whether the operative date is 1975, 1976, or 1978, and the Commission's own documents are not consistent; the Federal Register citation above is the one to rely on. https://www.federalregister.gov/documents/2011/05/13/2011-11742/requirements-for-bicycles.

Continental Paper Bag Co. v. Eastern Paper Bag Co., 210 U.S. 405 (1908). Decided June 1, 1908. Held that a patentee who has never used its invention may still enjoin others from using it: the right to exclude is "the very essence of the right conferred by the patent," and nonuse alone is no ground to deny an injunction. Justice McKenna wrote for the Court; Justice Harlan dissented. Cited in the Preface, and in the chapter *Four Problems, One Remedy*, as the third authority for the book's title: a monopoly lies in the power to exclude, which needs no production, and the patent's power to exclude is what outlives the term. https://supreme.justia.com/cases/federal/us/210/405/.

Corner Post, Inc. v. Board of Governors of the Federal Reserve System, 603 U.S. 799 (2024). Decided July 1, 2024, three days after Loper Bright and in the same term. Held, 6–3, that a claim under the Administrative Procedure Act accrues when the plaintiff is injured by an agency rule rather than when the rule was issued, so the six-year limitations period of 28 U.S.C. § 2401(a) runs from injury and not from promulgation. Justice Barrett wrote for the Court; Justice Kavanaugh concurred; Justice Jackson dissented, joined by Justices Sotomayor and Kagan. Cited in the filable-case chapter as the companion rule to the end of Chevron deference: the regulatory wall was built between 1970 and 1987, and under the older reading of the limitations period most of it was beyond challenge by anyone not already in court decades ago. https://supreme.justia.com/cases/federal/us/603/22-1008/.

Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2031. Final Rule, 89 Fed. Reg. 52396 (June 24, 2024). National Highway Traffic Safety Administration. https://www.federalregister.gov/documents/2024/06/24/2024-13243/corporate-average-fuel-economy-standards-for-passenger-cars-and-light-trucks-for-model-years.

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Cummings, Mary L. (“Missy”). Quoted in “Driverless Big Rigs Are Coming to American Highways, and Soon,” distributed by the New York Times News Service and syndicated at Journal-News (Cox Media Group), April 6, 2026. Cummings is director of the Mason Autonomy and Robotics Center at George Mason University and a former senior safety advisor to the National Highway Traffic Safety Administration. The article surfaces her phantom-braking warning — “There is no identified solution on the horizon for the problem of phantom braking. And it will not be addressed soon, because nobody wants to admit that it’s happening” — cited in the autonomous-driving chapter. https://www.journal-news.com/driverless-big-rigs-are-coming-to-american-highways-and-soon/article_f2d5f6e0-bfd5-4917-8ba0-17828a7c9492.html.

Daily Signal. “A New Car Will Cost You at Least $3,800 Extra Because of Government Regulation.” Heritage Foundation analysis, March 28, 2016. Estimates the minimum federal-regulatory cost addition on a new American vehicle at $3,800 in 2016 dollars; the analysis flags the figure as understated and likely higher with subsequent inflation. The cars-and-trucks chapter uses this as a floor and sanity check on the more detailed pre-1971 versus post-1970 cost-attribution figures derived from the Journal of Benefit-Cost Analysis study. https://www.dailysignal.com/2016/03/28/a-new-car-will-cost-you-at-least-3800-extra-because-of-government-regulation/.

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https://www.deere.com/en/harvesting/s-series-combines/.

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https://archive.detroitreman.com/about-us/history.

Diamond Alternative Energy, LLC v. Environmental Protection Agency, 606 U.S. 100 (2025). Seven-to-two decision, June 20, 2025, holding that fuel producers had Article III standing to challenge EPA’s approval of California regulations requiring automakers to build more electric vehicles, although the producers were not the regulated parties and their injury ran through the automakers’ responses. Justice Kavanaugh’s opinion for the Court held that commonsense economic principles, supported by the record, establish causation and redressability where the plaintiff shows a predictable chain of events, and stated that “the government generally may not target a business or industry through stringent and allegedly unlawful regulation, and then evade the resulting lawsuits by claiming that the targets of its regulation should be locked out of court as unaffected bystanders.” Justices Sotomayor and Jackson dissented. Cited in the chapter on objections considered as the answer to the objection that a buyer’s injury depends on the independent choices of remanufacturers not before the court.
https://supreme.justia.com/cases/federal/us/606/24-7/.

Diesel Engine Consent Decrees, 1998. United States v. Caterpillar Inc., Cummins Engine Company, Detroit Diesel Corporation, Mack Trucks Inc., Navistar International

Transportation Corporation, Renault Véhicules Industriels, and Volvo Truck Corporation. Settlements lodged October 22, 1998, in the U.S. District Court for the District of Columbia (Judge Henry H. Kennedy Jr.), entered as final judgments July 1, 1999. The decrees resolved Clean Air Act claims that the seven manufacturers had installed undisclosed defeat-device software (altering fuel injection timing) in heavy-duty diesel engines from the late 1980s through 1998 that allowed engines to pass EPA emissions tests while emitting up to two to three times the NOx standard during normal highway operation. The combined civil penalty was $83.4 million — the largest civil penalty in environmental enforcement history at the time — with twenty-five percent of each company's share directed to the California Air Resources Board in parallel settlements. Including injunctive relief (accelerated compliance with stricter NOx-plus-NMHC standards by October 2002, new not-to-exceed and emission surface limits, supplemental testing, recalls and fixes for in-use engines, and offset projects), the total settlement cost the manufacturers well over one billion dollars. Source for the consent-decree reference in the trucking chapter and the assertion that the 1998 settlements stripped heavy-duty engine manufacturers of credibility to resist tougher standards.
https://www.epa.gov/enforcement/heavy-duty-diesel-engine-settlements-1998. Companion primary source: U.S. Department of Justice Press Release No. 499 (October 22, 1998), https://www.justice.gov/archive/opa/pr/1998/October/499_enr.htm.

DieselNet. "United States: Heavy-Duty Onroad Engines." Emission standards reference compiled by Ecopoint Inc. The authoritative public summary of U.S. EPA particulate matter, NOx, and hydrocarbon standards for heavy-duty highway diesel engines from 1988 through the present, cross-referenced against 40 C.F.R. Part 86 and the underlying Federal Register notices. Documents the PM standard progression: 0.60 g/bhp-hr (model years 1988–1990), 0.25 g/bhp-hr (1991–1993), 0.10 g/bhp-hr (1994–2006), and 0.01 g/bhp-hr (2007–onward); and the in-use unregulated baseline of approximately 1.0 g/bhp-hr for pre-1988 engines. Source for Table 16.1 in the trucking chapter and for the chapter's reconciliation of the eighty-three-percent reduction from the 1988 federal baseline with the approximately ninety-percent reduction from the unregulated pre-1988 in-use level.
https://dieselnet.com/standards/us/hd.php.

DOE, “The History of the Light Bulb.” U.S. Department of Energy. Documents that the first spiral compact fluorescent lamp was created by engineer Edward E. Hammer at General Electric in 1976, in response to the 1973 oil crisis, and that early compact fluorescents reached the consumer market in the 1980s (the first successful integral product being the Philips Model SL, 1980). General Electric patented Hammer’s spiral work; those foundational compact-fluorescent patents, and the integral-lamp, ballast, and phosphor patents underlying the ordinary screw-base compact fluorescent, have long since expired — the latest clearly relevant foundational U.S. patent identified in the author’s review reaching expiration in 2015, and most expiring well before that; that expiration-date synthesis is this argument’s own, drawn from the underlying U.S. patent records rather than from the DOE history. The compact fluorescent was therefore public-domain technology when the April 2024 general service lamp rule raised the efficiency threshold beyond its reach. Cited in the healthy-lighting chapter for the recurrence of the withdrawal pattern in the technology that replaced the incandescent.
https://www.energy.gov/articles/history-light-bulb

DOE Energy Conservation Standards for Consumer Products and Industrial Equipment, 10 C.F.R. Parts 430 and 431. Department of Energy regulations implementing the energy-efficiency provisions of the Energy Policy and Conservation Act of 1975 and its amendments, including the National Appliance Energy Conservation Act of 1987. Cover refrigerators, freezers, room and central air conditioners, water heaters, furnaces, boilers, dishwashers, ovens and ranges, and a long list of other consumer durables.
https://www.ecfr.gov/current/title-10/chapter-II/subchapter-D.

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operative rule that pulled the ordinary incandescent light bulb from the United States residential market and is the principal subject of the healthy-lighting chapter. https://www.federalregister.gov/documents/2022/05/09/2022-09477/energy-conservation-program-energy-conservation-standards-for-general-service-lamps.

DOE General Service Lamps — Definitions Final Rule (2022). Energy Conservation Program: Definitions for General Service Lamps. 87 Fed. Reg. 27461 (May 9, 2022). U.S. Department of Energy (RIN: 1904-AF22). Final rule effective July 8, 2022. Reinstated the Obama-era GSL and GSIL definitions DOE had withdrawn in September 2019, expanding the regulatory scope to include LED, OLED, decorative candle and globe lamps, and reflector lamps that had been exempt from federal coverage. https://www.federalregister.gov/documents/2022/05/09/2022-09480/energy-conservation-program-definitions-for-general-service-lamps.

DOE General Service Lamps — 120 Lumens-per-Watt Final Rule (2024). Energy Conservation Program: Energy Conservation Standards for General Service Lamps. 89 Fed. Reg. 28859 (April 19, 2024). U.S. Department of Energy. Final rule effective July 3, 2024; compliance required on and after July 25, 2028. Raises the minimum efficacy threshold from 45 to more than 120 lumens-per-watt, effectively eliminating compact fluorescent lamps as compliant alternatives alongside the ordinary incandescent already eliminated by the 2022 rules. https://www.federalregister.gov/documents/2024/04/19/2024-07831/energy-conservation-program-energy-conservation-standards-for-general-service-lamps.

DOE Lighting Market Characterization 2020. U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Solid-State Lighting Program, published April 2024 (baseline year 2020). The most recent comprehensive federal characterization of the U.S. lighting market by lamp type, sector, technology, and installed-stock distribution. Documents approximately 819 million incandescent lamps installed in the U.S. residential sector at the 2020 baseline (approximately 12.6 percent of the residential installed lighting stock, comprising approximately 161 million A-shape general service incandescent lamps, 294 million decorative lamps, 334 million reflector lamps, and miscellaneous types).

Companion to the 2015 Lighting Market Characterization report. Source for the healthy-lighting chapter's installed-stock figure documenting the public-right being exercised at the moment the 2022 backstop rule withdrew the manufacturing future of those lamps.
https://www.energy.gov/sites/default/files/2024-08/ssl-lmc2020_apr24.pdf.

DOE Process Rule Notice of Proposed Rulemaking (2026). U.S. Department of Energy, Office of Critical Minerals and Energy Innovation. Notice of Proposed Rulemaking issued July 2, 2026, announced by Secretary of Energy Chris Wright. The Department's public announcement described the action as moving to end home-appliance and equipment efficiency mandates; the proposed rule itself amends the Department's internal "Process Rule" governing how future energy-conservation standards are evaluated and does not repeal any existing standard, including the general service lamp standards. Trade-press and legal analysis noted that any subsequent attempt to repeal an existing standard would implicate the anti-backsliding provision at 42 U.S.C. § 6295(o)(1) and would likely be challenged in litigation. Cited in the healthy-lighting chapter as the second test, after Executive Order 14154, of the executive path's closure.
https://www.energy.gov/documents/process-rule-nopr

Doenges, J. L. Autopsy series reporting spiral organisms in forty-three percent of 242 human stomach specimens, 1938. Cited in the chapter on outsiders and blind doctrine for the fact that the organisms had been counted in a published human series and connected to no disease. Known here through later reviews of *Helicobacter pylori*; the full citation has not been confirmed against the original.
See Endnote Commentary.

Dolan v. City of Tigard, 512 U.S. 374 (1994). The Supreme Court's holding that regulatory conditions imposed on the use of property must be "roughly proportional" to the impact of the proposed use, building on *Nollan v. California Coastal Commission* and supplying the second prong of the modern proportionality test reaffirmed in *Sheetz v. County of El Dorado* (2024).
https://supreme.justia.com/cases/federal/us/512/374/.

Drais, Karl von. The running machine Karl von Drais first rode in 1817 and protected by a ten-year Baden privilege in 1818. Baden had no patent law at the time, so what Drais received on 12 January 1818 was a *Großherzogliches Privileg* — a grand-ducal privilege running ten years and effective only within Baden, which the German Patent and Trade Mark Office describes as the equivalent of a patent today. His first public ride on the *Laufmaschine* was at Mannheim on 12 June 1817, and a French *brevet* followed the Baden grant. Cited in the chapters on pre-1900 American prosperity and on the sectors where the bargain still works for the proto-bicycle, or "dandy horse," that opens the historical lineage running forward through the safety bicycle of the 1880s and 1890s; the design was widely copied in the United States and Britain once the brief and territorially limited term ended. Deutsches Patent- und Markenamt, "World Day of the Bicycle," last updated 16 February 2026.
https://www.dpma.de/english/our_office/publications/background/sports-leisure-ip/worlddayofthebicycle/index.html.

Drug Price Competition and Patent Term Restoration Act of 1984 (the Hatch-Waxman Act), Pub. L. No. 98-417, 98 Stat. 1585. Codified in part at 21 U.S.C. § 355(j). Created the Abbreviated New Drug Application pathway for generic drug manufacturers, requiring proof of bioequivalence rather than full safety and efficacy review, and restored some patent term consumed by the FDA pre-market approval process for originators. Co-sponsored by Senator Orrin Hatch (R-Utah) and Representative Henry Waxman (D-California). Cited in the policy-framework chapter as the American precedent for opening a blocked patent commons through statutory action; the chapter contrasts the statutory mechanism Hatch-Waxman required with the constitutional remedy this book argues SCOTUS can deliver without congressional action.
https://www.law.cornell.edu/uscode/text/21/355.

Dun & Bradstreet. "Fuel Dealers in the United States — Business Directory." D&B Business Directory. Source for the count of fuel-dealer firms in the United States, listing roughly sixteen thousand two hundred company profiles; cited in the trucker chapter's fuel-

hauler subsection as evidence that the trade is dominated by small, independent, frequently multigenerational family operators rather than national chains. https://www.dnb.com/business-directory/company-information.fuel_dealers.us.html.

Duniho, Fergus, Jean-Louis Cazaux, and Hans Bodlaender. “Janggi — 장기 — Korean Chess.” The Chess Variant Pages. The principal English-language scholarly reference page on Janggi maintained by the chess-variants research community since the late 1990s; current version mostly rewritten by Duniho, based on earlier text by Cazaux and Bodlaender, and ultimately on Chapter LXXIV (“Tjyang-Keui—Chess”) of Stewart Culin’s *Korean Games* (1895). The page acknowledges, on its face, that “certain parts of Cazaux’s text were taken from Roleigh Martin and from David B. Pritchard” — placing Martin’s 1995 web page alongside Pritchard’s *Encyclopedia of Chess Variants* (1994) as a canonical English-language source for the rules of the game. The page further notes that “an earlier version had lifted large portions of text from Roleigh Martin’s page describing Korean Chess, and the rule in question was the interpretation of Martin.” The site’s persons index separately credits Martin as the author of “Korean Chess” (an extensive description) and “Chinese and Korean Chess Links,” with reference to his *ShareDebate International*. Cited in the preface as the independent third-party corroboration of the author’s claim to have published the first comprehensive English-language webpage on Korean Chess. https://www.chessvariants.com/oriental.dir/koreanchess.html.

Dunlop, John Boyd. Pneumatic tire. U.S. Patent 435,995, granted 1890. The specification describes an expansible rubber tube inside a protective cover reinforced with cloth or canvas. Dunlop had patented the design in Great Britain in December 1888, and British courts invalidated that patent in 1890 on Robert William Thomson’s prior art — Thomson having patented the same principle in France in 1846 and the United States in 1847. The American patent is the one that matters for this chapter’s argument. A Scottish veterinary surgeon living in Belfast, holding no American rights by virtue of his British filing, came to the United States Patent Office and accepted the same terms the Constitution offers every inventor. Seventeen years from grant, the design belonged to the American public. https://patents.google.com/patent/US435995A/en.

Eastern Railroad Presidents Conference v. Noerr Motor Freight, Inc., 365 U.S. 127 (1961), and *United Mine Workers v. Pennington*, 381 U.S. 657 (1965). Together the source of the Noerr-Pennington doctrine, under which concerted efforts to petition the government for favorable legislation or regulation are immune from antitrust liability, even where the petitioner's purpose is to disadvantage a competitor. Cited in *Objections Considered* as the doctrine that would defeat an antitrust theory built on the lobbying activity described in the chapter, *Why Nobody Noticed*.
https://supreme.justia.com/cases/federal/us/365/127/.

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https://www.amazon.com/dp/080951205X.

EdisonReport. "About That Executive Order…" *EdisonReport*lighting trade-press analysis, January 27, 2025. Reports the National Electrical Manufacturers Association's confirmation, in the week following Executive Order 14154 (Unleashing American Energy), that the executive order could not override the statutory anti-backsliding lock at 42 U.S.C. § 6295(o)(1) on the federal general service lamps efficiency standards. Source for the healthy-lighting chapter's account of the executive path's closure.
https://edisonreport.com/2025/01/27/about-that-executive-order/.

Eldred v. Ashcroft, 537 U.S. 186 (2003). Seven-to-two decision upholding the 1998 Sonny Bono Copyright Term Extension Act against a "limited Times" challenge under Article I, Section 8, Clause 8. Justice Ginsburg's majority opinion held that an extension is permissible as long as the resulting term remains "limited, not perpetual." Justices Stevens and Breyer dissented; Breyer's dissent argued that the extension produced a term worth 99.8 percent of the value of a perpetual copyright. Cited in the chapter on objections considered as the leading adverse precedent for the Patent Clause structural prong, and distinguished

there as a copyright case in a calendar-adjustment posture rather than a patent case in a structural-ratchet posture.
https://supreme.justia.com/cases/federal/us/537/186/.

Electronic Code of Federal Regulations. “49 CFR Part 562 — Lighting and Marking on Agricultural Equipment.” National Highway Traffic Safety Administration. Codifies the lighting and marking standards adopted by NHTSA in 2016 under the Moving Ahead for Progress in the 21st Century Act (MAP-21) and discussed in the farm-equipment chapter as the third of the three regulatory layers that captured the American tractor.
https://www.ecfr.gov/current/title-49/part-562.

Electronic Stability Control Rule (FMVSS No. 126). Final Rule, “Federal Motor Vehicle Safety Standards; Electronic Stability Control Systems; Controls and Displays,” 72 Fed. Reg. 17236 (April 6, 2007). National Highway Traffic Safety Administration.
https://www.federalregister.gov/documents/2007/04/06/E7-5921/federal-motor-vehicle-safety-standards-electronic-stability-control-systems-controls-and-displays.

Elster, Jon, ed. *Secrecy and Publicity in Votes and Debates*. Cambridge: Cambridge University Press, 2015. Edited volume of essays by political theorists on the effects of voting transparency in deliberative bodies. Includes Daniela Giannetti’s chapter documenting the use of *voto segreto* in the post-war Italian parliament as a strategic tool of legislators against external coercive pressures, the scholarly basis for the Hybrid Voting chapter’s reference to the parliament’s 119-year use of the practice.
https://www.amazon.com/dp/1107084334. See Endnote Commentary.

Emissions standards, 2010 versus current. A vehicle certified around 2010 met the federal Tier 2 program, whose fleet-average standard was 0.07 grams per mile NOx (Bin 5); California’s LEV II program reached full phase-in the same year, placing much of the fleet at ULEV levels. The current Tier 3 program sets a fleet-average NMOG-plus-NOx target of 0.030 grams per mile, reached in model year 2025, and lowered gasoline sulfur from 30 to 10 parts per million beginning in 2017, while extending required emission-control durability from 120,000 to 150,000 miles. A remanufactured vehicle built on a roughly 2010-era

patent-expired platform is therefore about one regulatory generation behind a new vehicle certified to the current standard, and is no dirtier than the vehicles of that generation already lawful on the road.
https://www.epa.gov/regulations-emissions-vehicles-and-engines/final-rule-control-air-pollution-motor-vehicles-tier-3.

Encyclopedia of Indianapolis. “Diamond Chain Company.” Indianapolis: Indiana Historical Society and IU Indianapolis. Hans Renold’s roller chain, developed in Manchester, made the chain-driven safety bicycle possible, and imported chain stayed beyond the reach of most buyers. On December 24, 1890, Arthur C. Newby, Edward C. Fletcher, and Glenn Howe invested five thousand dollars to begin manufacturing bicycle chain in Indianapolis, starting with four machines and four operators in rented rooms above a tinner’s shop. The encyclopedia states the consequence plainly: manufacturing the drive chains in Indianapolis made bicycles affordable to the masses in the 1890s. By 1904 the firm produced two-thirds of the bicycle chain used in America, and it grew to supply roughly sixty percent of all bicycles sold in the country. Wilbur and Orville Wright, running a bicycle shop in Dayton, were agents for the company; seven specially designed chains drove the propellers of their flying machine at Kitty Hawk in 1903. The company was later renamed Diamond Chain. No license was sought and none was needed.
https://indyencyclopedia.org/diamond-chain-company/.

Energy Independence and Security Act of 2007, Pub. L. No. 110-140, 121 Stat. 1492. Codified as amended at scattered sections of 42 U.S.C., including the lighting-efficiency provisions amending the Energy Policy and Conservation Act at 42 U.S.C. §§ 6291 et seq.
https://www.law.cornell.edu/uscode/text/42/chapter-77.

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(2025) — a fifteen-percent increase across the period in which DOE's general-service-lamp rule savings were projected to accrue.
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tive 2.5 g/bhp-hr with a 0.5 g NMHC cap) for heavy-duty diesel engines beginning with the 2004 model year — a fifty-percent reduction from the 1998 NOx standard of 4.0 g/bhp-hr. The October 1998 Consent Decrees pulled the 2004 standard forward by fifteen months to October 2002, the implementation point that pushed every signatory toward cooled exhaust gas recirculation. Source for the trucking chapter's account of the first wave of post-1999 regulatory tightening that ended the mechanical-engine era of heavy-duty diesel and defines the 1999 model year as the dream-truck cutoff.
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Public Law 103-298, which restored Cessna's production through liability reform and waived no airworthiness standard. The continuity runs with the certificate, which is transferable property; a third party gets it only by acquiring the certificate or licensing it from the holder under 14 C.F.R. § 21.55, and the 172S carries a model-specific basis at several Part 23 amendment levels rather than the 1955 rules alone. https://drs.faa.gov/browse/excelExternalWindow/DRSDOCID1538016001202506271647 31.0001; https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_21.101-1B.pdf; https://www.law.cornell.edu/cfr/text/14/part-21/subpart-B

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https://fred.stlouisfed.org/.

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laxation of its conditions of use. Justice Kavanaugh's opinion for the Court held that an unregulated plaintiff challenging the government's regulation of others must show a predictable causal link rather than a speculative one. Cited in the chapter on objections considered as the government's strongest recent authority against a buyer's standing. https://supreme.justia.com/cases/federal/us/602/23-235/.

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tion rule: the lineage runs from Friedman's case against licensure, through Svorny's proposal that liability insurers do the gatekeeping, to the author's own 2025 argument, cited separately.
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Motion Picture Patents Co. v. Universal Film Manufacturing Co., 243 U.S. 502 (1917). Held (6—3) that a patentee may not, by notice attached to a patented machine, extend the monopoly to unpatented supplies used with it; an unconditional sale exhausts the patent right. In the course of that holding the Court stated, as a rule settled since *Pennock v. Dialogue* (1829), that the primary purpose of the patent laws is not the creation of private fortunes for patent owners but the promotion of the progress of science and the useful arts. Cited in the Patent Clause prong chapter as the decision elevating that reasoning toward settled rule. https://supreme.justia.com/cases/federal/us/243/502/.

Müller, Julian F., and Jan Gogoll. "Should Manual Driving Be (Eventually) Outlawed?" *Science and Engineering Ethics* 26, no. 3 (June 2020): 1549–1567. A peer-reviewed paper in applied ethics arguing that, in a future in which autonomous cars are significantly safer than human drivers, manually driven cars should for moral reasons be banned from ordinary traffic. Cited in the autonomous-vehicle chapter on the documented movement to outlaw human driving. https://doi.org/10.1007/s11948-020-00190-9.

Muller v. Oregon, 208 U.S. 412 (1908). Justice Brewer, for a unanimous Court, upheld an Oregon statute limiting women in factories and laundries to ten hours a day. The decision is

cited in this book for the brief rather than the holding; see the entry under Brandeis and Goldmark for the filing that gave the Brandeis brief its name.
https://supreme.justia.com/cases/federal/us/208/412/.

Murray, H. J. R. *A History of Chess*. Oxford: Clarendon Press, 1913. The classical and exhaustive scholarly reference on the history of the chess family, supporting the long-standing mainstream view that the game originated as Indian *chaturanga* around the sixth century CE, spread westward via Persian *shatranj*, and that Chinese Xiangqi and Korean Janggi developed as eastward derivatives or parallels of the same ancestor. Cited in the preface as the principal representative of the mainstream Indian-origin position against which Li (1998) argues. Public-domain digital edition via the Internet Archive.
https://archive.org/details/historyofchess0000hjrm.

Natanson, Hannah, Jeff Stein, Dan Diamond, and Rachel Siegel. “DOGE Builds AI Tool to Cut 50 Percent of Federal Regulations.” *The Washington Post*, July 26, 2025. Reports, from a July 1 PowerPoint and four officials, that the “DOGE AI Deregulation Decision Tool” was built to analyze roughly 200,000 federal regulations for those no longer required by law, had made decisions on 1,083 regulatory sections at the Department of Housing and Urban Development, and had written the Consumer Financial Protection Bureau’s deregulations; HUD employees said it misread the language of the law on several occasions. Cited in the preface as the nearest existing use of AI on the regulatory state — asking whether a statute still requires a rule, not whether the statute and rule together do what the Constitution forbade.
https://www.washingtonpost.com/business/2025/07/26/doge-ai-tool-cut-regulations-trump/.

National Aeronautics and Space Administration. “Artificial Intelligence, NASA Data Used to Discover Eighth Planet Circling Distant Star.” News release, December 14, 2017. Reports the discovery of Kepler-90i by a neural network built by Christopher Shallue, a software engineer at Google, trained on fifteen thousand previously vetted Kepler signals and applied to 670 star systems already examined by professional astronomers. Shallue originated the approach and was not an astronomer; the astronomy was supplied by Andrew Vanderburg, then a NASA Sagan postdoctoral fellow at the University of Texas at Austin,

who worked the problem with him and co-authored the resulting paper. The finding tied Kepler-90 with the solar system for the most planets known to orbit a single star. Cited in the preface as an instance in which an outsider supplied the method and specialists inside the field supplied the domain judgment.
https://www.nasa.gov/news-release/artificial-intelligence-nasa-data-used-to-discover-eighth-planet-circling-distant-star/

National Appliance Energy Conservation Act of 1987, Pub. L. No. 100-12, 101 Stat. 103 (March 17, 1987). Codified primarily at 42 U.S.C. §§ 6291–6309, with the anti-backsliding provision at § 6295(o)(1). Senate Roll Call 28, February 17, 1987 (89–6); House voice vote, March 3, 1987.
https://www.law.cornell.edu/uscode/text/42/chapter-77.

National Association of Home Builders (NAHB). *Study of Life Expectancy of Home Components*. Washington, D.C.: NAHB and Bank of America Home Equity, February 2007. Industry-standard reference for residential appliance and component lifespans, compiled from a 2006 survey of manufacturers, trade associations, and researchers. The figures represent industry-wide averages assuming normal usage and adherence to maintenance schedules; the study notes that actual service life depends substantially on installation quality, usage frequency, and climate conditions, and that appliances are often replaced for reasons of technological advancement or changing consumer preferences well before they cease to function. Cited in the cars-and-trucks chapter for the appliance replacement-cycle figures, including the refrigerator (13 years), dishwasher (9 years), clothes dryer (13 years), microwave oven (9 years), gas range (15 years), and electric range (13 years). The chapter's per-appliance figures are conservative midpoints calibrated against the NAHB single-point estimates and the InterNACHI ranges.
https://hcmuddox.com/wp-content/uploads/2022/10/nahb20study20of20life20expectancy20of20home20components.pdf.

National Association of Landscape Professionals (NALP). Organizational homepage. The national trade association for the landscape, lawn care, irrigation, and tree care trades, head-

quartered in Fairfax, Virginia. Formed January 1, 2005 by the merger of the Associated Landscape Contractors of America (founded 1961) and the Professional Lawn Care Association of America as the Professional Landcare Network, and rebranded to its present name April 1, 2015. Identified in the plaintiff-categories chapter as the trade body for the landscaping and grounds-maintenance operator, whose governing hook is EPA Part 1054, the standard for small spark-ignition engines. The association has actively opposed restrictive small-engine emissions rules and state-level gasoline-equipment bans. Membership figures appear in two forms and should not be conflated: roughly 100,000 landscape industry professionals in membership, within an industry the association describes as approaching one million professionals.
https://www.landscapeprofessionals.org/.

National Automobile Chamber of Commerce. *Automobile Facts and Figures*, 1926 edition. Industry trade-association annual compilation of U.S. motor-vehicle production, sales, and registration data. Source for the 1925 U.S. motor-vehicle production figures cited in the cars-and-trucks chapter: 3,735,171 passenger cars and 4,265,830 total motor vehicles (passenger cars plus 530,659 trucks). The NACC was the primary industry body collecting and publishing automotive production statistics in the 1920s; comprehensive retail-sales data was not compiled and made public until after World War II, so the chapter's reference to four million new cars sold in 1925 represents factory production, which the industry of the period used interchangeably with factory sales. Archived by the Federal Reserve Bank of St. Louis, FRASER digital library, alongside related commercial-car registration series.
https://fred.stlouisfed.org/series/M0146DUSM543NNBR.

National Electrical Manufacturers Association v. United States Department of Energy, No. 17-1341 (4th Cir.). Petition for review filed by NEMA in 2017 challenging the U.S. Department of Energy's January 2017 expanded definitions of general service lamp and general service incandescent lamp. Settled by agreement on July 11, 2017; the settlement led directly to the Trump-administration September 2019 rollback of the expanded definitions (84 Fed. Reg. 46661, September 5, 2019). Cited in the healthy-lighting chapter's bootleggers-and-Baptists section as documentary evidence that NEMA's posture across the

2007–2024 arc included active litigation against DOE on the scope of the rule, complicating any straight-line account of industry support for the regulatory architecture.
https://www.law.nyu.edu/sites/default/files/settlement-agreement.pdf.

National Family Farm Coalition (NFFC). Organizational homepage. Coalition representing family farmers across dozens of state-level farmer organizations; multi-state structure provides the geographic and circuit diversity that strengthens a constitutional filing. Identified in the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing.
https://nffc.net/.

National Farmers Union (NFU). Organizational homepage. Founded 1902; represents more than 200,000 family farms and ranches across 33 state divisions through a grassroots policy process. President Rob Larew, Vice President Jeff Kippley. The NFU 124th Anniversary Convention (March 7–9, 2026, New Orleans) highlighted the 46 percent year-over-year increase in farm bankruptcy filings reported by the American Farm Bureau Federation. Identified in the farm-equipment chapter as the strongest single candidate plaintiff among the five farm associations developed under associational standing.
https://nfu.org/.

National Federation of Independent Business v. Sebelius, 567 U.S. 519 (2012). Five Justices held that the Commerce Clause and the Necessary and Proper Clause did not authorize the Affordable Care Act's individual mandate, because Congress may regulate commerce but may not compel it; a different five upheld the mandate's payment as a tax; and seven held the Medicaid expansion's threatened loss of all existing Medicaid funds unconstitutionally coercive under the Spending Clause. Cited in The Patent Clause Prong as the hardest case for the argument: the Court refused one power and accepted another for the same payment. The distinction the chapter draws from it is that the taxing power produced no outcome any clause forbids, while "limited Times" forbids the outcome itself; that distinction is the book's, not the Court's.
https://supreme.justia.com/cases/federal/us/567/519/.

National Highway Traffic Safety Administration. “Bicycle Safety.” U.S. Department of Transportation. Official federal portal for bicycle safety statistics, educational materials, and crash data drawn from the agency’s Fatality Analysis Reporting System (FARS). Cited in the bicycles section of the modern-bargain chapter as the federal-data anchor for cyclist fatality trends and as the agency-level confirmation that no federal product safety regulation operates on the bicycle itself.
https://www.nhtsa.gov/road-safety/bicycle-safety.

National Institute of Mental Health. “Seasonal Affective Disorder.” NIMH Health Publications, U.S. Department of Health and Human Services. The current authoritative federal mental-health publication on Seasonal Affective Disorder (SAD), a recurring depressive disorder linked to seasonal variation in light exposure. NIMH states that millions of Americans experience SAD, with prevalence rising at higher latitudes and higher rates in women. Treated with controlled-spectrum light therapy that requires careful filtering against the blue-shifted output of standard residential LED lighting. Cited in the healthy-lighting chapter’s coalition section as the federal-source anchor for the SAD-affected population.
https://www.nimh.nih.gov/health/publications/seasonal-affective-disorder.

National Psoriasis Foundation. “Phototherapy.” NPF patient-resource page. Authoritative patient-organization summary of phototherapy as an approved treatment modality for psoriasis. Documents that ultraviolet B (UVB) light, present in natural sunlight and delivered via therapeutic lamps, is an effective treatment for psoriasis because it penetrates the skin and slows the excessive growth of affected skin cells. Psoriasis affects approximately 7.5 to 8 million American adults per the NPF’s prevalence data, of whom a non-trivial fraction depends on UVB phototherapy delivered through specialized incandescent and halogen lamp architectures. Cited in the healthy-lighting chapter’s coalition section.
https://www.psoriasis.org/phototherapy/.

National Sustainable Agriculture Coalition (NSAC). Organizational homepage. Coalition of grassroots organizations advocating for sustainable and organic farmers; published analyses document that four firms control 45 percent of global farm equipment sales. Identified in

the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing.
https://sustainableagriculture.net/.

National Technology Transfer and Advancement Act of 1995, Pub. L. No. 104-113, 110 Stat. 775. Directs federal agencies to use technical standards developed by private-sector voluntary consensus standards bodies (such as SAE, ASTM, UL, and ANSI) where practicable, in lieu of developing federal standards from scratch. Implemented through OMB Circular A-119.
https://www.law.cornell.edu/uscode/text/15/272.

Neri, Cassio. “Shorter and Faster than Sort3AlphaDev.” arXiv:2307.14503 (2023). Supplies counterexamples to the claim that no shorter three-element sorting program exists.

New Civil Liberties Alliance (NCLA). Pro bono litigation organization founded in 2017, headquartered in Washington, DC. Focused on constitutional challenges to administrative-state overreach. NCLA was lead counsel on *Relentless, Inc. v. Department of Commerce*, which the Supreme Court consolidated with *Loper Bright Enterprises v. Raimondo* — the 2024 decision that ended Chevron deference. Cited in the trucking chapter as one of three public-interest legal foundations carrying constitutional cases of the kind the Patent Clause argument requires at no cost to the plaintiff.
https://nclalegal.org/.

Newbold, Charles. U.S. Patent for the cast-iron plow, granted 1797 (term expired 1811 under the fourteen-year rule of the Patent Act of 1790). The single-piece cast-iron moldboard plow that, after expiration, became the dominant American farming implement of the early-to-mid nineteenth century. The cast-iron plow entry in the chapter’s eight-product table.
https://en.wikipedia.org/wiki/Charles_Newbold.

Nichia Corporation and the LED-to-laser lighting succession. The efficient blue gallium-nitride light-emitting diode was developed by Shuji Nakamura at Nichia Corporation of Japan in the early 1990s; Nichia paired a blue chip with a yellow phosphor to produce the first commercial white LED in 1996, the architecture in most household LED bulbs today.

Nakamura shared the 2014 Nobel Prize in Physics for the blue LED. Nichia held the core white-LED patents, earned a reported fortune from them, and litigated to defend them; those patents, filed in the 1990s, have only recently begun to expire. As the LED portfolio lapses, the same inventor and the same company have moved to laser-based white light: Nakamura's post-LED work through SoraaLaser (SLD Laser) and Nichia's GigaWhite blue-laser-plus-phosphor module, developed for automotive, specialty, and industrial lighting, with Nichia in 2016 publicly prioritizing laser diodes as the next lighting source. Cited in the healthy-lighting chapter for the observation that the regulatory withdrawal pattern recurs one technology at a time, each expired-patent design displaced as its patented successor arrives.
https://led-ld.nichia.co.jp/en/product/ld_gigawhite.html. Nobel Prize background:
https://en.wikipedia.org/wiki/Shuji_Nakamura

Nixon v. United States, 506 U.S. 224 (1993). The Supreme Court's leading modern application of the political-question doctrine to internal Congressional procedure: the Senate's procedure for trying an impeachment was held nonjusticiable because the Constitution textually commits the matter to the Senate and the courts lack judicially manageable standards for second-guessing the chamber's procedural choices. Cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that federal courts have shown strong reluctance to police the chamber's internal voting mechanics, including the interpretive scope of the Yeas-and-Nays override in Article I, Section 5, Clause 3.
https://supreme.justia.com/cases/federal/us/506/224/.

Nollan v. California Coastal Commission, 483 U.S. 825 (1987). The Supreme Court's holding that regulatory conditions imposed on the use of property must bear an "essential nexus" to the harm the regulation claims to address, supplying the first prong of the proportionality test later completed in *Dolan v. City of Tigard* (1994) and reaffirmed in *Sheetz v. County of El Dorado* (2024).
https://supreme.justia.com/cases/federal/us/483/825/.

Nordlinger v. Hahn, 505 U.S. 1 (1992). The Supreme Court's decision upholding California's acquisition-value property-tax scheme under Proposition 13, which assessed

newly purchased property at its purchase price while capping annual increases on long-held property at two percent. Justice Blackmun wrote for the Court, which found the classification rationally related to legitimate state interests and confined *Allegheny Pittsburgh Coal Co. v. County Commission of Webster County* (1989) to the rare case in which the facts precluded any plausible inference that an acquisition-value policy explained the disparity. Justice Thomas concurred in part while stating that *Allegheny Pittsburgh* could not be distinguished; Justice Stevens dissented. The decision is the strongest authority against pleading this book's equality argument as a standalone equal-protection claim, and it is taken up in that posture in the objections chapter.
https://supreme.justia.com/cases/federal/us/505/1/.

Nori, Harsha, et al. "Sequential Diagnosis with Language Models." *Science* 394, no. 6822 (October 22, 2026): 423–428.
https://www.science.org/doi/10.1126/science.aeh7930. Peer-reviewed study of Microsoft's Diagnostic Orchestrator, a panel of AI agents tested against generalist physicians on 304 *New England Journal of Medicine* cases and on emergency-department admissions; source for the figures and the authors' stated limits in the chapter *When Outsiders Advance Constitutional Law and Other Disciplines*. See Endnote Commentary.

Northeast States for Coordinated Air Use Management (NESCAUM). "Heavy-Duty Omnibus Low-NOx Rule: Frequently Asked Questions." September 2024. The current authoritative public summary of state adoption of the California Air Resources Board Heavy-Duty Omnibus (Low-NOx) regulation under Clean Air Act § 177. Documents the nine states that have specifically adopted the CARB Heavy-Duty Omnibus standards as of 2024–2025: Colorado (model year 2027), Massachusetts (MY 2025), New Jersey (MY 2027), New Mexico (MY 2027), New York (MY 2026), Oregon (MY 2025), Rhode Island (MY 2027), Vermont (MY 2026), and Washington (MY 2026). The broader seventeen-state Section 177 group that adopted CARB light-duty (LEV/ZEV/ACC II) or general vehicle standards is distinct from the nine-state Heavy-Duty Omnibus group. Source for the truck-

ing chapter's reference to the nine states whose Heavy-Duty Omnibus adoption layered the federal mandate with a parallel state regime.
https://www.nescaum.org/documents/HD-Low-NOx-Omnibus-FAQ_website-version_clean_09-23-24.pdf.

Novak Conversions. *The Novak Guide to the Spicer Model 18 Transfer Case*. Reference history of the Jeep transfer case. The civilian Jeep's core drivetrain was supplied and patented by independent component makers rather than by Willys–Overland or its successors: the Borg-Warner T-90 three-speed transmission and the Spicer (Dana) Model 18 transfer case and Dana axles. The underlying utility patents on these mid-century components expired under the seventeen-year-from-grant term then in force, placing the CJ drivetrain in the public domain by the late 1980s.
https://www.novak-adapt.com/knowledge/transfer-cases/model-18/.

Nunes, Ashley, and Kristen D. Hernandez. "Autonomous Vehicles and Public Health: High Cost or High Opportunity Cost?" Massachusetts Institute of Technology, 2019. A full-cost study finding that a conventional human-driven vehicle costs about $0.72 per mile to operate, while a robotaxi—once licensing, insurance, cleaning, remote oversight, and the low utilization of the taxi fleet are counted—costs between $1.58 and $6.01 per mile. Cited in the autonomous-vehicle chapter on robotaxi economics.
https://www.researchgate.net/publication/332743767_Autonomous_Vehicles_and_Public_Health_High_Cost_or_High_Opportunity_Cost.

Occupational Safety and Health Act of 1970, Pub. L. No. 91-596, 84 Stat. 1590. Codified as amended at 29 U.S.C. §§ 651 et seq. Establishes federal workplace safety standards through the Occupational Safety and Health Administration; the General Duty Clause is at 29 U.S.C. § 654.
https://www.law.cornell.edu/uscode/text/29/chapter-15.

Occupational Safety and Health Administration. "29 CFR 1928.51 — Roll-Over Protective Structures for Agricultural Tractors." U.S. Department of Labor. Source for the first of the three regulatory layers discussed in the farm-equipment chapter: the 1976 OSHA rollover

rule requiring rollover protective structures and seatbelts on tractors manufactured after October 25, 1976, when operated by employees.
https://www.osha.gov/laws-regs/regulations/standardnumber/1928/1928.51.

Oil States Energy Services, LLC v. Greene's Energy Group, LLC, 584 U.S. 325 (2018). Seven-to-two decision holding that inter partes review, the Patent Office proceeding for reconsidering the validity of an issued patent, violates neither Article III nor the Seventh Amendment. Justice Thomas's majority opinion characterized a patent as a public franchise and patent validity as a matter of public rights that Congress may assign to an executive agency; it described the holding as narrow, declined to decide whether inter partes review may be applied to patents issued before the Leahy-Smith America Invents Act, and stated that the decision should not be read as holding that patents are not property for purposes of the Due Process Clause or the Takings Clause. In the same passage the majority repeated, from its own older cases, that the grant of a patent is a matter between the public, who are the grantors, and the patentee, and that in issuing a patent the Patent Office takes from the public rights of immense value and bestows them upon the patentee. Justice Breyer, joined by Justices Ginsburg and Sotomayor, concurred. Justice Gorsuch, joined by Chief Justice Roberts, dissented on the ground that an issued patent is private property revocable only by an Article III court. Cited in the chapter on objections considered as adverse authority on the government's power over the patent grant, and distinguished there as a decision about the grant side of the Clause that says nothing about the invention after its patent expires.
https://supreme.justia.com/cases/federal/us/584/16-712/case.pdf.

Old Tacoma Marine. "Fairbanks-Morse." Marine engine history page. States that after the expiration of Rudolf Diesel's American licence in 1912, Fairbanks-Morse entered the large-engine business, its Model Y semi-diesel becoming a standard workhorse in sugar, rice, timber, and mine mills. The page's 1912 date refers to Diesel's American licence; the underlying patent, U.S. Patent No. 608,845, ran to 1915 (see that entry). Cited in the trucking chapter for Fairbanks-Morse as an American firm that entered the diesel-engine business after Diesel's American patent rights ran out.
https://www.oldtacomamarine.com/engines/fairbanks/index.html.

Olson, Mancur. *The Logic of Collective Action: Public Goods and the Theory of Groups*. Cambridge, MA: Harvard University Press, 1965. https://www.hup.harvard.edu/books/9780674537514.

Open Source Initiative. “The Open Source Definition.” Adopted 1998; derived by Bruce Perens from the Debian Free Software Guidelines. The ten-point definition that fixed the term’s meaning: free redistribution, available source, permitted derived works, and no discrimination against persons, groups, or fields of endeavor. Cited in How a Remake Is Built and Certified as part of the lineage credited for the public-record procedure. https://opensource.org/osd.

Open-source software methodology. Topical entry. The procedures the remake chapter borrows, each with its origin: publication of the design (Richard Stallman’s GNU project, announced September 27, 1983, and the Free Software Foundation, 1985); proof at scale (the Linux kernel, 1991, which underlies most of the world’s servers and Android phones); review of each change in the open rather than re-examination of the whole (the practice Eric S. Raymond described in *The Cathedral and the Bazaar*, cited separately); a listed bill of materials with each component’s license status (the SPDX standard, Linux Foundation); verification that a shipped build matches its published source (the Reproducible Builds project); and government certification attached to a public artifact, usable by anyone who uses it unchanged and revalidated when it changes (the FIPS 140 validations of the OpenSSL cryptographic module under NIST’s Cryptographic Module Validation Program). Safety certification of open-source software for vehicles remains a frontier rather than a finished record; the Linux Foundation’s ELISA project is the active effort. Cited in How a Remake Is Built and Certified for the sentence crediting open-source software with the public-record procedure. https://www.gnu.org/gnu/initial-announcement.html; https://spdx.dev/; https://reproducible-builds.org/; https://csrc.nist.gov/projects/cryptographic-module-validation-program; https://elisa.tech/. *See Endnote Commentary.*

Ostrom, Elinor. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press, 1990. https://doi.org/10.1017/CBO9780511807763. See Endnote Commentary.

Owner-Operator Independent Drivers Ass'n, Inc. v. United States Department of Transportation, No. 15-3756 (7th Cir. Oct. 31, 2016), cert. denied, 137 S. Ct. 2246 (2017). Seventh Circuit decision rejecting OOIDA's Fourth Amendment challenge to the FMCSA Electronic Logging Device rule. The court held that the trucking industry's status as a "pervasively regulated industry" under *New York v. Burger*, 482 U.S. 691 (1987), permits the warrantless inspections the ELD rule effectively requires. Cited in the trucking-section of the surveillance chapter as the doctrinal anchor for the proposition that the Fourth Amendment frame has lost on the question of continuous federally-mandated driver surveillance — and as the open door for the Patent Clause and Takings Clause prongs this book develops. https://caselaw.findlaw.com/court/us-7th-circuit/1752477.html.

Owner-Operator Independent Drivers Association (OOIDA). "2024 Freight Rate Survey: Final Report." OOIDA Foundation, January 2025. The current authoritative source for OOIDA's self-described membership: approximately 150,000 members in all fifty states, collectively owning and operating more than 240,000 individual heavy-duty trucks. Source for the trucking chapter's 150,000-member figure used in the standing section. https://www.ooida.com/wp-content/uploads/2025/01/2024-Freight-Rate-Survey-Final-Report.pdf.

Pacific Legal Foundation. Pro bono constitutional litigation organization founded in 1973, based in Sacramento, California. Has won eighteen of its last twenty Supreme Court cases, including *Sheetz v. County of El Dorado* (2024), cited above. Focused on property rights, economic liberty, and constitutional limits on regulatory action. Cited in the trucking chapter as one of three public-interest legal foundations carrying constitutional cases of the kind the Patent Clause argument requires at no cost to the plaintiff. https://pacificlegal.org/.

Palmer, E. D. Histological study of gastric biopsies from a thousand subjects concluding that spiral organisms were contaminants introduced from the mouth, 1954. Cited in the chapter on outsiders and blind doctrine as the study that closed the question for thirty years. Known here through later reviews; the full citation has not been confirmed against the original.

Pardee, Joseph T. “Unusual Currents in Glacial Lake Missoula, Montana.” *Geological Society of America Bulletin* 53, no. 11 (1942): 1569-1600. Identified the water source that resolved the Scablands dispute in Bretz’s favour.

Patent Act of 1790, ch. 7, 1 Stat. 109 (April 10, 1790). Federal Reserve Archival System for Economic Research (FRASER), Federal Reserve Bank of St. Louis. The First Congress did not merely permit an exclusive right; it deployed national authority to secure one. Section 1 directed that letters patent issue “in the name of the United States,” signed by the President. Section 4 armed the right with a cause of action founded on the Act — damages assessed by a jury and forfeiture of the infringing articles to the person aggrieved; the Patent Act of 1793, § 5, placed that action in the circuit courts of the United States. This is the affirmative, coercive sense of the Constitution’s verb “securing.” Note one correction often made in secondary summaries: the familiar treble-damages remedy (three times the price the patentee usually charged) is not in the 1790 Act; it belongs to the Patent Act of 1793, § 5, 1 Stat. 318.
https://fraser.stlouisfed.org/title/patent-act-1790-5734/fulltext.

Patent Act of 1861, ch. 88, 12 Stat. 246 (March 2, 1861). Extended the U.S. utility patent term from fourteen years to seventeen years from the date of grant. The seventeen-year rule remained in force until 1995, when the Uruguay Round Agreements Act harmonized U.S. patent terms with TRIPS at twenty years from the date of filing.
https://en.wikipedia.org/wiki/Term_of_patent_in_the_United_States.

Patent Clause Letter Campaign. Hub page for the campaign’s free, browser-based letter and brochure generators, which produce mailing packages addressed to county sheriffs, gover-

nors and state attorneys general, and tribal nations; readers can consult the hub to see whether a generator has been added for a given plaintiff category. https://marti124.github.io/Patent_Letters/.

Patterson v. Kentucky, 97 U.S. 501 (1878). Foundational Supreme Court decision holding that the existence of a federal patent does not exempt the tangible product embodying the invention from state police-power regulation aimed at the safety of the public. The Kentucky statute at issue banned the sale of a patented "improved burning oil" that the state inspector had condemned as unsafe; the patentee's argument that his federal patent placed the oil beyond state regulatory reach was rejected unanimously. Cited in the chapter on objections considered as the principal authority for the proposition (which the book accepts) that patents do not confer immunity from regulation. https://supreme.justia.com/cases/federal/us/97/501/.

Peace Corps, Establishment of. Executive Order 10924, "Establishment and Administration of the Peace Corps in the Department of State," March 1, 1961, signed by President John F. Kennedy; authorized by Congress in the Peace Corps Act, Pub. L. No. 87-293, 75 Stat. 612, approved September 22, 1961. Cited in the remedy chapter as the verified historical instance of a President standing up a federal institution by executive order, later ratified by statute. National Archives, "Executive Order 10924: Establishment of the Peace Corps." https://www.archives.gov/milestone-documents/executive-order-10924.

Penn Central Transportation Co. v. City of New York, 438 U.S. 104 (1978). https://supreme.justia.com/cases/federal/us/438/104/.

Penn State Extension. "A Summary of Laws and Regulations Related to Agricultural Safety and Machinery." Synthesizing source for the federal regulatory landscape surrounding agricultural equipment manufacture and operation, drawn upon in the farm-equipment chapter's diagnostic of the three regulatory layers (OSHA, EPA, NHTSA). https://extension.psu.edu/a-summary-of-laws-and-regulations-related-to-agricultural-safety-and-machinery.

Pennock v. Dialogue, 27 U.S. (2 Pet.) 1 (1829). Held that an inventor who permits public use or sale of his invention before applying for a patent forfeits the right to one (an ancestor of the on-sale bar). In reasoning the Court has treated as settled ever since, Justice Story explained that the main object of the patent system is to promote the progress of science and useful arts, best served by giving the public at large the right to make, construct, use, and vend the invention at as early a period as possible, with due regard to the rights of the inventor. The opinion construes the Patent Act of 1793. Cited in the Patent Clause prong chapter for the public's half of the bargain, stated early and by a Madison appointee. https://supreme.justia.com/cases/federal/us/27/1/.

Pennsylvania Coal Co. v. Mahon, 260 U.S. 393 (1922). https://supreme.justia.com/cases/federal/us/260/393/.

Perkins, Jacob. U.S. Patent for a nail-making machine, granted 1795 (term expired 1809 under the fourteen-year rule of the Patent Act of 1790). The Perkins machine and its post-expiration competitors collapsed the American price of nails by roughly seventy percent across the nineteenth century. The nail-making entry in the chapter's eight-product table. https://en.wikipedia.org/wiki/Jacob_Perkins.

Pipeline and Hazardous Materials Safety Administration. "Specification for DOT-406 Cargo Tank Motor Vehicles and Hazardous Materials Transportation Requirements." Codified at 49 C.F.R. Parts 171–180. U.S. Department of Transportation. Source for the distinct federal hazardous-materials regime governing fuel-tanker operators: bulk packaging above 119 gallons triggers hazmat and tanker-endorsement requirements, the cargo tank itself must meet the DOT-406 construction specification, and the driver must hold a hazardous-materials endorsement and clear a Transportation Security Administration background check. Cited in the trucker chapter's fuel-hauler subsection. https://www.ecfr.gov/current/title-49/subtitle-B/chapter-I/subchapter-C.

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Reid v. Covert, 354 U.S. 1 (1957). Held that civilian dependents of servicemembers abroad could not be tried by court-martial for capital offenses under executive agreements with the host nations, because the Fifth and Sixth Amendments guaranteed them indictment and jury trial; the treaty power and the Necessary and Proper Clause could not supply what the Bill of Rights withheld. Justice Black's plurality opinion, for four Justices, states the principle in its broadest form: no agreement with a foreign nation can confer on Congress or any branch a power "free from the restraints of the Constitution." Justices Frankfurter and Harlan concurred in the result on narrower grounds; the judgment is binding, the plurality's wording is not. Cited in The Patent Clause Prong as the clearest statement that a power granted in one place cannot escape a limit written in another.
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Pony Express — with photographs from the Center's collection. The Buffalo Bill Center of the West is a Smithsonian Affiliate. https://centerofthewest.org/2021/08/20/parts-of-a-western-saddle-and-the-variations/. See Endnote Commentary.

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S. 1568 (119th Cong., 2025) — Liberating Incandescent Technology (LIT) Act of 2025. Introduced May 1, 2025 by Senator Mike Lee (R-UT), Chairman of the Senate Committee on Energy and Natural Resources, with five original cosponsors — Senators John Curtis (R-UT), Jim Justice (R-WV), Tom Cotton (R-AR), Bill Hagerty (R-TN), and Eric Schmitt (R-MO) — joined after introduction by Senator Bernie Moreno (R-OH), bringing the total to six; House companion (H.R. 3341) introduced May 13, 2025 by Representative Craig Goldman (R-TX), with cosponsors Representatives Michael Rulli (R-OH) and Robert Onder (R-MO). The bill repeals Section 325(i) of the Energy Policy and Conservation Act, repeals the two May 2022 DOE general service lamps final rules (87 Fed. Reg. 27439 and

87 Fed. Reg. 27461), and repeals the April 2024 DOE final rule (89 Fed. Reg. 28859). Referred to the Senate Committee on Energy and Natural Resources and the House Committee on Energy and Commerce; no hearing held, no committee mark-up, no floor action through the period covered by the healthy-lighting chapter. https://www.energy.senate.gov/2025/5/chairman-lee-introduces-bill-to-end-biden-era-lightbulb-ban.

S. 3007 (119th Cong., 2025) — Diesel Truck Liberation Act. Introduced October 2025 by Senator Cynthia Lummis (R-Wyoming). Companion bill introduced in the House in March 2026. Removes Environmental Protection Agency authority over emission-control devices on heavy-duty diesel trucks. Status as of this writing: in committee. Source for the legislative-parallel reference in the trucking chapter. https://www.congress.gov/bill/119th-congress/senate-bill/3007.

Sackett v. Environmental Protection Agency, 598 U.S. 651 (2023). A landowning couple challenged the reach of a federal Clean Water Act rule over their own property and prevailed at the Supreme Court without first incurring a penalty. Cited in the plaintiff chapter as an example of a directly regulated party reaching the Court through a pre-enforcement challenge — the party the rule injured, suing in its own name, without first breaking the rule. https://supreme.justia.com/cases/federal/us/598/651/.

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Self-certification and certification by simulation. The United States does not pre-approve motor vehicles before sale; the National Traffic and Motor Vehicle Safety Act establishes a self-certification process under which manufacturers certify that their vehicles comply with all applicable Federal Motor Vehicle Safety Standards. NHTSA has confirmed that a manufacturer exercising reasonable care may certify compliance through simulation or engineering analysis, not only through the agency's physical test procedures. Independent engineering and homologation firms perform such certification work today, at both the whole-vehicle and component level. NHTSA, Notice Regarding the Applicability of NHTSA FMVSS Test Procedures to Certifying Manufacturers, 85 Fed. Reg. 83,502 (Dec. 21, 2020). Cited in How a Remake Is Built and Certified for two points: that certification by engineering analy-

sis is already lawful, so the analysis may use any tool, and that the federal statute asks no state license of the engineer who certifies, so the remedy adds none.
https://www.federalregister.gov/documents/2020/12/21/2020-28107/notice-regarding-the-applicability-of-nhtsa-fmvss-test-procedures-to-certifying-manufacturers.

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See Endnote Commentary.

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https://supreme.justia.com/cases/federal/us/601/22-1074/.

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https://supreme.justia.com/cases/federal/us/299/476/.

Shoshone Tribe v. United States (1937; covered in the Takings Clause prong chapter). The federal government had settled another tribe on land guaranteed to the Shoshone by treaty, diminishing their interest. The Supreme Court held the loss compensable. Justice Cardozo's opinion supplied the phrase this book relies on — "Spoliation is not management" — and

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https://saddleschool.wordpress.com/2012/10/18/some-history-of-the-western-saddle/. See Endnote Commentary.

Signify N.V. Annual Report 2025. Signify N.V. (formerly Philips Lighting), Eindhoven, Netherlands. The current annual report of the lighting company that emerged from the May 2016 separation of Philips’ Lighting division. Documents that Signify holds more than twenty thousand patents (with the cumulative portfolio reflecting decades of Philips Lighting investment in light-emitting-diode and earlier lighting technologies; the foundational compact-fluorescent patents dating to the 1980 Philips SL lamp have long since expired, while the in-force portfolio is concentrated in solid-state lighting); that annual research and development spending is reported at approximately 250 to 271 million euros, the great majority of it on solid-state lighting and adjacent technologies. Cited in the healthy-lighting chapter’s bootleggers-and-Baptists section as the primary-source anchor for the patent-protected business model that benefits structurally from the displacement of the unpatentable incandescent.
https://www.signify.com/static/2025/signify-annual-report-2025.pdf.

Simon v. Eastern Kentucky Welfare Rights Organization, 426 U.S. 26 (1976). Holding that indigent plaintiffs lacked standing to challenge an Internal Revenue Service ruling that relaxed the charity-care condition for hospitals’ tax exemption, because whether the hospitals would serve them depended on the independent decisions of hospitals not before the court.

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https://www.gutenberg.org/ebooks/140.

Singer Manufacturing Co. v. June Manufacturing Co., 163 U.S. 169 (1896).
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Sinotruk (Hong Kong) Limited. Corporate page and HOWO 6×4 tractor product listings. The Sinotruk corporate page reports product sales in more than ninety countries; the HOWO 6×4 tractor is the company's current-production heavy semi-truck platform, with verified export pricing from suppliers and international trade-listing sites in the range of $28,000 to $35,000 FOB China for standard 371–420 horsepower configurations. The price comparison is cited in the trucking chapter as evidence of what the global market delivers for a no-aftertreatment Class 8 platform built to Euro II emission standards — not as a recommendation to import.
https://www.sinotrukgroup.com/.

Skidmore v. Swift & Co., 323 U.S. 134 (1944).
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Skilsaw and the portable circular saw, 1924. Edmond Michel, a French immigrant living in New Orleans, built the first electric handsaw in 1923 by mounting a six-inch blade on a wooden frame and driving it with a motor taken from a malted-milk mixer. Joseph W. Sullivan read a newspaper account of the invention, sought Michel out, and the two

founded the Michel Electric Handsaw Company in Chicago in 1924; they named the saw the Skilsaw. The firm was renamed Skilsaw Incorporated in 1926 after Michel left it, and the die-cast aluminum Model E followed in 1928. Cited in the preface for the date the portable circular saw reached the market, which the trade did not adopt in quantity until after the Second World War. Skil, “History” (corporate timeline). https://www.skileurope.com/uk/history. Companion trade-press account: Industrial Equipment News, “IEN 75th Anniversary Perspective & Timeline: Skil Power Tools,” December 2, 2008, https://www.ien.com/operations/news/21098125/ien-75th-anniversary-perspective-timeline-skil-power-tools (metered access).

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https://web.mit.edu/ebj/www/doc/JAPAv61n1.pdf.

Sperling, Daniel, Ethan C. Abeles, David S. Bunch, Andrew Burke, Belinda S. Chen, Kenneth S. Kurani, and Thomas S. Turrentine. “The Price of Regulation.” *ACCESS Magazine*, no. 25 (Fall 2004): 9–18. University of California Transportation Center, summarizing research prepared for the California Air Resources Board under contract 02-310. The article estimates total federal regulatory cost (safety, emissions, and fuel-economy compliance) on a 2001 passenger vehicle at $2,500–$4,020 in then-current dollars, against approximately $11 for the same regulatory categories on a 1967 vehicle. The proportional finding — that more than 99 percent of the cumulative federal regulatory cost on a new vehicle by 2001 was added after 1970 — is the source the cars-and-trucks chapter relies on for the post-1970 attribution. The authors’ preferred estimate is the lower figure ($2,500, of which approximately $1,000 is emissions compliance and less than $1,500 is safety compliance); the higher figure ($4,020) is the BEA/BLS estimate compiled in Ward’s Automotive Yearbook.
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https://supreme.justia.com/cases/federal/us/221/1/.

Star Athletica, L.L.C. v. Varsity Brands, Inc., 580 U.S. 405 (2017). Establishes the governing separability test under 17 U.S.C. § 101: a feature incorporated into the design of a useful article is eligible for copyright protection only if it can be perceived as a two- or three-dimensional work of art separate from the useful article and would qualify as a protectable pictorial, graphic, or sculptural work on its own or in another medium if imagined separately from the article. The Court noted that the test does not render the underlying useful article itself eligible for copyright. Cited in the chapter *Objections Considered* for the limit it places on copyright claims over vehicles: a separable badge or graphic may qualify, the vehicle may not.
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https://www.legislation.gov.uk/aep/Ja1/21/3.

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vote on a law and in all cases of a personal nature. Article 63 is the constitutional foundation for the Italian parliament's *voto segreto* practice from 1848 to 1922 under direct constitutional mandate — the first span within the 119 years the Hybrid Voting chapter refers to. The post-war Republican Constitution's Article 64 transferred chamber voting procedure to each chamber's own rules of procedure, under which *voto segreto* continued in practice from 1948 until the October 1988 chamber-rules reform under the De Mita government. https://en.wikisource.org/wiki/Statuto_Albertino. See Endnote Commentary.

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https://apps.abacus.ai/chatllm/share/conversation/18d6a5d7:13f78811d8 ;
https://datamp.org/patents/advance.php?pn=187X&id=36942&set=3

Stigler, George J. "The Theory of Economic Regulation." *Bell Journal of Economics and Management Science* 2, no. 1 (Spring 1971): 3–21.
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TrafFix Devices, Inc. v. Marketing Displays, Inc., 532 U.S. 23 (2001). Holds that a functional product feature cannot receive trade dress protection under the Lanham Act, and that

an expired utility patent is strong evidence that the features it claimed are functional. Where the expired patent claimed the feature in question, the party seeking trade dress protection carries a heavy burden of showing that the feature is merely ornamental, incidental, or arbitrary. Cited in the chapter *Objections Considered* as the doctrinal bar that prevents trade dress from being used to recapture a functional design after the patent on it has expired — the trademark-side counterpart to the patent-side rule of *Bonito Boats, Inc. v. Thunder Craft Boats, Inc.* and *Sears, Roebuck & Co. v. Stiffel Co.*.
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United Food and Commercial Workers Union Local 751 v. Brown Group, Inc., 517 U.S. 544 (1996). Construing the third prong of the associational-standing test of *Hunt v. Washington State Apple Advertising Commission*. The Court explained that individual participation by members is not normally necessary when an association seeks prospective or injunctive relief on their behalf, but would be required in an action for damages, so that an association's damages action running solely to its members is barred for want of standing. Cited in the citizen's-playbook chapter and in *Objections Considered* for the proposition

that a membership association is available as plaintiff for the remedy this book seeks and unavailable for a damages claim.
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United States Code. “17 U.S.C. § 101 — Definitions (‘useful article’ and ‘pictorial, graphic, and sculptural works’).” Copyright Act of 1976, as amended. Defines a useful article as one having an intrinsic utilitarian function that is not merely to portray the appearance of the article or to convey information, and provides that the design of a useful article is protected only to the extent it incorporates features that can be identified separately from, and are capable of existing independently of, the utilitarian aspects of the article. Cited in the chapter *Objections Considered*.
https://www.law.cornell.edu/uscode/text/17/101.

United States Code. “17 U.S.C. § 302(c) — Duration of Copyright: Anonymous Works, Pseudonymous Works, and Works Made for Hire.” Provides that copyright in a work made for hire endures for ninety-five years from the year of first publication or one hundred twenty years from the year of creation, whichever expires first. Cited in the chapter *Objections Considered* as the source of the ninety-five-year figure a correspondent attributed to whole vehicles; the term is correctly stated but does not attach to a useful article.
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United States Code. “17 U.S.C. § 1201 — Circumvention of Copyright Protection Systems.” Enacted by the Digital Millennium Copyright Act, Pub. L. No. 105-304, 112 Stat. 2860 (1998). Forbids circumventing a technological measure that controls access to a copyrighted work, and trafficking in tools that do so; narrow exceptions cover reverse engineering for interoperability (§ 1201(f)), encryption research, and security testing. Fair use, a defense to infringement, has been held no defense to circumvention: Universal City Studios, Inc. v. Corley, 273 F.3d 429 (2d Cir. 2001). The Federal Circuit reads the ban more narrowly, requiring a link to infringement: Chamberlain Group, Inc. v. Skylink Technologies, Inc., 381 F.3d 1178 (Fed. Cir. 2004). Cited in the chapter *Objections*

Considered for the digital lock that can bar a remanufacturer from reading the original engine-control firmware.
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United States Code. “42 U.S.C. § 7521(a)(1) — Emission Standards for New Motor Vehicles or New Motor Vehicle Engines.” Cornell University Legal Information Institute. The Clean Air Act provision granting EPA authority to prescribe emission standards “applicable to the emission of any air pollutant from any class or classes of new motor vehicles or new motor vehicle engines.” Cited in the trucking chapter as the statutory grandfathering principle: EPA’s authority reaches engines at the time of manufacture, fixing each model year’s emission certification as the perpetual standard for the life of that engine. A 1999 model-year heavy-duty diesel certified to the 1998 standards (4.0 g/bhp-hr NOx, 0.10 g/bhp-hr PM) remains subject to those standards, and only those standards, for its operational life.
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United States Code. “49 U.S.C. § 30112(a) — Prohibitions on Manufacturing, Selling, and Importing Noncomplying Motor Vehicles” and “49 U.S.C. § 30115 — Certification of Compliance.” Cornell University Legal Information Institute. Section 30112(a) forbids any person to manufacture for sale, sell, offer for sale, introduce or deliver for introduction into interstate commerce, or import a motor vehicle manufactured on or after the date an applicable Federal Motor Vehicle Safety Standard takes effect unless the vehicle complies with the standard and is covered by a § 30115 certification. Cited in the chapter on objections considered to establish that a faithful new copy of a patent-expired vehicle, built to its original platform-year design, cannot be lawfully manufactured for sale or sold today — the standards binding by date of manufacture, so a vehicle built now must meet the standards in

force now. Cited also in How a Remake Is Built and Certified, where the date-of-manufacture rule supplies the only sense of "unsafe" the remedy recognizes.
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United States Code. "49 U.S.C. § 30118 — Notification of Defects and Noncompliance" and "49 U.S.C. § 30120 — Remedies for Defects and Noncompliance." Cornell University Legal Information Institute. Section 30118 provides for a recall when a motor vehicle contains a defect related to motor vehicle safety or does not comply with an applicable Federal Motor Vehicle Safety Standard — applicable meaning the standard in force on the vehicle's date of manufacture under § 30112(a), so that a later standard never recalls an earlier design. Section 30120 fixes the remedies: repair, replacement, or refund of the purchase price. Cited in How a Remake Is Built and Certified as the source of the only sense of "unsafe" the remedy uses, and of the three-way rule that a recalled-and-repaired design enters the commons as repaired, while a design recalled and bought back, or banned, is excluded. The chapter's rule, drawn from these sections: a design recalled and repaired is not excluded; the remake is certified to the design as it stood at the end of its model year, with every repair made by then, and to every standard in force at any time during that year; repairs ordered after that date are handled under recall law as any maker handles them. For this pur-

pose the model year ends on the day the first unit of the next model year of that make and model was delivered to a buyer in the United States.
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United States Department of Agriculture v. Moreno, 413 U.S. 528 (1973).
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United States International Trade Commission. *Remanufactured Goods: An Overview of the U.S. and Global Industries, Markets, and Trade*. Investigation No. 332-525, USITC Publication 4356. Washington, DC: USITC, October 2012. Source for the figure cited in the chapter on the wall that the United States is the world’s largest producer of remanufactured goods, valued at no less than $43 billion in annual production over 2009–2011 and supporting some 180,000 full-time jobs, with motor-vehicle parts among the three largest sectors. The chapter cites the report to distinguish the thriving trade in remanufactured *components* below the certification gate from the assembled patent-expired *products* the wall blocks at the gate.
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discussed in this chapter expired no later than the late 1980s. For applications filed on or after June 8, 1995, the term is twenty years from filing; Chapter 3 relies on that rule in dating tire-pressure-monitor patents filed before 2009 as expired by the end of 2028, subject to any term adjustment on an individual patent. Accessed May 9, 2026. https://www.uspto.gov/web/offices/pac/mpep/s2701.html.

United States v. 50 Acres of Land, 469 U.S. 24 (1984). https://supreme.justia.com/cases/federal/us/469/24/.

United States v. Ballin, 144 U.S. 1 (1892). The closest Supreme Court case to address the Journal Clause of Article I, Section 5, Clause 3. In *Ballin* the Court held the House Journal to be conclusive evidence of what occurred during a contested quorum vote, but did not interpret whether the Yeas-and-Nays override ("at the Desire of one fifth of those Present, be entered on the Journal") constitutionally requires individual named records or only an aggregate tally. The case is cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that no Supreme Court holding has squarely decided the interpretive scope of the Yeas-and-Nays override, leaving the question one of first impression. https://supreme.justia.com/cases/federal/us/144/1/.

United States v. Lopez, 514 U.S. 549 (1995). Five-to-four decision holding the Gun-Free School Zones Act of 1990 beyond the commerce power. Cited for the shape of an enumerated-powers question: the Court asked whether the regulated activity fell within the power, not whether Congress could rationally have thought the statute wise. https://supreme.justia.com/cases/federal/us/514/549/.

United States v. Moghadam, 175 F.3d 1269 (11th Cir. 1999), cert. denied, 529 U.S. 1036 (2000). The Eleventh Circuit's decision upholding the federal anti-bootlegging statute under the Commerce Clause while assuming that the Copyright Clause could not sustain it, live performances being unfixed and so outside the term "Writings." The court held that the commerce power may not be used where the legislation is fundamentally inconsistent with the limitations of Article I, Section 8, Clause 8, and its footnote identified the statute's lack

of any express time limit as the likely site of that inconsistency. The court did not decide the point; the appellant had not preserved it. https://law.justia.com/cases/federal/appellate-courts/F3/175/1269/637322/.

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icy on human cell, tissue, and cellular and tissue-based products considered “more than minimally manipulated.” https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/regenerative-medicine-advanced-therapy-designation.

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patents, copyrights, trademarks, and trade secrets. Cited in the chapter *Why Nobody Noticed*, and in the Preface for the 13 percent figure. https://static1.squarespace.com/static/5ed4066e709a614ffb39f898/t/634d92189b1e4d7e226418d1/1666028057285/USIPA_Survey_on_Intelectual_Property_REPORT_Q422.pdf.

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U.S. Senate Committee on Agriculture, Nutrition, and Forestry. “Revisiting Farm Production Expenses.” Minority Blog. Source for the figure of approximately $500 billion in U.S. farm production expenditures in 2023 and the 28 percent two-year increase (the sharpest on record) cited in the farm-equipment chapter’s treatment of how equipment cost flows through the food supply chain. https://www.agriculture.senate.gov/newsroom/minority-blog/revisiting-farm-production-expenses.

Vargo, Jason, Benjamin G. Gerhardstein, Geoffrey P. Whitfield, and Arthur Wendel. “Bicyclist Deaths Associated with Motor Vehicle Traffic — United States, 1975–2012.” *Morbidity and Mortality Weekly Report* 64, no. 31 (August 14, 2015): 837. Centers for Disease Control and Prevention. Analyzes FARS bicycle fatality data for the period 1975–2012 and reports cyclist mortality on a per-population rate basis (a methodology that addresses denominator concerns inherent in absolute-count comparisons). Cyclist mortality rate fell from 0.41 to 0.23 deaths per 100,000 across the study period — a 44 percent decline. The steepest decline occurred among children aged under 15; age-specific mortality rates for adults aged 35–74 increased over the same period. The bicycles section of the modern-bargain chapter cites the Vargo et al. rate figure as the population-adjusted counterpart to the chapter’s absolute-count IIHS figure, making possible an apples-to-apples comparison against the per-vehicle-mile fatality rate the chapter cites for the automobile. https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6431a1.htm.

Vaughn, Mark. “FCA Wins Legal Battle Against Mahindra and Roxor.” *Autoweek*, June 12, 2020. https://www.autoweek.com/news/industry-news/a32851849/fca-wins-legal-battle-against-mahindra-and-roxor/. The dispute, before the U.S. International Trade Commission, turned on Jeep’s trade dress—the vehicle’s ornamental appearance, including its boxy body, flat vertical sides, and round headlights—and not on the vehicle’s mechanical design or on Mahindra’s underlying license. The Commission found the 2018–2019 Roxor infringed that trade dress while violating none of FCA’s registered trademarks. Mahindra restyled the front end, and in December 2020 the Commission cleared the redesigned vehicle, which resumed sale in the United States mechanically unchanged.

Verge, The. Trade-press report on General Motors CEO Mary Barra’s October 2025 confirmation that the company will phase out support for Apple CarPlay and Android Auto across its entire vehicle lineup — internal-combustion and electric — in favor of a proprietary infotainment system built on Google’s Android Automotive operating system, with full transition expected by the 2028 model year. October 22, 2025. https://www.theverge.com/transportation/804562/gm-apple-carplay-android-auto-gas-cars-mary-barra.

Vesuvius Challenge. “First Word Discovered in Unopened Herculaneum Scroll by 21yo Computer Science Student.” 12 October 2023. https://scrollprize.org/firstletters. Records the First Letters Prize of forty thousand dollars to Luke Farritor for more than ten characters recovered from roughly four square centimetres of scroll, the word πορφύρας, the second-place award to Youssef Nader, and the First Ink Prize to Casey Handmer whose crackle pattern supplied Farritor’s training data. The seven-hundred-thousand-dollar grand prize followed in December 2023 to Farritor, Nader and Julian Schilliger for about two thousand characters. Scans produced by Brent Seales and EduceLab at the University of Kentucky; readings confirmed by papyrologists.

Vesuvius Challenge. Project site and results archive. Documents the open competition to read the carbonized Herculaneum papyri without unrolling them. In August 2023 Casey Handmer, a former Jet Propulsion Laboratory physicist, identified a faint “crackle” texture

in the scan data as ink — the first ink found inside an unopened scroll — and took the First Ink Prize for it. Building on Handmer's discovery, Luke Farritor, then a twenty-one-year-old undergraduate in computer science with no training in papyrology or classical Greek, trained a model on the crackle shapes and in October 2023 recovered the Greek word for purple from inside an unopened scroll, winning the First Letters Prize; papyrologists confirmed the reading. Farritor subsequently shared the competition's $700,000 Grand Prize with Youssef Nader and Julian Schilliger for recovering roughly fifteen columns of a previously unread philosophical text. Cited in the preface as an instance in which an outsider supplied the method and specialists inside the field supplied the domain judgment. https://scrollprize.org/grandprize See Endnote Commentary.

Village of Arlington Heights v. Metropolitan Housing Development Corp., 429 U.S. 252 (1977). Holding that a nonprofit developer that had contracted for a specific site and planned a specific low- and moderate-income housing project there had standing to challenge the village's refusal to rezone the site, although the project still depended on financing and other contingencies, and that a prospective resident who sought housing near his job had standing as well; on the merits the Court found no proof of discriminatory purpose. Justice Powell wrote for the Court; Justice Marshall, joined by Justice Brennan, concurred in part and dissented in part; Justice White dissented; Justice Stevens took no part. Cited in the chapter on objections considered as the authority for the standing of a would-be builder with a definite project blocked by law. https://supreme.justia.com/cases/federal/us/429/252/.

von Neumann, John, and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton: Princeton University Press, 1944. https://press.princeton.edu/books/paperback/9780691130613/theory-of-games-and-economic-behavior.

Walk-Behind Power Lawn Mower Safety Standard. Consumer Product Safety Commission, 16 C.F.R. Part 1205. Final Rule, 47 Fed. Reg. 1880 (January 14, 1982). Mandatory product-safety standard issued under the Consumer Product Safety Act, 15 U.S.C. §§ 2051 et seq. https://www.ecfr.gov/current/title-16/chapter-II/subchapter-B/part-1205.

Walterscheid, Edward C. *The Nature of the Intellectual Property Clause: A Study in Historical Perspective*. Buffalo, NY: William S. Hein, 2002. https://www.amazon.com/Nature-Intellectual-Property-Clause-Perspective/dp/1575887096.

Warth v. Seldin, 422 U.S. 490 (1975). Five-to-four decision holding that low- and moderate-income residents of the Rochester, New York, area and allied organizations lacked standing to challenge the zoning ordinance of the suburb of Penfield as excluding them, because they had not shown that any specific project would have been built absent the ordinance or that relief would let them live there. Justice Powell wrote for the Court; Justice Douglas dissented, and Justice Brennan, joined by Justices White and Marshall, dissented. Cited in the chapter on objections considered as the government's closest authority against a buyer whose relief depends on a producer's decision to build. https://supreme.justia.com/cases/federal/us/422/490/.

Washington Examiner. "Industry, Not Environmentalists, Killed Incandescent Bulbs." Washington Examiner opinion analysis. Documents the role of patent-protected lighting incumbents — through the National Electrical Manufacturers Association, representing GE Lighting, Signify (formerly Philips), Osram, Cree, and Acuity — in lobbying for the federal efficiency standards that displaced their unpatentable incandescent competitor. Cited in the bootleggers-and-Baptists section of the healthy-lighting chapter. https://www.washingtonexaminer.com/opinion/2372297/industry-not-environmentalists-killed-incandescent-bulbs/.

Webber v. Virginia, 103 U.S. 344 (1880). Supreme Court decision reaffirming the Patterson v. Kentucky principle that letters patent granted by the United States do not exclude from state tax or license law the tangible property in which the invention or discovery is embodied. Cited in the chapter on objections considered as part of the Supreme Court line establishing that patents and regulation coexist. https://supreme.justia.com/cases/federal/us/103/344/.

West Virginia v. Environmental Protection Agency, 597 U.S. 697 (2022). https://supreme.justia.com/cases/federal/us/597/20-1530/.

Westerman, Wayne, and John G. Elias. “Method and Apparatus for Integrating Manual Input.” U.S. Patent 6,323,846, filed January 25, 1999, issued November 27, 2001. Assigned to the University of Delaware. Foundational multi-touch capacitive touchscreen patent; expired 2019. Cited in the surveillance chapter as the foundational expired multi-touch patent from which a remanufactured vehicle’s in-dash capacitive touchscreen can be built. https://patents.google.com/patent/US6323846B1.

Wheaton v. Peters, 33 U.S. (8 Pet.) 591 (1834). Foundational Supreme Court decision holding that there is no common-law copyright after publication; intellectual-property rights exist only as Congress grants them under the Patent and Copyright Clause. Cited in the chapter on objections considered as the holding that establishes the Patent Clause as the sole constitutional source of intellectual-property rights, foreclosing the defense that a patent-equivalent commercial protection without an expiration date is something other than a patent right. https://supreme.justia.com/cases/federal/us/33/591/.

Whitney, Eli. U.S. Patent for the cotton gin, granted 1794 (term expired 1808 under the fourteen-year rule of the Patent Act of 1790). The cotton gin entry in the chapter’s eight-product table. Whitney’s during-patent commercial model was service-based at a rate equivalent to roughly nineteen hundred dollars per gin per year; after expiration, mass-produced gins were selling for about twelve hundred dollars by the 1830s and roughly five hundred dollars by the 1880s, a fifty-six percent real-terms drop. https://en.wikipedia.org/wiki/Cotton_gin.

Whittemore, Amos. U.S. Patent for the cotton carding machine, granted 1797 (term expired 1811 under the fourteen-year rule of the Patent Act of 1790). The cotton carding entry in the chapter’s eight-product table. The Whittemore design fell from a during-patent price of about thirty cents per pound of carded cotton to ten cents by the 1880s, a fifty percent real-terms drop. https://en.wikipedia.org/wiki/Amos_Whittemore.

Wickard v. Filburn, 317 U.S. 111 (1942). The Supreme Court held that Congress may, under the Commerce Clause, regulate even wheat a farmer grew for consumption on his own farm, on the ground that the aggregate effect of such activity on interstate commerce brought it within the commerce power. The decision marks the modern outer reach of that power. Cited in the chapter on the Patent Clause prong as the authority the regulatory wall's defenders rely on for the breadth of federal regulatory jurisdiction, and as the case the prong answers: the commerce power, however broad, does not extend to nullifying what the Patent Clause commands at the end of the patent term.
https://supreme.justia.com/cases/federal/us/317/111/.

Wilkinson, David. U.S. Patent for a screw-cutting machine, granted 1798 (term expired 1812 under the fourteen-year rule of the Patent Act of 1790). The screw-cutting entry in the chapter's eight-product table. The Wilkinson design fell from a during-patent price of about four dollars per pound of cut screws to roughly sixty cents per pound by the 1880s, a sixty-four percent real-terms drop, as automated threading entered competing American shops.
https://en.wikipedia.org/wiki/David_Wilkinson_(mechanic).

Williamson v. Lee Optical of Oklahoma, Inc., 348 U.S. 483 (1955). The unanimous Supreme Court decision establishing the modern rational-basis floor for federal economic classifications: a court asks only whether the classification is rationally related to some conceivable legitimate government interest, not whether the stated rationale is the real one. The decision upheld an Oklahoma statute requiring eyeglass replacement lenses to be made only from prescriptions written by licensed optometrists or ophthalmologists, against the challenge that the statute was economic protectionism dressed as health regulation. Cited in the chapter on objections considered as the doctrinal wall that almost every Fifth Amendment equal-protection challenge to federal economic regulation runs into and breaks on, and the principal reason equality under the law cannot serve as a standalone constitutional prong of the book's argument.
https://supreme.justia.com/cases/federal/us/348/483/.

World Trade Organization. Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). Annex 1C of the Marrakesh Agreement Establishing the World Trade

Organization, signed at Marrakesh on 15 April 1994. Article 33 establishes the minimum patent term as twenty years from the filing date. The Agreement binds all 166 WTO member nations.
https://www.wto.org/english/docs_e/legal_e/27-trips_04c_e.htm.

Wyden, Ron, and Edward J. Markey. Letter to FTC Chair Lina M. Khan. July 26, 2024. United States Senate. Joint senatorial inquiry to the Federal Trade Commission urging investigation into automaker disclosures of connected-vehicle driving-behavior data to consumer-reporting brokers; draws on oversight responses obtained directly from General Motors, Honda, and Hyundai documenting per-vehicle broker payments on the order of twenty-six cents (Honda) and sixty-one cents (Hyundai). Source for the surveillance chapter's documentation of the broker-payment magnitudes.
https://www.wyden.senate.gov/imo/media/doc/wyden-markey_auto_privacy_letter_to_ftc.pdf.

XBOW. First non-human to top HackerOne's United States leaderboard, June 2025, as reported in "How an AI-Based 'Pen Tester' Became a Top Bug Hunter on HackerOne," *Dark Reading*, August 2025; HackerOne co-founder Michiel Prins confirmed the ranking to Bloomberg. Roughly 1,060 vulnerability reports submitted over a few months, including a previously unknown flaw in Palo Alto Networks' GlobalProtect affecting more than 2,000 hosts. Cited in the preface.
https://www.darkreading.com/vulnerabilities-threats/ai-based-pen-tester-top-bug-hunter-hackerone.

Yandle, Bruce. "Bootleggers and Baptists: The Education of a Regulatory Economist." *Regulation* 7, no. 3 (May/June 1983): 12–16.
https://www.cato.org/sites/cato.org/files/serials/files/regulation/1983/5/v7n3-3.pdf.

Yao, Xiaoxi, et al. "Artificial Intelligence-Enabled Electrocardiograms for Identification of Patients with Low Ejection Fraction: A Pragmatic, Randomized Clinical Trial." *Nature Medicine* 27, no. 5 (2021): 815-819. One hundred twenty primary care teams, 22,641 adults; new diagnoses rose from 1.6 to 2.1 percent, an odds ratio of 1.32.

Zoltek Corp. v. United States, 442 F.3d 1345 (Fed. Cir. 2006) (per curiam), reh'g en banc denied, 464 F.3d 1335 (Fed. Cir. 2006), cert. denied, vacated in part, 672 F.3d 1309 (Fed. Cir. 2012) (en banc). The panel held that a patent holder whose invention the United States uses may not bring a Fifth Amendment takings claim, and that the statutory remedy Congress supplied is the only one available. Judge Plager dissented from the panel opinion; Judge Newman dissented from the denial of rehearing en banc. In 2012 the court sitting en banc vacated the 2006 opinion, giving as its grounds the panel's construction of that statute. Commentators divide on what the vacatur did to the constitutional holding: some treat the takings analysis as vacated with the rest, others report that the reasoning was never repudiated and that later filings continue to cite the case for it. A Federal Circuit decision, not a Supreme Court decision. Cited in the objections chapter.
https://law.justia.com/cases/federal/appellate-courts/F3/442/1345/642048/.

Zwicky, Fritz. "Die Rotverschiebung von extragalaktischen Nebeln." *Helvetica Physica Acta* 6 (1933): 110-127. Applied the virial theorem to the Coma cluster and named the discrepancy *dunkle Materie*.
See Endnote Commentary.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "Corporate Introduction." ZX Auto Global Official Site. Source for ZX Auto's claim of operations in more than 90 countries and regions, cited in the farm-equipment chapter alongside Mahindra's Global Presence page as one of two manufacturer pages from which 43 countries can be individually extracted by name against the manufacturers' own claimed totals.
https://en.zxauto.com.cn/CORPORATE.html.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "GRANDTIGER." ZX Auto Global Official Site. Manufacturer product page for the ZX Grand Tiger pickup truck. Specifications include 2.4L gasoline and 2.8T diesel engine options, 5-speed manual transmission, 2WD/4WD, double cab, and a 5310×1812×1800mm body. The chapter on patent-expired international vehicles cites the Grand Tiger as a working pickup truck derived from the 1997-to-2004 Toyota Hilux platform whose core mechanical patents have been in the public domain for more than a decade. Export pricing runs roughly $13,000 to $18,000 in

the buyer markets the chapter discusses. ZX Auto, founded in 1949, is described on its own page as China's first independent pickup brand.
https://en.zxauto.com.cn/GRANDTIGER.html.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "TERRALORD." ZX Auto Global Official Site. Manufacturer product page for the ZX Terralord pickup truck. Specifications include 4WD diesel powertrain, 6AT/6MT transmission, double cab, three trim levels (Standard, Luxury, Supreme Luxury), four-wheel disc brakes, and 265/65R17 tires. The chapter on patent-expired international vehicles cites the Terralord alongside the Grand Tiger as a working pickup truck derived from late-1990s Toyota Tacoma platform technology, exported by ZX Auto to more than ninety countries including Kenya, Ghana, Nigeria, the Philippines, Bolivia, and across Latin America.
https://en.zxauto.com.cn/TERRALORD.html.

Endnote Commentary

(Ordered to match the Endnotes section, keyed to the corresponding entry above. Substantive notes only; not every Endnote has a Commentary entry.)

***The Chicago Manual of Style*. 17th ed.** The book follows the 17th edition (2017) because it is the most widely used scholarly standard for U.S. nonfiction trade and academic publishing, and because most U.S. literary agents and academic-trade publishers expect endnote apparatus formatted according to its conventions. The 18th edition, published in 2024, refines a number of digital-citation conventions but does not alter the core bibliographic forms the book's apparatus uses. Where this book's conventions diverge from strict Chicago style, the divergence is intentional and noted. The principal divergence is that every Endnote entry concludes with a working URL pointing to the source itself — the publisher's canonical page for periodicals, books, and reports; Justia for U.S. Supreme Court cases; the Cornell University Legal Information Institute for federal statutes; Founders Online for Founding-era documents — rather than to a citation database or aggregator. The aim is to put the reader in front of the authoritative location of each source. Some

publisher canonical pages sit behind paywalls; the book does not provide workarounds for those paywalls, but it does name the canonical location so that any reader with institutional access, or any reader who chooses to purchase, can verify the citation directly at the source. Where the publisher's canonical URL is unavailable, the URL hierarchy falls back through DOI or scholarly database, with Amazon as a last resort for older works without stable publisher URLs. Court cases are linked to Justia. Federal statutes are linked to the Cornell University Legal Information Institute. Founding-era documents are linked to Founders Online at the National Archives.

See Endnote for full citation.

Boulware, David R., et al. "A Randomized Trial of Hydroxychloroquine as Postexposure Prophylaxis for Covid-19." The Minnesota trial is the most-cited randomized evidence against hydroxychloroquine, and the chapter offers it as an exhibit rather than as a result. The critique that follows is this book's own and is not the journal's: that the dose administered exceeded the range the early-treatment protocols specify, that zinc was not co-administered although the protocols treat it as necessary, and that participants were enrolled past the window in which early treatment is said to work. On the last point the published record supplies a mechanism: the drug was shipped to participants, and the shipping interval meant a portion of the treatment arm did not begin dosing inside the four-day post-exposure window the trial reports. Readers should weigh the trial and the critique together. The point the chapter draws is narrower than either — that when a molecule is off patent, no sponsor has a commercial interest in designing the trial that would vindicate it.

See Endnote for full citation.

Class 8 tractor, remanufactured: the $80,000 should-cost figure. The book prices a faithfully remanufactured pre-2000 Class 8 highway tractor — a Peterbilt 379 or Kenworth W900 class chassis around a Caterpillar 3406E, first-year C-15, Cummins N14, or Detroit Series 60 certified to 1999 standards — at $80,000 in 2025 dollars, against $172,900 for a 2025 Kenworth T680 sleeper (see the Kenworth Truck Company endnote). The figure is a projection, and the chapters that use it say so; no such market exists while the wall stands. It is built from three components. The floor is the bare platform: a new Sinotruk HOWO

6×4 tractor of comparable horsepower sells for $28,000 to $35,000 FOB China with no aftertreatment (see the Sinotruk endnote), which is what a public-domain heavy-truck platform costs to build where no patent royalty, no diesel particulate filter, no selective catalytic reduction system, and no post-2000 electronics are carried. The remainder is American: domestic labor and materials, the independent engineering certification the remedy requires, and a domestic margin. The result sits at less than half the new-truck price rather than at one-fifth, because the book prices an American-built truck at American wages, not an import. The chapters that carry the figure — the opening chapter, the environmental chapter, the trucking chapter, the autonomous-vehicle chapter, the plaintiff chapter, the plaintiffs' recovery chapter (which compares against the $161,850 T680 day cab), and the Epilogue — all use $80,000, and the per-mile arithmetic in the autonomous-vehicle chapter (about eleven cents of capital cost over 750,000 miles) is derived from it. The methodology chapter states the book's base year: unless a year is given, prices are in 2025 dollars.

Code of Federal Regulations. "40 C.F.R. § 1068.240 — Exempting New Replacement Engines." The replacement-engine exemption is the nearest thing outside aviation to the time-shift this book proposes, and its limits are the point. Under it a newly manufactured engine may be built to the standards of an earlier model year. Four conditions confine it. Only a certificate holder may build the engine. The engine may only replace one already placed into service. The equipment may be no more than forty years old. And the maker must first determine that no engine certified to current standards will fit; where an engine of an intermediate tier fits, that one must be used. Section 1068.120 does the parallel work for the rebuilder, who must hold a rebuilt engine to the certified configuration of its own model year or a cleaner one, not to today's standards. So the EPA already accepts that an engine of an earlier year's design may lawfully be made new and run. What it does not accept is that engine in a newly built machine, or a new engine of that kind made by anyone but a certificate holder. Compare the Cessna record (see the Federal Aviation Administration, Type Certificate Data Sheet 3A12 endnote): there the time-shift reached the whole airplane, but again only the certificate's holder and those it licensed. In neither field does the time-shift reach the public. The comparison is this book's own; no court has drawn it.

See Endnote for full citation.

Coin Laundry Association (CLA). The laundromat is the cleanest statutory hook in the American economy, and the reason is that no gap exists between the regulated equipment and the business model. A federal efficiency standard naming clothes washers as a covered product governs the only capital asset the business owns, so damages trace to the regulation at essentially one hundred percent — a posture no other category in the plaintiff chapter can claim. The arithmetic is unglamorous and that is its strength. A commercial washer-extractor runs from roughly three thousand dollars for a mid-size unit to well above twenty thousand for a high-capacity machine, and a working store is a floor of twenty, thirty, or more such units, so a full equipment retool is a six-figure capital event spent entirely on machines a federal standard governs. The patent-expiration anchor is equally uncontroversial: the foundational washing-machine patents date to the early twentieth century and expired generations ago. One caution on the trade body. It is a smaller organization than the restaurant or hotel associations and carries less Washington presence, and laundromat ownership is a small-business field, frequently immigrant-owned and pragmatically cost-driven, which makes it genuine but resource-limited as a litigation partner.
See Endnote for full citation.

Competitive-discount floor of fifteen percent. Two different fifteen-percent figures appear in the book, and they describe two different things. In the methodology chapter, the retrospective calculation of the $15.73 trillion floor applied a competitive-pressure discount in a range of eight to fifteen percent — component two of that calculation, the price reduction felt by the buyers who still purchase a new item rather than a remanufactured one. That range is a minor adjustment riding atop the dominant component, the direct savings of roughly forty to sixty percent realized by the buyers who purchase remanufactured goods, conservatively drawn from an observed international spread of fifty to eighty-eight percent. The fifteen-percent figure used in the recovery chapters is a forward-looking estimate, not a restatement of that historical range: it is the conservative floor of the competitive discount a reopened commons would deliver after a favorable ruling. Three independent reasons make fifteen percent conservative rather than optimistic. First, in a post-ruling economy no durable good sits atop a fully patented bill of materials; somewhere in every supply chain remanufacturing introduces discounting, so the effect compounds through the chain rather

than capping at the final assembly. Second, the roughly ninety countries where patent-expired remanufacturing is already lawful show real-dollar prices well below a fifteen-percent discount off the American equivalent — often a half or a third of the U.S. price — so fifteen percent understates what those markets actually achieve. Third, the methodology's "flowing" tier deliberately excludes a large mass of goods that would in fact be remanufactured once the fifty-eight percent blocked tier is freed, so the conservative base itself omits real savings. A wholly independent verification path points the same way: applying a twenty-five to forty percent savings rate (a thirty-four percent midpoint) to an $81.1 trillion present-value base of consumer durable spending yields roughly $28 trillion, comfortably above the $15.73 trillion floor and confirming the floor is conservative from two directions. See the methodology chapter for the full calculation and the recovery chapters for the per-household figures.

Doenges, J. L. The chapter compresses a ninety-year sequence into a paragraph, and the sequence deserves its dates. Bizzozero (1893) described spiral organisms living in the glands of the stomach — in dogs, a qualification the popular retelling drops. Doenges (1938) examined 242 human stomachs after death, found the same organisms in forty-three percent of them, and connected them to no disease. Freedberg and Baron (1940) found them in living patients; the result was never repeated, and nobody published a failure to repeat it either. Then Palmer (1954) studied stomach tissue from a thousand people and concluded the organisms were contamination carried in from the mouth. Medicine already believed the stomach was sterile, because stomach acid kills bacteria. Palmer's study told the field what it already believed, on a sample too large to argue with, and the question closed for thirty years. That is why the case is labeled mixed rather than filed as a clean unasked question. The slides were an unasked question for ninety years. The answer, once Warren and Marshall reopened the question in 1979 and 1981, came from the better tool: the bacterium had to be grown, the classical test for proving that a microbe causes a disease had to be met, and clinical trials had to follow. Both halves are true and the chapter needs both. The sharper point survives the labeling. The organisms were not merely present and unre-

marked. They had been counted, in a published series, in nearly half a large sample — and then explained away.

See Endnote for full citation.

Elster, Jon, ed. *Secrecy and Publicity in Votes and Debates*. The Italian parliament's *voto segreto* practice operated under two distinct legal regimes across its 119-year history, a feature of the record the chapter's single-sentence reference does not set out. The complete timeline:

Period	Authority for voto segreto
1848 – 1922	Constitutional mandate under Article 63 of the Statuto Albertino: secret balloting was the constitutionally required method for every final vote on a law and for all votes of a personal nature. Seventy-four years on direct constitutional authority.
1922 – 1943	Fascist suspension. Mussolini's March on Rome (October 1922) did not formally abolish the Statuto, but parliamentary practice was suspended in substance over the twenty-one years of Fascist rule.
1943 – 1948	Transition. Mussolini fell July 1943; the September 1943 armistice opened a five-year transitional period; a June 1946 referendum abolished the monarchy; a Constituent Assembly drafted a new Constitution.
1948 – 1988	Chamber rule under the *Regolamento*. The 1948 Republican Constitution formally abolished the Statuto Albertino in its entirety. Its Article 64 transferred voting procedure from constitutional authority to each chamber's own rules, where *voto segreto* continued for the next forty years on chamber-rule authority alone.
1988 – present	Reformed. On October 13, 1988, the De Mita government repealed *voto segreto* as the default method for final votes through amendments to the Camera and Senato *Regolamenti* — no constitutional amendment required, because the practice no longer rested on constitutional ground. *Voto segreto* survives today for "conscience" votes (life-and-death, family, individual moral status) and personal-nature votes (individual members or ministers) only.

Article 63 of the Statuto Albertino, in English translation: *Votes shall be taken by rising, by division, and by secret ballot. The latter method, however, shall always be employed for the final vote on a law and in all cases of a personal nature.* That is a direct constitutional mandate, not a chamber preference. The 1848 framers of the Statuto thought secret balloting on final votes was important enough to put into the constitutional text itself, not merely into chamber rules.

How the practice lost constitutional rank is a story in itself, and it has no American analogue. The Statuto was a *flexible* constitution. It set out no amendment procedure, and Article 65 gave the interpretation of laws to Parliament rather than to any court, so it carried no judicial review. Everything in it, Article 63 included, could be altered by ordinary legislation. Nothing was entrenched in the sense an American reader assumes.

The demotion therefore did not happen by amendment. It happened at a refounding. The 1948 Republican Constitution replaced the Statuto entirely, and the Constituent Assembly had to decide, provision by provision, what to carry forward. On 14 October 1947 the Christian Democrat leader Aldo Moro moved to strike secret voting from draft Article 69 of the new text, arguing that rules governing the legislative process belong in each chamber's own rules of procedure and not in a constitution. The Assembly approved the change the next day. A Communist amendment to keep the provision was rejected 135 to 160 — by secret ballot. The motives were strategic on both sides. The Christian Democrats, having learned from the June 1946 election that they could expect to govern, wanted open voting to enforce party discipline; the Communists, expecting permanent exclusion from government, wanted secret voting precisely because it would let them exploit divisions in future majorities. Both read the future correctly.

That single drafting decision is why October 1988 required no constitutional amendment. Because Article 64 of the 1948 Constitution left voting procedure to each chamber, the repeal needed only an amendment to the Regolamento, carried by absolute majority. It passed 323 in favour, 58 against, with 222 abstaining — seven votes above the 316 required. The constitutional objection was raised and nearly prevailed: three motions asked whether Parliament had the right under the Constitution to reform secret voting at all, and the floor

resolved the question by secret ballot, 258 holding that reform violated no constitutional principle against 231 holding that it did.

Two points of precision. The Fascist interruption is dated here from 1922 because parliamentary practice was hollowed out from that year, but the formal abolition came later: Law No. 129 of 19 January 1939 created the Camera dei Fasci e delle Corporazioni and provided at Article 15 that voting is always open, the chamber rules of December 1938 having already said so. And the 119-year figure used in the chapter is the 140-year span from 1848 to 1988, which is the figure Giannetti uses, less the 21 years of the Fascist interruption. Both numbers describe the same record.

One further point belongs on the record, because it is the strongest argument against the reform this chapter proposes, and it comes from the Italian precedent itself. Italy abolished *voto segreto* because the secret ballot had become the instrument of the *franchi tiratori* — the snipers, factions inside governing parties who anonymously killed their own government's bills. Prime Minister Cossiga won an open confidence vote 329 in September 1980 and lost the secret vote on the same decree minutes later, 297 to 298, and resigned. Craxi won confidence 338 in June 1986 and lost the secret vote 266, a twenty-one percent desertion, and resigned. Governments fell not despite winning in the open but immediately after winning in the open.

The objection is real, and the answer is sequence. The Italian secret ballot came *second*, and destroyed what the open vote had just approved. Hybrid Voting reverses the order: the open vote comes first, and the secret ballot is reached only on a bill that has already failed. It cannot be used to sink a measure that passed in the open, because a measure that passed in the open never reaches it. The two systems deploy the same pair of instruments in opposite order, and the order is what determines whether the secret ballot destabilises a government or rescues a bill from the lobby.

The Hybrid Voting chapter's argument that the United States could adopt the procedure by simple chamber-rule majority under Article I § 5 cl. 2 finds its closest historical analogue not in the Statuto-era mandate but in the post-1948 chamber-rule regime: forty years of

working secret-ballot practice, in a major Western parliamentary democracy, on chamber-rule authority alone, terminated when the chamber chose to terminate it. The American reform proposed in the Hybrid Voting chapter requires neither a constitutional amendment nor a chamber-rule mandate at the level the Statuto provided — only the simple-majority chamber-rule adoption the post-1948 Italian parliament already operated under for forty years.

See Endnote for full citation.

Erlich, Mark. *With Our Hands: The Story of Carpenters in Massachusetts*. Three things in Erlich's record deserve a longer note than the preface's use of it can carry. The first is that the carpenters' resistance was not simple Luddism. Oscar Pratt, one of Erlich's interview subjects, acknowledged that the men could not refuse to use the new tools even while they could see what those tools would do to the trade — which is to say the doubters misjudged how fast the tool would improve and read its effect on their craft correctly. The analogy in the preface is drawn to the first error and not to the second. The second is that the 1932 Springfield item was a proposal rather than an imposed penalty. Erlich documents a fine that did land: in 1964 a member of Local 685, Philador Lemay, was found to have violated union rules and fined twenty-five dollars after it emerged that his greater speed rested on supplying his own truck, four-foot level, and electric saber saw for his employer's use. By then union by-laws placed the duty to furnish power tools on the employer, which was the point of the 1932 proposal in the first place. The third is a textual variance worth recording. The United Brotherhood of Carpenters and Joiners of America retells the roof story on its own history pages, citing Erlich at page 146 and attributing it to a Massachusetts carpenter who joined the union in 1946; the union's version renders the carpenter's line as a fear of cutting his hand, where Erlich's text has him declaring he will cut by hand. This book follows Erlich, whose reading is the primary one and the one that parses with the man's refusal. United Brotherhood of Carpenters and Joiners of America, "Tool Revolutions."

https://www.carpenters.org/tool-revolutions/. See Endnote for full citation.

Expired Patent Commons. This is the central term of art of this book, and the reason the book carries the title it does. When a patent reaches the end of its limited term, the invention it covered is meant to pass to the public — to be made, copied, improved, and remanufactured by anyone. The Supreme Court's cases call that destination the public domain. This book calls it the Expired Patent Commons, and the choice of the word "commons" is deliberate. A commons is not an empty space or the mere absence of a right; it is a shared resource the public can actually reach and actually use, as villagers once walked onto common land and grazed it. An invention has entered the Expired Patent Commons only when an American may in fact pick it up and build it — the term is defined by use, not by paper. A patent that has lapsed on paper, but that federal regulation forbids anyone to manufacture, has not entered the commons; it is merely a patent that has expired. The Expired Patent Commons is the public's half of Jefferson's bargain — the forgotten half of the Patent Clause — and the book's argument is that the post-1970 regulatory wall does not raid this commons but prevents it from ever forming. The term is defined in full in the chapter developing the Patent Clause prong.

Fawzi, Alhussein, et al. "Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning." AlphaTensor is often folded into the same story as the attention results in the chapter's third movement, and it should not be. The system found a way to multiply certain small tables of numbers using fewer multiplications than the best method known since 1969. One qualifier is not optional: the headline four-by-four result holds in arithmetic modulo two, not in ordinary arithmetic, where Strassen's method still stands. Kauers and Moosbauer (2023) answered within about a week, and they did not merely match the machine. They found further four-by-four schemes at the same length, then went past it, cutting the five-by-five case from AlphaTensor's ninety-six multiplications to ninety-five and a rectangular format from sixty-three to sixty-two. Their method, which they call flip graphs, is their own. Their computer was small and the search took seconds. And they said plainly that they had used the machine's output as their starting point. The sorting case ran the same way by a different route. AlphaDev (Mankowitz 2023) found faster routines for sorting short lists at the level of individual machine instructions, and they were adopted into a standard library that a great deal of software uses. The paper also

claimed that for the three-element case no shorter program existed. Neri (2023) produced counterexamples to the impossibility claim, and a plain implementation that ran faster besides. Two machine results, then, and two human responses. The machine results are the better tool: a machine built something that had not existed before, and that indicts nobody. The human responses are the unasked question. AlphaDev asserted an impossibility and was wrong, and being wrong in public is what provoked somebody to check. AlphaTensor asserted nothing of the kind. It reported an improved upper bound, which is all anyone reports in that field, and in doing so showed that ground the discipline had stopped digging in, sixty years after Strassen, was not exhausted. Within days, mathematicians who had held the capability the whole time went further than the machine had. What the machines dislodged was not a technical barrier. In one case it was a false claim. In the other it was a settled belief that a question was closed.
See Endnote for full citation.

Federal court procedure: party labels at each level. A reader who has followed the book through its chapters on the filable case and on plaintiff categories will at some point notice an apparent inconsistency in how parties to the Supreme Court's cases are named. The chapters on filability and on who can sue use *plaintiff* and *defendant* throughout. The cases themselves — *Loper Bright*, *Cedar Point*, *Sheetz*, *Massachusetts v. EPA*, *Horne* — are captioned with the constitutional challenger sometimes first and sometimes second, and the Court's opinions speak of *petitioner* and *respondent* rather than of plaintiff and defendant. The inconsistency is not editorial. It is the way the federal court system labels the parties at each layer, and the labels shift as the case moves upward.

The terminology runs as follows. In federal district court, where every constitutional challenge to a federal regulation begins, the party who files the complaint — the grandfather-exemption advocate, in the case this book describes — is the *plaintiff*; the federal agency named as defendant (the EPA, NHTSA, or DOE, represented in court by the Department of Justice) is the *defendant*. The chapters of this book that describe who can file the case and what the complaint will ask for use those labels because that is where the case begins. At the United States Court of Appeals, whichever side lost in the district court and appealed

becomes the *appellant*; the other side is the *appellee*. At the Supreme Court, the party who sought review by filing the petition for a writ of certiorari is the *petitioner*; the other side is the *respondent*. The case caption at the Supreme Court level always lists petitioner first.

Whether the grandfather-exemption advocate stands at the Supreme Court as petitioner or as respondent depends entirely on who won at the Court of Appeals. If the advocate loses at the circuit court and petitions the Supreme Court for review, the advocate is the *petitioner*. That is the procedural posture of every modern Supreme Court win this book builds on — *Loper Bright Enterprises v. Raimondo* (fishermen petitioned; won), *Cedar Point Nursery v. Hassid* (growers petitioned; won), *Sheetz v. County of El Dorado* (landowner petitioned; won), *Massachusetts v. EPA* (states petitioned; won), *Horne v. Department of Agriculture* (raisin growers petitioned; won). The constitutional challenger as petitioner is the modal posture in the cases the book relies on. If the advocate instead wins at the circuit court and the federal government petitions the Supreme Court, the advocate is the *respondent*. *Ruckelshaus v. Monsanto Co.* is the canonical example on the Takings Clause side: Monsanto had won below; the EPA Administrator petitioned; Monsanto stood as respondent and won again.

The plaintiff and defendant labels do not disappear when the case reaches the Supreme Court. They remain attached to the case's history, and the briefs and the Court's opinion will refer back to "the plaintiff below" or "the district-court defendant" when recounting the procedural posture. But they are not the active labels at the Supreme Court. The advocates speak as petitioner and respondent. The caption reads "Petitioner v. Respondent."

One usage note for the reader who reads journalism or general-interest writing on Supreme Court cases. Both routinely use *plaintiff* loosely to mean "the original challenger" all the way through the litigation, including at the Supreme Court. The usage is imprecise but common, and a reader who carries the imprecise usage into the Court's own opinions will misread some of the citations. In a book written as a brief in waiting for an actual Supreme Court case, the precise terminology — plaintiff at the trial court, appellant or appellee at the

circuit, petitioner or respondent at the Supreme Court — will read more credibly to the lawyer who eventually picks up the brief and works from it.
See Endnote for full citation.

Ford Motor Company. "2025 Ford F-150 Lariat 4×4 SuperCrew 5.5′ Box." The chapter's title figure of $72,495 is the out-the-door total a typical American family pays to drive a new F-150 home in 2025. It reflects the Lariat 4×4 SuperCrew 5.5′ Box — Ford's mid-tier family trim, positioned above the work-truck XL (base $37,250 MSRP) and the volume-trim XLT (approximately $47,590 MSRP) but below the luxury-tier Platinum ($74,905 base MSRP) and Raptor ($79,005 base MSRP). The Lariat is what Ford positions as the family truck: the trim that combines hauling capacity with the interior appointments a household financing a new vehicle expects.

The components of the $72,495 total reconstruct as follows. The Lariat 4×4 SuperCrew 5.5′ Box base MSRP for model year 2025 was $65,255 at announcement, revised marginally during the production year as Ford adjusted incentives and option-package pricing. The destination and delivery charge added $1,995, producing a manufacturer's invoice subtotal of $67,250. State and local sales tax, computed at approximately 7 percent on the manufacturer's invoice subtotal — the U.S. weighted average across state and local jurisdictions in 2024–2025, per the Tax Foundation — added $4,708. Title, license, and registration fees added approximately $200, varying by state. The dealer documentation fee, which varies by state and dealership but commonly runs $300 to $700, added approximately $337 at the median. The total at the dealer's cashier — the amount the buyer's lender finances or the buyer writes in a check — is $72,490, within rounding of the $72,495 the chapter uses.

The figure converges from a second verification path. TrueCar dealer-inventory listings show real-world 2025 Lariat 4×4 SuperCrew 5.5′ Box PowerBoost Hybrid configurations at MSRP-with-destination totals of $72,480, $72,580, and $72,690 — before tax and fees. The hybrid drivetrain adds approximately $5,000 to the base Lariat's MSRP, but the verification is about the figure, not the engine: the base Lariat's out-the-door total and the hybrid Lariat's bare MSRP-with-destination converge at the same number, $72,495, from two different directions. The convergence is what gives the chapter title figure its specificity.

The $25,000 should-cost baseline against which the $72,495 is compared is the book-canonical figure established in the wealth-transfer chapter. The $47,495 per-vehicle wall premium is the arithmetic difference between what the family pays and what the family would pay for an equivalent-utility patent-expired vehicle absent the post-1970 federal regulatory architecture this book has been tracing. The reader who wants to reconstruct the out-the-door figure independently can do so from Ford's published 2025 MSRP schedule on shop.ford.com, the destination charge listed on every Ford window sticker, the reader's own state and local sales tax rate (or the Tax Foundation's national average), and a representative dealer documentation fee for the reader's state.
See Endnote for full citation.

Gautret, Philippe, et al. "Hydroxychloroquine and Azithromycin as a Treatment of COVID-19." The retraction is itself part of the record this chapter maps. The paper was published in March 2020, cited thousands of times, criticized on methodological grounds within months — the journal printed Rosendaal's review and the authors' replies in the same volume — and then retracted in January 2025, four and a half years later. A retraction that arrives half a decade after publication, in a journal that had already printed the criticism and the reply, is a different act from a retraction that corrects a fresh error. The chapter cites the paper for what it reported and records the retraction so that a reader can weigh both. The URL given in the Endnote is the authors' own institutional copy, with an archived snapshot, because the publisher's copy now carries the paper only behind the retraction notice.
See Endnote for full citation.

Irons, Peter. *Justice at War*. Aiko Herzig-Yoshinaga had no law degree, no university post, and no institutional standing of any kind. As a teenager she had been imprisoned at Manzanar, one of the camps where the United States held Japanese Americans during the Second World War. In the late 1970s she was working in the National Archives in Washington, and she found what forty years of litigation had not: the government's own original copy of General DeWitt's Final Report on the removal, containing passages that a later, revised version had taken out. The deleted material showed the government had known there was no time to hold individual loyalty hearings, and that the claim of military

necessity made to the Supreme Court in 1944 had been made while contrary evidence sat in the file. Irons, a lawyer, carried the finding into court. Fred Korematsu had been convicted for refusing to go, and the Supreme Court had upheld the conviction in Korematsu v. United States (1944). In 1983 a federal court in San Francisco vacated it on a writ of coram nobis — an old and rarely used procedure that lets a court reopen a case when it learns the original proceeding was tainted. In Trump v. Hawaii (2018) the Supreme Court said Korematsu had been gravely wrong the day it was decided. The chapter calls this the most damning unasked question in its set, and the reason is the location of the evidence. It was not merely available somewhere. It was in the government's own files, put there by the government, kept by the government, for forty years. The finding and the argument sat in two people, which is the pattern the chapter's second movement is about.
See Endnote for full citation.

Jefferson, Thomas. "Letter to Isaac McPherson, August 13, 1813." Chapter 2 quotes a single line from this letter — the line in which ideas spread "like fire, expansible over all space, without lessening their density in any point." That line is one of three poetic images Jefferson stacks in a single passage, and it is not the most famous of the three. The full passage runs as follows:

> *If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of every one, and the receiver cannot dispossess himself of it. Its peculiar character, too, is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening me. That ideas should freely spread from one to another over the globe, for the moral and mutual instruction of man, and improvement of his condition, seems to have been peculiarly and benevolently designed by nature, when she made them, like fire, expansible over all space, without lessening their density in any point, and like the air in which we breathe, move, and have our physical being, incapable of confinement or exclusive appropriation.*

Three images stand out. The first is the thinking-power image: an idea, once divulged, cannot be reclaimed. The second is the taper image — he who lights his taper at mine receives light without darkening me — which is the line most often quoted in patent scholarship and in economic writing on non-rival goods. The third is the fire-and-air image, which is the line Chapter 2 quotes. Each of the three has at one time or another been treated as the heart of the passage. The reader who wants Jefferson's full argument — the part the chapter pivots to — will find the entire letter at the URL listed in the corresponding Endnote.
See Endnote for full citation.

John Deere W70. The W70 is the largest combine Deere sells in India, a 100-horsepower straw-walker machine; the S7 600, a 334-horsepower rotary machine, is the smallest Deere sells in the United States, which is why the chapter sets the two against each other and compares them by daily harvesting capacity rather than sticker price. Deere's only U.S. straw-walker combine, the 402-horsepower T6 800, would match the W70 on threshing design but is larger still in power, grain tank, and weight, carries no published U.S. price, and appears in Deere's U.S. catalogue only inconsistently, so it cannot serve as the comparison. A same-company pair closer in size does not exist: Deere India sells only the W50 (75 horse-

power) and the W70, and Mahindra sells no combine in the United States. Search conducted October 22, 2026, across deere.co.in, deere.com, mahindratractor.com, and mahindrausa.com.

***Kahle v. Gonzales*.** Brewster Kahle, the engineer and archivist who built the Internet Archive, made a structural observation of exactly the kind this book makes. When Congress removed the requirement that copyright owners renew their copyrights, it turned an opt-in system into an opt-out one. On paper, no individual term changed. In practice, works stopped entering the public domain by default, because nobody now had to do anything to keep them out of it. That is an observation about how a system behaves, not about what a statute says. He lost. The Ninth Circuit told him he was making the argument the Supreme Court had already rejected in Eldred v. Ashcroft (2003), wearing different clothes. Two things in the opinion matter to this book, and they cut opposite ways. The court admitted an opening. Eldred, it said, had not directly decided whether the words "limited Times" put a real ceiling on how long a copyright can last; the Supreme Court had expressly left that question open. Then the Ninth Circuit answered it, and answered it badly for anyone following. It held that the ceiling is found by weighing the encouragement longer terms give authors against the benefit shorter terms give the public, and that the weighing belongs to Congress, subject only to a test courts almost never fail anyone. One sentence keeps the door from shutting: future extensions may or may not survive review, but the current term is constitutional. The distinction this book rests on is that Kahle was about extending a term. The argument here alleges no extension of anything. The patent term ran and expired on schedule; the wall stands in the regulation that outlived it. Whether that distinction holds is for a court to say. It is stated here as this argument's own, not as something Kahle decided.

See Endnote for full citation.

Knorozov, Yuri V. This is the purest case in the chapter, and the correction is what makes it so. In the sixteenth century the Spanish friar Diego de Landa wrote down what he took to be the Maya alphabet. His manuscript was published in Paris in 1864, and photographic copies of the surviving Maya books were in print by the end of that century. Everything

needed was public. Scholars used it and failed, and not for want of trying. De Landa's list is inconsistent, some of its signs never appear in real inscriptions, and — above all — everyone read it as an alphabet. In 1952 Knorozov, working in Leningrad, read it as a syllabary instead. When de Landa asked his Maya informant for the letter B, the informant gave him the syllable be. Knorozov added a rule about how syllables pair, and the writing began to open. The popular version says nobody asked. Many asked. What Knorozov brought was not evidence and not an instrument. It was a different idea about what kind of writing it was. That is why this case, of all the cases in the chapter, is the one that matches the Patent Clause. The Clause has been read continuously since 1787. Reading a text wrongly is not the same as never reading it, and that is why forgetting is the word the book uses for what happened to the Clause's second half.
See Endnote for full citation.

Kory, Pierre, et al. "Review of the Emerging Evidence Demonstrating the Efficacy of Ivermectin." The Expression of Concern the journal attached to this paper repays reading closely, because of what it declines to say. It states that it "does not imply that the methodology used by Dr. Pierre Kory and his collaborators was incorrect," and that "the use of summary data published by others is a generally accepted approach in biomedical metanalytic research." A journal that concedes the method was sound, and attaches a notice anyway, has recorded something about the pressure on the literature rather than about the paper. Kory is a critical-care physician, a founder of the Front-Line COVID-19 Critical Care Alliance, and a co-author of the curated cancer-care standard this chapter cites elsewhere. The chapter takes no position on ivermectin's clinical efficacy; it records that the hindrance pattern it describes reaches the published record and not only the pharmacy counter.
See Endnote for full citation.

Magill, Gord. The foreword author's media reach, referenced in the Preface. Gord Magill is a third-generation trucker with more than twenty-five years behind the wheel. He has written for *Newsweek*, *The American Conservative*, and *American Affairs*; publishes the *Autonomous Truck(er)s* Substack; and hosts the *Voice of GO(r)D* podcast. He has carried the trucking story onto the two largest platforms of the populist right. He appeared on *The*

Tucker Carlson Show twice — on January 24, 2024, on the imprisoned "Coutts Four" and the 2022 Freedom Convoy, and on June 1, 2026, in a roughly sixty-five-minute conversation built around *End of the Road* and the long deregulation of trucking. The reach figures are drawn from contemporaneous reporting. CNN reported in June 2025 that Carlson's program averaged roughly 651,000 views an episode on YouTube in May 2025 and roughly 1.06 million in June 2025, with individual interviews drawing several million and a single teaser clip on X reaching 39.9 million. Magill also appeared on Steve Bannon's *WarRoom* on April 15, 2025, on foreign labor in the trucking workforce; *WarRoom* streams several times each weekday and has ranked for years among the most popular political podcasts in the United States, described by NBC News as routinely near the top of Apple's charts.
See Endnote for full citation.

Markopolos, Harry. *No One Would Listen*. Markopolos's own account gives the timing the chapter compresses. His firm asked him to copy a competitor's investment returns. The competitor was Bernard Madoff. Markopolos looked at the numbers and could not copy them, because they were not possible; he has said it took him about five minutes to see that, and about four hours to build the model that proved it. Those figures are his own, from the time, not a story improved in the retelling. He told the Securities and Exchange Commission in 2000. He told them again in 2001. In 2005 he sent the agency a document titled "The World's Largest Hedge Fund is a Fraud." The agency did not act, and the scheme collapsed in December 2008, taking roughly sixty-five billion dollars of client money with it. The chapter opens its first movement with this case because it has the same shape as the book's argument. Nothing was hidden. Markopolos did not find a document. He read numbers the agency already had, that anyone could have read, and he was not an insider at the agency in any sense — no standing, no channel, no way to make anyone look. The parallel does not need labor. The rules in this book are published in the Federal Register. The patent expirations are docketed. Nobody has to find anything.
See Endnote for full citation.

Martin, Roleigh. "Towards an Holistic View of Social Policy Problems." The author's 1977 master's thesis examined three computer-assisted group-decision methods of the

mainframe era, each tested in its own literature against the same kind of problem: a ranking task with a known optimal answer, the best known being the fifteen-item “Lost on the Moon” exercise Jay Hall devised for NASA, in which small groups using a structured method often out-ranked their most capable member (Hall and Watson, 1970, cited separately). SPAN, the Social Participatory Allocative Network developed by William J. MacKinnon at the University of Arizona in the mid-1960s, let members allocate weighted influence to one another as well as to the options. ISM, Interpretive Structural Modeling, developed by John N. Warfield at Battelle Memorial Institute in Columbus, Ohio, in 1973–74, built a group’s shared structure of a problem from pairwise judgments. ELECTRE, Bernard Roy’s outranking method from SEMA in Paris, 1965–68, a multi-criteria tool that serves a group once the group supplies the criteria, was tested head to head against SPAN in a 1977 Portland State University systems-science dissertation by Perihan Soliman Tawfik, a neighbor of the author’s, who found that the structured methods worked and that SPAN was the stronger on the task used (cited separately). The author visited MacKinnon in Tucson and knew Robert E. Good, Tawfik’s advisor, at Portland State. The thesis recommended running all three methods together, because they complemented rather than competed with one another. The personal computer ended that line of work: the methods ran on terminals tied to a mainframe, and interest in computer-mediated group decision moved elsewhere once the machine sat on one desk. The relevance to this book is the design. Eleven models given the same examination, one legal model assigned to argue against the thesis, and a human deciding what counted, is a group whose members’ strengths and weaknesses are known, using complementary methods, measured against a problem with a fixed record. The author did not plan the parallel; he noticed it after the fact. *See Endnote for full citation.*

Martin, Roleigh, ed. *ShareDebate International*. *ShareDebate International* was a small-press project at the leading edge of pre-Web electronic publishing. The magazine was a freeware diskette publication — which in 1990 meant a 1.44-megabyte 3.5-inch disk written in IBM PC format, copied by thousands of bulletin-board system operators, and read by their users one at a time on personal computers running DOS. After Issue 6 the publication moved from shareware to copyrighted freeware, eliminating the suggested subscription fee

while preserving the distribution structure. Contributors across the magazine's run included Milton Friedman, Murray Rothbard, George Gilder, Doug Bandow, Eamonn Butler, Jerry Pournelle, Leonard Peikoff, Ben Bova, J. Neil Schulman, Victor Koman, Joseph Sobran, George Benston, George Kaufman, Lt. Gen. Daniel O. Graham, Paul Levinson, Douglas Ostrom of the Japan Economic Institute, Masanobu Taniguchi, the Libertarian Party, the Republican Liberty Caucus, and the League for Programming Freedom (with Richard Stallman). Ben Bova praised the magazine as "a bold, exciting venture" deserving of broad attention; J. Neil Schulman called it the arrival of "grassroots mass-market publishing."

Friedman's 1990 letter bears on the remedy chapter as well, for a reason that has nothing to do with the proposal it declined. Asked to endorse a scheme that would have used government to promote an objective he agreed was "urgent and desirable," Friedman wrote that "if any such program is adopted, it will not be organized the way you want. It will not work the way you want. Political pressure will be brought to bear on it." He proved the point with his own experience: he had persuaded Treasury Secretary George Shultz to try a Dutch auction for long-term securities, the experiment saved money, and the day Shultz left office the Treasury dropped it and for years prevented publication of the findings. The reason, he wrote, was that "the system I proposed eliminated any discretion by Treasury officials," whereas the method they preferred made them "important people sought out by important people" with expertise "that has a value in the market if they choose to leave the government." He enclosed his columns "Barking Cats" and "Laws That Do Harm," and added that a new agency sweeps clean at first, "but then it was a cat and so it meowed." That is the design criterion the remake chapter claims for its mechanism: the underwriting gate, the public model, and recall law's own definition of "unsafe" leave the administering officials no discretion to exercise, which is why the chapter declines to route the certifier's qualification through a licensing board or the compliance year through the agencies that built the wall. Bears on the capture paragraph of "Who Can Order This Remedy" in How a Remake Is Built and Certified. The letter, addressed to "Raleigh Martin," is reproduced in full, as scanned images and in retyped text, in an X Article by the author published October 27, 2024,
https://x.com/RoleighMartin/status/1850587071204741532.

Two threads in the magazine bear directly on the present book. The first is Debate Topic #4, the "Japanese VS American Patent Law Debate," opened by the editor in Issue 1 (1990) with the framing: *Is the Japanese Patent Law more just and more conducive to economic growth than America's Patent Laws? A side aspect to this topic is the fairness and economic impact of America's patent laws.* That side aspect — the fairness and economic impact of America's patent laws — is a variant of the question this book takes up thirty-five years later in a deeper constitutional form. Issue 3 carried a four-page engagement letter from Nobel Laureate Milton Friedman, dated July 26, 1990, on Hoover Institution letterhead, praising the magazine's aims, declining to support a proposal of the editor's, and wishing the editor well. Friedman returned to the magazine across multiple subsequent issues. The second relevant thread is Debate Topic #6, "Government's Impact on Health Costs," which prefigures the broader cost-of-living argument the author's MALA (Making American Living Affordable) project has taken up in the years since. Third-party commentary on the magazine appears in Easton (2006), cited separately above. *See Endnote for full citation.*

Martin, Roleigh. "The Forgotten Half of the Patent Clause: What was Promised When Patents Expired." The $30 to $45 trillion range this book gives as the plausible true total rests on seven categories of cost the consumer-expenditure floor calculation excludes. Each is named here so that a reader can see what the floor leaves out rather than take the range on trust.

First, compounding economic effects: reduced business formation, lost innovation, and diminished entrepreneurship in sectors where the cost of regulatory compliance puts new entrants out of the market before they can begin. Second, indirect costs through the rest of the economy: higher housing costs caused by expensive construction equipment, higher food costs caused by expensive farm equipment, and higher transportation costs that ripple into the price of everything moved by truck. Third, the time value of money — the floor calculation performs no present-value adjustment, and one would raise the figure substantially before any other exclusion is added. Fourth, specialized industrial-equipment categories: mining, oil and gas, semiconductor manufacturing, pharmaceutical production, textile ma-

chinery, packaging equipment, and a long tail of others the floor did not attempt to price. Fifth, commercial vehicles beyond those counted: delivery vans, buses, and specialized trucks outside the categorical breakdown. Sixth, building materials and systems: windows, insulation, and roofing affected by post-1970 energy codes. Seventh, the innovation tax — products never developed because regulatory barriers made market entry impossible. See Endnote for full citation.

Martin, Roleigh. A note from the author, on the path that led to this book. The recognition that the post-1970 federal regulatory wall around expired-patent commerce constitutes a constitutional violation, rather than merely an unfortunate policy outcome, did not come to me from a law degree or from a career in patent litigation. It came from two professional habits of mind I have spent a working life inside.

The first is game theory. My academic background trained me to look for the structure underneath an outcome — to ask, of any policy outcome whose stated rationale and observed effect do not match, what set of player incentives, what set of payoff matrices, and what set of moves the rules permit and forbid, would produce the observed effect from rational actors. Applying that lens to the post-1970 expansion of federal energy, safety, and emissions regulation produced, after some years of reading and re-reading the Patent Clause against the regulatory record, the recognition the book develops in detail: that the wall converts the patent bargain from a contract with two halves to a contract with one, and that the constitutional text already names the missing half. The half is in the words *limited Times.*

The second is the working habit of a man who spent thirty-four years in software — first as a programmer, then as a database programmer, inside the technical organization of a Fortune 25 company. The first joke I ever heard about my occupation, the one I have kept all the way through, is that *the worst programmer is the one who does exactly what you ask them, and nothing more.* The joke is funny because it is precisely right. A compiler does not interpret intent. It reads what you wrote, applies the language's rules, and returns either an executable result or a compilation error. It does not consult the spirit of the document; it consults the syntax of the document. A working life shaped by that machine produces a habit of reading any structured text as if it were code — including the Constitution.

Article I, Section 8, Clause 8, read by the compiler-reading mind, behaves like any other line of structured text. The phrase that arrests the reading is *limited Times.* It has a type signature. It is a duration. Durations have a start and an end. For a database programmer in particular, the same clause reads as a foreign-key constraint: the patent points at a defined term, the federal regulations that the rest of this book examines point at the patent-expired product, and the two tables join on the patent. When the patent's term expires, the parent row reaches end of life. Any regulation that continues to reference the patent-protected status as if it were live is holding a foreign key to a row that no longer exists in the form the constraint requires. A database with referential integrity enabled refuses the operation. The federal government has been operating with referential integrity disabled since approximately 1970. The book is, in one of its readings, a fifty-year audit log of constitutional referential-integrity violations.

The two habits reinforce each other. Game theory asks what incentive structure would produce the wall as the equilibrium outcome of rational actors. Compiler-reading asks what the constitutional text, read as written, requires of the actors regardless of their incentives. The first habit explains why no one stopped the wall from being built. The second habit explains why the Constitution did not authorize what was built. The book is the long form of both observations.

The same compiler-reading habit, applied to the constitutional clause this book's Hybrid Voting chapter spends most of its constitutional argument on, returns a result the constitutional consensus has not. Article I, Section 5, Clause 3 reads in part: "the Yeas and Nays of the Members of either House on any question shall, at the Desire of one fifth of those Present, be entered on the Journal." Two phrases in that sentence admit ambiguities that two centuries of practice have papered over.

The first is *the Yeas and Nays of the Members.* Structurally a noun phrase, it could resolve to "the count of yeas and the count of nays" — a numeric tally — or to "the named position of each member" — a per-member roster. The text does not specify which. A compiler asked to resolve a noun phrase that admits two materially different parsings emits an error and demands clarification from a higher authority. The second is *on any question.*

Structurally adverbial, it could resolve to “on every vote a chamber takes on the question” or to “on the question, once” — meaning a chamber that votes a bill twice could satisfy the clause by recording the named yeas and nays on either round. Again, the text does not specify. The constitutional consensus on both phrases is supplied by two centuries of legislative practice; legislative practice is not the type of authority a compiler accepts. A compiler accepts the text, the rules of the language, and binding constructions from a higher level of authority. Above the chamber, two such layers exist: the constitutional text itself, which on these two questions is ambiguous, and the Supreme Court’s construction of the text, which on these two questions does not exist. With both higher layers silent, the compiler accepts the next available authority — and the Constitution itself names what that authority is. Article I, Section 5, Clause 2 grants each chamber the power to *determine the Rules of its Proceedings*, and the rules of each chamber are, in practice, set by the procedural majority the chamber’s elected leadership commands. The Speaker of the House and the Majority Leader of the Senate, each elected by the chamber that operates under the Yeas-and-Nays clause, hold that procedural majority. The compiler-reading version of the political-question doctrine, then, is not an evasion the courts performed; it is a delegation the Constitution wrote, with two ambiguous phrases handed downward through the layers of authority until they come to rest with the leadership the chamber elected. The identity of that leadership is therefore not procedural housekeeping. It is the runtime variable on which the meaning of two constitutional ambiguities is set, and the chamber that elects its leader is, in the act of that election, choosing which resolution the Yeas-and-Nays clause will receive.

The Hybrid Voting chapter develops both of these textual ambiguities at greater length and closes with a line from Bernardo Bertolucci’s *The Last Emperor* — Reginald Johnston, played by Peter O’Toole, instructing the young Emperor Pu Yi: “If you cannot say what you mean, Your Majesty, you will never mean what you say. And a gentleman should always mean what he says.” The Patent Clause meant what it said, on both halves of the bargain it described. The Yeas-and-Nays clause, on these two questions, did not.

See Endnotes for full citations to the author’s articles and working paper.

Nori, Harsha, et al. "Sequential Diagnosis with Language Models." The chapter *When Outsiders Advance Constitutional Law and Other Disciplines* compresses this paper into two paragraphs; the paper's own abstract and stated limits are reproduced here so a reader can weigh them uncompressed. The benchmark holds 304 *New England Journal of Medicine* clinicopathological conference cases and 325 MIMIC-IV-ED emergency-department admissions. The physicians were generalists — primary-care, hospital, and acute-care — tested on 56 of the conference cases held out from development, closed-book and open-book: 22.3 percent and 36.1 percent accuracy. The best single physician reached 51.8 percent. Before publication the authors reran the full evaluation with GPT-5.5 without changing the prompts or the orchestration, and the system's margin over the plainly prompted model widened. The abstract, verbatim:

> *We explored the capabilities of frontier language models in sequential diagnosis, a diagnostic paradigm that captures the iterative nature of clinical reasoning. Given a brief clinical presentation, clinicians selectively acquired information to reach a diagnosis, enabling assessment of diagnostic accuracy and workup cost. We developed an interactive benchmark and an artificial intelligence (AI) system that simulates a panel of physicians, continually revising the differential diagnosis and selecting cost-effective tests. Using this benchmark, the system achieved 83.2% accuracy on New England Journal of Medicine clinicopathological conference cases and 80.3% on MIMIC-IV-ED emergency department admissions, outperforming generalist physicians (36.1 and 46.8%, respectively). Although these findings were obtained in a controlled benchmark setting, they highlight the promise of language models for supporting more accurate, lower-cost diagnoses.*

The authors' limits, verbatim: "These comparisons do not establish superiority over specialists, multidisciplinary teams, or real-world clinical practice." And: "We would expect the performance gap to narrow when comparing DxO with expert specialists or multidisciplinary specialist teams selected to match the content of each case." On the panel design: "improvements in model capability do not necessarily diminish the benefits of careful or-

chestration." The chapter labels the result the better tool for the first two reasons and cites it for the third.

See Endnote for full citation.

Open-source software methodology. The remake chapter's public-record procedure maps onto open-source practice nearly step for step, and the mapping is the reason the borrowing is credited rather than merely asserted. The published specification model is the source code. Re-examining only the changed and affected areas of an improved design is code review of a diff rather than a re-read of the whole program, and it is also the logic of the FAA's changed-product rule, 14 C.F.R. § 21.101, so the two citations reinforce each other. The rule that a remake contain no still-patented component is the software bill of materials, in which every component is listed with its license status. Confirming the physical build against the certified model is the reproducible build, which verifies that a shipped binary matches its published source. One accountable signature over many contributors' work is the maintainer's signed release. Improvement by anyone is the fork. The closest precedent for the chapter's rule that certification runs with the public record rather than with its first holder is the FIPS 140 validation of the OpenSSL module: a federal certification attached to a public artifact, usable by anyone who uses it unchanged, and requiring revalidation when the code changes. Two limits keep the comparison honest. Open source supplies the procedures, not a track record of certifying vehicles; safety certification of open-source software is a live frontier, not a finished one. And a method is not property — *Baker v. Selden* settles that — so no credit is legally owed; it is given because a named forty-year lineage is a precedent and an unnamed "open-source approach" is a slogan. The comparison is this book's own; no court or agency has drawn it. *See Endnote for full citation.*

Ostrom, Elinor. *Governing the Commons*. Elinor Ostrom was awarded the 2009 Nobel Prize in Economics for her work on the management of common-pool resources — the first woman to receive the prize, and one of the few political scientists ever to receive it. The award recognized a body of work spanning thirty years and hundreds of empirical cases: fisheries, irrigation systems, forests, grazing lands, groundwater basins. Her 1990 book *Governing the Commons* is the synthesis. Its central claim is that human communities,

given the right institutional design, have repeatedly and successfully governed shared resources without resort to either state ownership or full privatization — the two solutions Garrett Hardin's "Tragedy of the Commons" had told mid-twentieth-century policy thinkers were the only options. Ostrom showed Hardin had been wrong about the empirical record. Real commons regimes, when they worked, depended on a small number of recurring design principles: clearly defined boundaries, rules matched to local conditions, collective-choice arrangements for those affected, monitoring, graduated sanctions, low-cost dispute resolution, and recognition by higher-level authorities. The reader who wants to see how those principles map onto the post-expiration patent commons described in this chapter — and why the post-1970 federal regulatory state has destroyed the higher-level recognition the commons depends on — will find the framework developed across the diagnostic and remedy chapters of the book.
See Endnote for full citation.

Poplin, Ryan, Avinash V. Varadarajan, et al. "Prediction of Cardiovascular Risk Factors from Retinal Fundus Photographs via Deep Learning." Doctors have been photographing the back of the eye for about a hundred years. The picture shows the retina and its blood vessels, and it is used to look for eye disease. Poplin and his colleagues trained a system on retinal photographs from 284,335 patients and tested it on two separate groups. From the photograph alone, the system estimated a patient's age to within about three and a quarter years and read blood pressure to within about eleven points. It predicted whether the patient smoked, and whether the patient would go on to have a major cardiac event. It also identified the patient's sex, almost perfectly. No ophthalmologist can do that, and none suspected it could be done; the authors say plainly that these signals were not previously thought to be present in retinal images at all. The chapter files this as an unasked question without qualification, and the reason bears stating. The instrument was new. The information was not. It had been in every such photograph taken for a century.
See Endnote for full citation.

Railway Labor Executives' Association v. Gibbons, 455 U.S. 457 (1982). In 1980 Congress passed a statute for one bankrupt railroad, the Rock Island, directing how its estate would

be liquidated and what its employees would be paid. The government defended the statute under the Commerce Clause, Article I, Section 8, Clause 3, which plainly reaches railroads. The Court held that the statute was in substance a bankruptcy law, that a law for a single debtor is not uniform, and that Congress could not escape the limit in Clause 4 by legislating under Clause 3: to allow it, Justice Rehnquist wrote, would eradicate from the Constitution the uniformity limitation the Bankruptcy Clause places on the bankruptcy power. The parallel to this book's argument is exact in form. Both limits sit in Article I, Section 8; both are a single word attached to a grant of power, "uniform" in Clause 4 and "limited" in Clause 8; and in *Gibbons* the Court refused to let a neighboring clause's general power do what the specific clause's word forbids. Two cautions. Justices Marshall and Brennan concurred in the judgment on a narrower ground and declined the majority's broad reading of the uniformity requirement, which is the reading this book relies on. And *Gibbons* concerned the Bankruptcy Clause; carrying its reasoning to the Patent Clause's "limited Times" is this book's step and not the Court's.

See Endnote for full citation.

Ranalli, Brent, James D'Angelo, and David King. *The Sunshine Reforms and the Transformation of Congressional Lobbying*. Brent Ranalli is a historian; James D'Angelo is a political scientist; David King is a senior lecturer at the Harvard Kennedy School. Their joint work at the Congressional Research Institute is the most rigorous body of empirical research available on the structural transformation of the U.S. Congress after the 1970 Legislative Reorganization Act. The 2019 *Foreign Affairs* essay "The Dark Side of Sunlight," co-authored by D'Angelo and Ranalli, is the landmark synthesis; access to that essay sits behind a paywall. The 2018 Congressional Research Institute paper *The Sunshine Reforms and the Transformation of Congressional Lobbying* develops the same empirical case at greater length and is freely available on the Congressional Research Institute website (
https://congressionalresearch.org). The empirical record this book uses on lobbyist growth, on the explosion of legislation since 1970, and on the relationship between the "sunshine" transparency reforms and the rise of lobbyist surveillance derives from their joint research at that institute. Three findings are central. First, the registered-lobbyist count in

Washington grew from a few hundred in 1970 to over thirteen thousand by 2024. Second, the volume of federal legislation tripled in the decade after the 1970 Act. Third, the federal government has not passed a budget that reduced the national debt in a single fiscal year since 1957. The reader who wants to understand why the modern American Congress produces the legislative outputs it does should begin with the Congressional Research Institute's published work. American constitutional self-government depends on citizens understanding what was changed in October of 1970 and what the change has cost. See Endnote for full citation.

Risner, Genevieve. "Parts of a Western Saddle and the Variations." The Buffalo Bill Center of the West, located in Cody, Wyoming, is a Smithsonian Affiliate that operates five museums under one roof — the Buffalo Bill Museum, the Plains Indian Museum, the Whitney Western Art Museum, the Cody Firearms Museum, and the Draper Natural History Museum. Risner's 2021 article on the parts and types of the Western saddle is the chapter's recommended one-stop online reference for visual identification of the saddle types this chapter touches. The article identifies and photographs four distinct American saddle types from the Center's own collection: the Vaquero saddle, the stock saddle, the McClellan, and the Pony Express saddle. Each is presented with photographs from multiple angles and a short narrative on the working role it served. The article's evolution timeline runs from the eighth-century Moorish saddle, through the sixteenth-century Mexican vaquero adaptations that added the roping horn, through the 1850s adoption of the McClellan by the U.S. Military, through the 1860–1861 Pony Express. Although the article does not address the patent-expiration economics this chapter foregrounds, it is the standard accessible reference for the working forms the post-1797 commons produced.
See Endnote for full citation.

Semmelweis, Ignaz. The chapter states the two numbers and the admission rule; the rest of the case belongs here. The Vienna General Hospital ran two maternity clinics. Doctors and medical students staffed the first; midwives staffed the second. From 1840 through 1846 the death rate from childbed fever in the doctors' clinic was 98.4 per thousand births. In the midwives' clinic it was 36.2. The figures were in the hospital's own ledgers and printed in

its annual returns. Women were admitted to the two clinics on alternate days, with no medical selection of any kind, so the obvious objection — that the doctors were taking sicker patients — is not available. The only systematic difference between the two groups of women was who delivered them. The hospital was running a controlled experiment without meaning to, publishing the results every year, and reading them as nothing. Semmelweis saw it in 1847, after his colleague Jakob Kolletschka cut himself during an autopsy and died of what Semmelweis recognized as the same disease that was killing the mothers. The doctors came to deliveries straight from the dissecting room. The midwives did not. He ordered chlorine handwashing that May, and deaths fell. The correction the chapter insists on is the one the popular version omits. Semmelweis was not an outsider. He was a credentialed physician, an assistant in the very clinic whose figures he read. If a theory of blindness requires an outsider to break it, that theory deletes the best case medicine has. The blindness here was not caused by a shortage of credentials. It was caused by a settled sense of what those mortality figures were about — the mechanism Kuhn named, and the one this book says operated on the Patent Clause.

See Endnote for full citation.

Shallue, Christopher J., and Andrew Vanderburg. “Identifying Exoplanets with Deep Learning.” NASA’s Kepler telescope watched roughly 150,000 stars at once, looking for the minute dimming that occurs when a planet crosses in front of one. It produced an archive that was public, enormous, and worked over for years by professional astronomers who had extracted thousands of planets from it. Shallue was a software engineer at Google, not an astronomer. The professionals had concentrated on strong signals, for a defensible reason: a weak dip in a star’s brightness is usually noise, and chasing weak signals wastes time on false positives. Shallue’s idea was that a neural network might sort the weak ones — that the constraint had been human attention rather than the data. He trained a network on fifteen thousand Kepler signals that human scientists had already labelled as real planets or false alarms; on a held-out test set it classified them correctly ninety-six percent of the time. Then he and Vanderburg, the astronomer on the paper, pointed it at 670 star systems already known to hold multiple planets, on the reasoning that a system with several planets was a likely place to find another. It found Kepler-90i: about thirty percent larger than

Earth, orbiting its star every 14.4 days, with a surface temperature above eight hundred degrees Fahrenheit. Kepler-90 had been identified in 2013 as the first seven-planet system Kepler found; the eighth planet's signal was too weak for the methods that found the other seven, and its discovery made Kepler-90 the first system known to match our own for planet count. The same run turned up Kepler-80g. Vanderburg supplied what Shallue could not: the judgment about where to look and which candidates survived scrutiny. The chapter labels the case a hybrid and separates the halves. The network was a new tool, and Shallue built it. But the tool is not what had been missing. The data was public and had been examined. What had not happened was anyone asking whether the signals the professionals had reasonably skipped were worth a second look. The question was the contribution. The network was how he could afford to ask it.

See Endnote for full citation.

Sierra Saddlery School. "Some History of the Western Saddle." The chapter does not lean on this source for the dates and patent figures it cites — those come from the patent record itself and from the standard scholarly references. The Sierra Saddlery School blog post earns its place in the apparatus for a different reason: it is a competent and accessible narrative of how the working saddle of the American West came to look the way it does. The trans-Atlantic line runs from the Moorish horsemen of the Dark Ages, who carried into the Iberian peninsula a high-cantle saddle designed for armored war, to the Spaniards who adapted it into the Spanish War Saddle and brought it to the New World, to the post-conquest evolution of the saddle into a working tool of the Mexican vaquero. The vaqueros, in turn, were the saddle-makers who taught the trade to the working cowboy of Texas, California, and the Great Basin, and the Texican-and-Californio split among working cowboys produced the two saddle silhouettes the modern reader still recognizes — the swell-fork, square-skirted, double-rigged Texican saddle suited to the heavy brush and "hard-and-fast" roping of the Texas range, and the slick-fork, round-skirted, center-fire-rigged Californio saddle suited to the milder climate, the dally style of roping, and the time the vaquero had for tooled-leather artistry. The reader who wants the longer saddle-history story than the chapter has room to tell will find a serviceable summary at the URL given in the corresponding Endnote, with the caveat that the post is a saddle-school blog rather than a

museum or scholarly source. The Buffalo Bill Center of the West (Risner 2021) is the better one-stop reference for visual identification of the saddle types the chapter discusses.
See Endnote for full citation.

Statuto Albertino. The full timeline of the Italian parliament's *voto segreto* practice under the Statuto and under the post-1948 chamber-rule regime, together with Article 63 in English translation, is set out in the Elster commentary entry above.
See Endnote for full citation.

Vesuvius Challenge. Two people acted on the same scans, and the chapter's third movement separates what each of them did, because the sequence is the point. In AD 79 Vesuvius buried Herculaneum and with it a villa library of roughly 1,800 papyrus scrolls. The eruption carbonized them, which is why they survived and why they were unreadable: the ink is carbon, the papyrus is carbon, and unrolling one destroys it. In 2019 Brent Seales and his laboratory at the University of Kentucky put two scrolls through a particle accelerator and produced X-ray scans at resolutions down to a few micrometres. In 2023 the scans were released publicly as an open competition with a grand prize of seven hundred thousand dollars, and hundreds of people downloaded them. Casey Handmer, a physicist formerly at the Jet Propulsion Laboratory with no papyrology and no Greek, downloaded them too, and built nothing. He opened the flattened scan segments and looked at them for hours, and he noticed a faint cracked-mud texture appearing in shapes that looked like Greek letters. He called it crackle and published it on his blog in August 2023. The carbon ink was invisible because it has almost the same X-ray density as the carbon papyrus. What Handmer saw was the disturbance the ink had left in the surface it dried on — the mark of a thing rather than the thing. It was the first substantial evidence of ink inside an unopened scroll, and the competition awarded him ten thousand dollars for it. Handmer has described his own part as modest. It was the hinge on which everything after it turned. Luke Farritor, a twenty-one-year-old computer science undergraduate at the University of Nebraska–Lincoln, read Handmer's post and did what Handmer could not. He took the crackle pattern as training data, built a machine-learning model to find more of it, fed the letters it found back in as further training, and iterated. The model returned ink strokes, then characters,

then, in a patch about four square centimetres across, more than ten of them in sequence: πορφύρας, purple. Farritor won forty thousand dollars in October 2023; Youssef Nader reached the same word independently and took ten thousand. By December, Farritor, Nader and Julian Schilliger had recovered about two thousand characters and shared the grand prize. Then papyrologists at Naples, at Pisa, and elsewhere read the Greek, confirmed the readings, and worked out what the text was. Four steps, and the labels matter. An institution supplied the capability. An outsider supplied the observation — the unasked question in its purest modern form, since no instrument was missing and the scans were in hundreds of hands. A second outsider supplied the machine — the better tool. Insiders supplied the judgment about whether any of it was true. That is the arrangement behind this book, and the chapter says so rather than leaving a reader to find it out. The middle step is the only one a machine performs. Somebody has to notice, and somebody else has to check.
See Endnote for full citation.

Zwicky, Fritz. Zwicky measured how fast galaxies were moving inside the Coma cluster and compared that with how much matter he could see. The galaxies were moving far too fast. At those speeds the cluster should have flown apart long ago; something invisible was holding it together, and there was a great deal of it. He applied the virial theorem to published measurements and named the missing material dunkle Materie. The calculation used standard physics, and any competent astronomer of 1933 could have run it. It was largely neglected for about forty years, and the chapter labels the case mixed for a reason. The 1933 result was an unasked question: the measurements were in print and the arithmetic was ordinary. The resolution took better tools. Acceptance came when Rubin and Ford (1970) measured how fast stars orbit inside individual galaxies, starting with Andromeda, and when radio astronomers reached further out than optical telescopes could. Both halves belong in the label, and the chapter’s rule is to say which part is which rather than round the case toward the answer it wants.
See Endnote for full citation.

Appendix A: SCOTUS Case Master Table

The Second Monopoly

Ninety-six rulings surveyed, 1829–2025 · cross-reference table, sorted by decision year

Scope. Every Supreme Court ruling this book surveys appears here: ninety-six rulings spanning one hundred ninety-six years, from Pennock v. Dialogue (1829) to Diamond Alternative Energy, LLC v. Environmental Protection Agency (2025). Inclusion is not endorsement, and survey is not citation. Some of these rulings are discussed in the running text. Others are here because a defender of the wall would raise them, because the open-book examination puts them in play, or because they bear on the argument at one remove. A row earns its place by belonging to the field the argument must cross, not by appearing in a chapter. The table lists every case the argument relies on, every case cited to concede a point, and every case a defender of the wall would cite in response — each described for what it actually holds. Several cut against parts of the argument, and the "Why Case is Relevant" note says so plainly. Each row also records how the Court divided, in the Score field explained in the key that follows the table. Cases decided on state-law preemption or copyright grounds are marked as such; they are cited for the principle they state, not for holdings they do not contain. Each row is one Supreme Court ruling. Thirteen of those rulings decided more than one separately docketed case in a single opinion — Shoshone Tribe of Indians v. United States (two), Brown v. Board of Education (four), Reid v. Covert (two), Graham v. John Deere Co. (three), Railway Labor Executives' Association v. Gibbons (two), Simon v. Eastern Kentucky Welfare Rights Organization (two), Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (three), Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (two), Printz v. United States (two), National Federation of Independent Business v. Sebelius (three), West Virginia v. Environmental Protection Agency (four), Food and Drug Administration v. Alliance for Hippocratic Medicine (two), and Loper Bright Enterprises v. Raimondo (two) — so the ninety-six rulings listed here resolved one hundred sixteen cases. Bolling v. Sharpe is listed on its own line because it was decided independently of Brown, on Fifth Amendment grounds, and was not among the four cases Brown consolidated. A companion case argued and decided the same day but carrying its own opinion is not part of the ruling it accompanies: Compco Corp. v. Day-Brite Lighting is a companion to Sears, Roebuck & Co. v. Stiffel Co.,

> *not part of it. Not one of the ninety-six holds that a federal agency may prohibit the remanufacture of a patent-expired design. The Court has never been asked.*

#	SCOTUS Case	Description
1	Pennock v. Dialogue (1829) 27 U.S. (2 Pet.) 1 (1829) **Score:** 6x0*8	**General Description:** An inventor who allowed the public to use and buy his hose invention for years before applying forfeited the patent. Justice Story, writing for a unanimous Court, held the patent void.
	In the reasoning the Court has treated as settled ever since, Story explained that the "main object" of the patent system is to promote the progress of science and useful arts, best served by giving the public at large the right to make, construct, use, and vend the invention at as early a period as possible, with due regard to the rights of the inventor. The opinion construes the Patent Act of 1793. The holding itself enforces the public's side of the bargain against a tardy inventor; holding and rationale point the same way. **Why Case is Relevant:** *The Patent Clause Prong* — The earliest judicial statement of the public's half of the bargain, from a Justice Madison appointed while framers still lived. Cited for the principle that the public's eventual right to make and vend the invention is the system's main object. Presented as the Court's reasoning, not as its holding.	
2	Wheaton v. Peters (1834) 33 U.S. (8 Pet.) 591 (1834) **Score:** —*10	**General Description:** Henry Wheaton, the Supreme Court's third Reporter of Decisions, sued his successor Richard Peters for republishing his volumes in condensed form. Wheaton claimed a perpetual common-law copyright in the reports.
	Justice McLean, for the Court, rejected a federal common-law copyright in published works: post-publication protection exists only by statute, and the statutory formalities must be strictly met. The Court added, unanimously on this point, that no reporter can hold any copyright in the written opinions of the Court, and that the Justices cannot confer such a right. Justices Thompson and Baldwin dissented from the judgment. The decision is a copyright case, not a patent case; it is cited for the constitutional structure the two clauses share — that the exclusive right is a statutory grant on the public's terms, not a natural right that precedes it. **Why Case is Relevant:** The Patent Clause Prong — Cited for the proposition that exclusive rights under the Patent and Copyright Clause are creatures of statute granted on the public's terms, not pre-existing natural rights. Copyright authority, offered for the shared clause structure; it is not a patent holding and is not presented as one.	

#	SCOTUS Case	Description
3	Patterson v. Kentucky (1878) 97 U.S. 501 (1878) **Score:** 8x0 **In Exam**	**General Description:** The State of Kentucky banned the sale of a patented "improved burning oil" that the state inspector had condemned as unsafe. The patentee argued that his federal patent placed the product beyond state regulatory reach.
	The Supreme Court rejected the argument unanimously. The patent secured the inventor's exclusive right to manufacture and sell against private competitors; it did not exempt the tangible product from state police-power regulation aimed at the safety of the public. The case is the foundational holding that a patent confers no immunity from generally applicable regulation — a holding the book accepts and builds on. The remanufacturer of a patent-expired design asks for the same regulation, not exemption from it. **Why Case is Relevant:** *Objections Considered* — The book's lead authority for the objection that a patent already carries no exemption from regulation — the remanufacturer asks for the same regulation everyone follows, not exemption from it. *Reason the Constitutional Argument Yourself* — Supplied among the Supreme Court holdings the reader reasons through: a patent does not exempt the patented article from generally applicable state safety regulation.	
4	Baker v. Selden (1879) 101 U.S. 99 (1879) **Score:** 9x0*20	**General Description:** Charles Selden copyrighted a book explaining a system of bookkeeping, with ruled forms for putting it into practice. Baker published a book using a similar system with differently arranged forms. Selden's estate sued for copyright infringement, contending that the copyright in the book carried with it the exclusive right to the system the book described.
	Justice Bradley, for the Court, held that the copyright secured the author's explanation, not the art it explained. A method or system may be secured, if at all, only by patent; describing it in a copyrighted book gives no exclusive right to use it, and the forms necessary to practice it are open to the public. The case is the root of the rule that copyright reaches expression and not function. **Why Case is Relevant:** *Objections Considered — The first of three reasons the copyright in engine-control firmware is narrower than its ninety-five-year term suggests: the maker's copyright covers its lines of code, not the job those lines do. A remanufacturer who writes its own program to run the same engine copies nothing copyright reaches.*	
5	Webber v. Virginia (1880) 103 U.S. 344 (1880) **Score:** —*7 **In Exam**	**General Description:** Two years after Patterson v. Kentucky, the Supreme Court reaffirmed the Patterson principle in the tax setting.
	Virginia imposed a license tax on the sale of sewing machines, and a patentee argued that his federal patent exempted the patented machines from the tax. The Court rejected the argument, holding that letters patent granted by the United States do not exclude the tangible property in which the invention is embodied from state tax or license law. The case is part of the Patterson line establishing that the existence of a patent does not place the patented product beyond ordinary state and federal regulation. **Why Case is Relevant:** *Objections Considered* — Cited as reaffirming the Patterson principle in the tax setting — a patent does not place the article beyond general state law. *Reason the Constitutional Argument Yourself* — Exam holding: a patent confers the right to make and sell, but does not place the article beyond the reach of general law.	

#	SCOTUS Case	Description
6	James v. Campbell (1882) 104 U.S. 356 (1882) **Score:** 6x1 **In Exam**	**General Description:** An inventor held patents on improvements used in machinery the federal government employed to print postage stamps and currency, and claimed the government's use infringed them.
	The Supreme Court held that the government's use of a patented invention is a taking of private property for which the inventor must be compensated — establishing that a patent is property in the full constitutional sense, not a privilege the government may draw on at will. **Why Case is Relevant:** *The Patent Clause Prong* — Establishes that a patent is property and that the government's own use of a patented invention is a taking — anchoring the property premise of the first prong. *The Takings Clause Prong* — Cited for the same property point on the takings side: when the federal government uses a patented invention without leave, it must pay. *Reason the Constitutional Argument Yourself* — Exam holding: a patent is private property; the government's use of a patented invention is a taking requiring compensation.	
7	United States v. Ballin (1892) 144 U.S. 1 (1892) **Score:** 8x0	**General Description:** The closest Supreme Court case to address the Journal Clause of Article I, Section 5, Clause 3. In Ballin the Court held the House Journal to be conclusive evidence of what occurred during
	a contested quorum vote, but did not interpret whether the Yeas-and-Nays override ("at the Desire of one fifth of those Present, be entered on the Journal") constitutionally requires individual named records or only an aggregate tally. The case is cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that no Supreme Court holding has squarely decided the interpretive scope of the Yeas-and-Nays override, leaving the question one of first impression. **Why Case is Relevant:** *Hybrid Legislative Voting* — The closest authority on the Journal Clause: the House Journal is conclusive evidence of what occurred, without resolving whether names or only totals must be recorded — leaving room for the chapter's hybrid-voting proposal.	

#	SCOTUS Case	Description
8	Singer Manufacturing Co. v. June Manufacturing Co. (1896) 163 U.S. 169 (1896) **Score:** —*7 **In Exam**	**General Description:** The Singer sewing machine had been built under patents that had long since expired. A competitor began making machines of the same form and selling them as "Singer" machines. Singer sued, claiming an exclusive right to both the design and the name.
	The Supreme Court refused. Once the patents expired, the Court held, "the right to make the patented article and to use the generic name passed to the public with the dedication resulting from the expiration of the patent"; and the name "Singer," having come to denote the type of machine rather than its single maker, passed to the public as a generic term, subject only to the competitor's duty to identify its own source of manufacture so buyers were not deceived. The case is cited for the Court's description of the post-expiration interest as property that becomes the public's, while the maker's trademark survives — the holding concerns patent expiration and trademark, not the Takings Clause. **Why Case is Relevant:** *The Takings Clause Prong* — Cited for the holding that at expiration "the right to make the thing formerly covered by the patent becomes public property" — the vested public right the prong protects. *Objections Considered* — Answers the "monopoly will be destroyed" objection: Singer kept its market lead for generations on its sales network, financing, and name — everything except the patent monopoly. *Reason the Constitutional Argument Yourself* — Exam holding: when the patents on a machine expire, its form becomes public property, free for anyone to make.	
9	McCormick Harvesting Machine Co. v. Aultman (1898) 169 U.S. 606 (1898) **Score:** —*7 **In Exam**	**General Description:** After a patent had been issued, its owner returned it to the Patent Office to have certain claims reissued; the Office instead treated the original patent as cancelled. The Supreme Court held the Patent Office had no such power.
	Once a patent issues, it becomes the property of the patentee, and only a court — not the executive agency that granted it — may set it aside. The case fixes the separation-of-powers line this chapter relies on. **Why Case is Relevant:** *The Patent Clause Prong* — Draws the line the chapter needs: once a patent issues it is the patentee's property, and only a court — not the granting agency — may set it aside. *Reason the Constitutional Argument Yourself* — Exam holding: once a patent issues it is the property of the patentee; only a court, not the agency, may set it aside.	
10	Muller v. Oregon (1908) 208 U.S. 412 (1908) **Score:** 9x0	**General Description:** Oregon limited women in factories and laundries to ten hours a day, and an employer challenged the law as an interference with freedom of contract. Justice Brewer, for a unanimous Court, upheld it.
	The decision is remembered less for its holding, which rested on reasoning about women that the law has long since abandoned, than for what the State filed. Of the one hundred thirteen pages of the brief prepared by Louis Brandeis and Josephine Goldmark, roughly ninety-eight consisted of factory-inspection reports, medical testimony, and foreign labor statutes. The Court took formal notice of the material, and the Brandeis brief has been the ancestor of every filing since that brings a court a body of evidence rather than a chain of citations. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The founding example of an outsider supplying the material a court needed. Goldmark had no law degree and did not argue the case; she assembled the record that changed how the case was decided.	

#	SCOTUS Case	Description
11	Continental Paper Bag Co. v. Eastern Paper Bag Co. (1908) 210 U.S. 405 (1908) **Score:** 8x1*22	**General Description:** Eastern Paper Bag held a patent on a self-opening bag machine it had never put into use, and sued Continental for infringing it. The Court held that a patentee who never uses its invention may still enjoin others from using it.
	Justice McKenna, for the Court, held that the right to exclude others is not lost by the patentee's nonuse: exclusion is "the very essence of the right conferred by the patent," and nonuse alone is no ground to deny an injunction. Justice Harlan dissented, arguing that public policy should bar relief. **Why Case is Relevant:** Preface — The third authority behind the book's title. A monopoly lies in the power to exclude, which needs no production, and that power is what outlives the patent: it passes to one federal rulebook that still decides who may build the expired design. *Four Problems, One Remedy* — The same point, by back-reference to the Preface.	
12	Standard Oil Co. of New Jersey v. United States (1911) 221 U.S. 1 (1911) **Score:** 9x0*1	**General Description:** The antitrust decision dissolving the Standard Oil trust, cited in the chapter on why nobody noticed as the legal endpoint of the public pressure that Ida Tarbell's McClure's series had
	built. **Why Case is Relevant:** *Why Nobody Noticed* — Cited as the historical trust that Ida Tarbell's reporting helped break up — an illustration of how public attention is mobilized.	
13	Motion Picture Patents Co. v. Universal Film Mfg. Co. (1917) 243 U.S. 502 (1917) **Score:** 6x3*9	**General Description:** A patent owner attached a notice restricting its patented projector to films it approved. The Court held the restriction void: an unconditional sale exhausts the patent, and the patentee may not extend the monopoly to unpatented supplies.
	Writing for the Court, Justice Clarke stated that since Pennock v. Dialogue (1829) the Court has consistently held that the primary purpose of the patent laws is not the creation of private fortunes for patent owners but the promotion of the progress of science and the useful arts. Holmes dissented, joined by McKenna and Van Devanter; McReynolds concurred in the result only. **Why Case is Relevant:** *The Patent Clause Prong* — The decision that elevated Pennock's public-interest reasoning toward settled rule, letting the argument say the Court has treated the public purpose as dominant "consistently" across a century. Cited for the ceiling on how far a patent's commercial benefit may be pushed, not for its exhaustion holding.	

#	SCOTUS Case	Description
14	Pennsylvania Coal Co. v. Mahon (1922) 260 U.S. 393 (1922) **Score:** 7x1 **In Exam**	**General Description:** A Pennsylvania statute forbade underground coal mining that would cause the surface above to collapse, even where the coal company had sold the surface rights while expressly keeping the right to mine the coal below.
	Justice Holmes, writing for the Court, held that the statute went too far and effected a taking. The decision is the origin of regulatory-takings law: it established that although property may be regulated, a regulation that goes too far will be recognized as a taking requiring compensation. **Why Case is Relevant:** *The Environmental Question* — Listed among the takings doctrines (regulatory takings) that the Patent Clause remedy does not disturb. *The Takings Clause Prong* — The origin of regulatory-takings doctrine — a regulation that goes "too far" is a taking — the doctrinal root the prong builds on. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation that diminishes property value too far may constitute a taking.	
15	J. W. Hampton, Jr., & Co. v. United States (1928) 276 U.S. 394 (1928) **Score:** 9x0	**General Description:** An importer challenged a tariff rate the President had raised under the flexible-tariff provision of the Tariff Act of 1922, arguing that Congress had unconstitutionally delegated its legislative power. Chief Justice Taft, for a unanimous Court, upheld the delegation.
	The opinion supplies the intelligible-principle test: Congress may authorize the Executive to act where it lays down by legislative act a principle intelligible enough to guide the exercise. The rule is a limit on how a granted power may be handed to an agency; it says nothing about whether the power existed to grant. **Why Case is Relevant:** *The Patent Clause Prong* — Cited for the principle that an agency cannot hold a power its creating Congress did not have. A delegation may be intelligible and still be void if what was delegated lay outside Congress's own authority.	
16	Shoshone Tribe of Indians v. United States (1937) 299 U.S. 476 (1937) **Score:** 8x0	**General Description:** The federal government had settled another tribe on land guaranteed to the Shoshone by treaty, diminishing their interest. The Supreme Court held the loss compensable.
	Justice Cardozo's opinion supplied the phrase this book relies on — "Spoliation is not management" — and the principle it carries: a collective, non-individually-titled interest is property the government must pay to take. The book cites the case for that principle, not for any holding about the patent commons. **Why Case is Relevant:** *The Takings Clause Prong* — Cardozo's "Spoliation is not management" — cited for the principle that a collective, non-individually-titled interest is still protected property, fitting the public's post-expiry right.	

#	SCOTUS Case	Description
17	Aetna Life Insurance Co. v. Haworth (1937) 300 U.S. 227 (1937) **Score:** 9x0	**General Description:** Aetna had issued disability-income policies to Edwin Haworth and had been paying benefits for several years when it concluded that Haworth was no longer disabled within the meaning of the contracts;
	Haworth disagreed but had not yet sued for past or future benefits. Aetna sought a federal-court declaration that the policies were void and no further benefits were owed. The Supreme Court held that the federal Declaratory Judgment Act of 1934 allowed Aetna to file: where a controversy is real and immediate, a federal court may declare the parties' legal rights without waiting for one of them to be sued or prosecuted. The case is the foundation of the constitutional-litigation path this book describes — the path on which the patent-bargain case will be filed before any plaintiff has been arrested, fined, or enjoined. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The leading early declaratory-judgment case: upheld the Declaratory Judgment Act against an Article III challenge, confirming a real controversy can be adjudicated before enforcement. *The Filable Case* — Cited for the same point in the filability chapter — a pre-enforcement dispute can be a genuine case or controversy.	
18	Kellogg Co. v. National Biscuit Co. (1938) 305 U.S. 111 (1938) **Score:** 6x2	**General Description:** National Biscuit claimed the exclusive right to the name "shredded wheat" and to the pillow shape of the biscuit. Both had been made under the Perky patents, which expired in 1912. Kellogg began making the same product in the same shape in 1922.
	Justice Brandeis, for the Court, held that on expiration of the patents there passed to the public "not only the right to make the article as it was made during the patent period, but also the right to apply thereto the name by which it had become known." The pillow shape went with it: upon expiration of the patents "the form, as well as the name, was dedicated to the public." The sole surviving obligation on the newcomer was to identify its own product so that buyers were not deceived as to source. The suit arose in diversity as a common-law unfair-competition claim, and the case is cited for what it holds about post-expiration rights, not for a constitutional proposition. Justices McReynolds and Butler would have affirmed the court below, stating that Kellogg was fraudulently seeking to appropriate a goodwill built up at great cost by the respondent. Eight Justices sat; Cardozo had died in July 1938 and Frankfurter was not yet confirmed. At 120 the Court rested the rule on *Singer*, at 185: "on the termination of the patent there passes to the public the right to make the machine in the form in which it was constructed during the patent." The Court then gave a second and independent reason the shape had to remain available. At 122 it found the pillow shape functional: the evidence was persuasive that the cost of the biscuit would be increased and its high quality lessened if some other form were substituted. Requiring a different form would therefore have imposed a penalty on the newcomer — an observation this argument draws from the Court's reasoning, not a holding the Court announced. **Why Case is Relevant:** *The Patent Clause Prong* — The Court's plainest statement that the form of the article, and not merely its underlying idea, enters the public domain when the patent expires. The faithful remake is the right the Court described, and Kellogg completes the line that runs from Singer (1896) through Scott Paper (1945) to Sears and Compco (1964). *Objections Considered* — The source-identification duty is the answer to the trademark objection: the remanufacturer must mark its own goods as its own, and that is all it must do. The functionality finding is the closer analogue still: the Court would not force a competitor into a different form when the original form was the one that worked.	

#	SCOTUS Case	Description
19	Betts v. Brady (1942) 316 U.S. 455 (1942) **Score:** 6x3	**General Description:** An indigent defendant tried for robbery in Maryland asked the court to appoint counsel and was refused. Justice Roberts, for a six-Justice majority, held that due process did not require a State to furnish counsel in a noncapital case. Justice Black dissented, joined by Justices Douglas and Murphy.
	The case stood for twenty-one years and was overruled in Gideon v. Wainwright (1963), where Justice Black wrote for a unanimous Court. The book cites it not for either holding but for the interval between them, and for the fact that the Court had been looking for a vehicle in which to reconsider Betts before Gideon supplied one. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited to show that the case a court needs and the theory that decides it rarely come from the same person. Gideon supplied the case; Abe Fortas supplied the argument.	
20	Wickard v. Filburn (1942) 317 U.S. 111 (1942) **Score:** 9x0	**General Description:** Roscoe Filburn grew wheat on his Ohio farm in excess of his federal allotment, intending the surplus for his own use, not for sale.
	The Supreme Court upheld the penalty against him, reasoning that even purely local activity may be regulated under the Commerce Clause when its aggregate effect on interstate commerce is substantial. The case is the high-water mark of the modern commerce power and the authority a defender of the regulatory wall will cite for the proposition that Congress may regulate the manufacture of vehicles in commerce. The book does not dispute that power. The Patent Clause prong answers it on a different ground: a power broad enough to reach the wheat on Filburn's farm is still not a power to cancel the public's ownership of an expired invention, because no grant of federal power authorizes Congress to nullify what another clause of the Constitution commands. **Why Case is Relevant:** *The Patent Clause Prong* — Cited to concede the breadth of the Commerce Clause power — the book does not dispute Congress's authority to regulate; its argument is about the Patent Clause limitation.	
21	Skidmore v. Swift & Co. (1944) 323 U.S. 134 (1944) **Score:** 9x0	**General Description:** Held that an agency interpretation is entitled to judicial respect proportional to its power to persuade — the "Skidmore deference" standard.
	Why Case is Relevant: *You No Longer Need to Break a Law to Challenge One* — Cited for "Skidmore weight" — after Loper Bright, agency interpretations earn respect only to the extent persuasive, not deference — which matters to the patent-bargain case. *The Filable Case* — Same point in the filability chapter: the agency's view earns respect only if persuasive; "the thumb came off the scale."	

#	SCOTUS Case	Description
22	Korematsu v. United States (1944) 323 U.S. 214 (1944) **Score:** 6x3	**General Description:** The Court upheld the wartime exclusion of Japanese Americans from the West Coast. Justice Black wrote for the majority; Justices Roberts, Murphy, and Jackson each dissented.
	The claim of military necessity was made to the Court while contrary evidence sat in the government's own files. In 1983 a federal court in San Francisco vacated Fred Korematsu's conviction on a writ of coram nobis after that record surfaced, and in Trump v. Hawaii (2018) the Court stated that Korematsu had been gravely wrong the day it was decided. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The most damning instance of a question nobody asked. The evidence was not merely available somewhere; it was in the government's files, put there by the government, for forty years.	
23	Scott Paper Co. v. Marcalus Manufacturing Co. (1945) 326 U.S. 249 (1945) **Score:** 6x2 **In Exam**	**General Description:** Marcalus had assigned a patent to Scott Paper, then later built a product using a design covered by a different patent that had already expired.
	Scott Paper invoked a contract doctrine — assignor estoppel — to stop its former assignor from competing. The Supreme Court refused. The public's right to use an invention whose patent has expired, the Court held, is a matter of patent policy that a private estoppel rule cannot override. The patent laws, the Court held, "dedicate to public use the invention of an expired patent," and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent." **Why Case is Relevant:** *Four Problems, One Remedy* — Listed among the cases affirming that what a patent covered belongs to the public at expiration. *The Patent Clause Prong* — Authority a plaintiff sets before the court: a patent holder cannot use a private contract to keep control after the patent expires. *The Takings Clause Prong* — Cited for describing the expired invention as dedicated to public use — the vested right the prong protects. *Reason the Constitutional Argument Yourself* — Referenced in the examination's synthesis of the public-domain cases (with Graham and Singer).	
24	Brown v. Board of Education (1954) 347 U.S. 483 (1954) **Score:** 9x0	**General Description:** Chief Justice Warren, for a unanimous Court, held that separate educational facilities are inherently unequal and that state-imposed school segregation violates the Equal Protection Clause of the Fourteenth Amendment.
	The work of Kenneth and Mamie Clark appears in footnote eleven of the opinion, in a list of social-science authorities. The book cites the case for what that footnote does and does not show: the Clarks were psychologists, neither was a lawyer, and their studies did not decide the case so much as supply a sentence the Court was prepared to write. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited against the popular version of its own story. Outsiders furnished evidence; lawyers and Justices carried it to a holding.	

#	SCOTUS Case	Description
25	Bolling v. Sharpe (1954) 347 U.S. 497 (1954) **Score:** 9x0 **In Exam**	**General Description:** The federal companion to Brown v. Board of Education, decided the same day. Brown held that state-mandated school segregation violated the Fourteenth Amendment's Equal Protection Clause.
	Bolling faced the parallel question for the District of Columbia, which is not a state and to which the Fourteenth Amendment does not apply. The Court held that the Fifth Amendment's Due Process Clause incorporates an equal-protection guarantee against the federal government. Federal action classifying similarly situated persons unequally is subject to constitutional review under the Fifth Amendment even where the Fourteenth Amendment has no purchase. The case is what makes any federal-equal-protection framing of the regulatory wall possible at all. **Why Case is Relevant:** *The Patent Clause Prong* — Cited for the equal-protection obligation the Fifth Amendment imposes on the federal government (the wall is federal), though economic classifications get only rational-basis review. *The Takings Clause Prong* — Supplies the bridge applying equal protection to the federal government, reaffirmed in Adarand. *Reason the Constitutional Argument Yourself* — Exam holding: the Fifth Amendment's Due Process Clause imposes a federal equal-protection obligation.	
26	Williamson v. Lee Optical of Oklahoma, Inc. (1955) 348 U.S. 483 (1955) **Score:** 8x0	**General Description:** Oklahoma forbade opticians to fit or duplicate lenses without a prescription. Justice Douglas, for the Court, upheld the statute; Justice Harlan took no part.
	The opinion is the canonical statement of rational-basis review for economic regulation: the law may be needless and wasteful, and it is for the legislature rather than the courts to balance advantage against disadvantage. A court need only be able to conceive of a state of facts that would justify the classification. **Why Case is Relevant:** *Objections Considered* — The reason equality cannot be pleaded as a standalone third prong. An equal-protection claim against the regulatory wall would meet rational-basis review and lose, which is why equality operates inside each of the two prongs rather than beside them.	
27	Reid v. Covert (1957) 354 U.S. 1 (1957) **Score:** 6x2	**General Description:** Civilian wives of servicemen stationed in England and Japan were tried by court-martial for murder under executive agreements with the host nations. Justice Black announced the judgment for a plurality of four; Justices Frankfurter and Harlan concurred in the result; Justices Clark and Burton dissented; Justice Whittaker took no part.
	The Court held that the Fifth and Sixth Amendments guaranteed the defendants indictment and trial by jury, and that neither the treaty power nor the Necessary and Proper Clause could authorize what those Amendments forbade. The plurality's formulation: no agreement with a foreign nation can confer on Congress or any other branch a power free from the restraints of the Constitution. **Why Case is Relevant:** *The Patent Clause Prong* — The clearest statement that a power granted in one clause cannot escape a limit written in another. Cited alongside Gibbons, Horne, and Printz for the proposition that the commerce power cannot produce the outcome "limited Times" forbids. The plurality's wording is relied on as a statement of principle, not as a majority holding.	

#	SCOTUS Case	Description
28	Eastern Railroad Presidents Conference v. Noerr Motor Freight, Inc. (1961) 365 U.S. 127 (1961) **Score:** 9x0	**General Description:** Twenty-four eastern railroads ran a publicity campaign against the trucking industry, and truckers sued under the Sherman Act. Justice Black, for a unanimous Court, held that the campaign was not an antitrust violation even though its purpose was to injure competitors.
	With United Mine Workers v. Pennington (1965) the case established the Noerr-Pennington doctrine, which immunizes petitioning the government from antitrust liability. The doctrine is why the regulatory wall cannot be attacked as a conspiracy: the lobbying that raised it is protected activity. **Why Case is Relevant:** *Objections Considered* — Cited to close an alternative theory rather than to open one. A plaintiff who reached for an antitrust account of how the wall was built would meet Noerr-Pennington, which is one reason the case must be constitutional.	
29	Aro Manufacturing Co. v. Convertible Top Replacement Co. (1961) 365 U.S. 336 (1961) **Score:** 6x3*2	**General Description:** The Supreme Court's repair-and-reconstruction decision, arising from replacement tops for Ford and General Motors convertibles.
	The Court held that replacing a worn, unpatented element of a patented combination is permissible repair, not infringing reconstruction, and that no unpatented element of a combination is entitled to the patent monopoly, however essential to the combination it may be. Cited for the principle that a combination patent reaches only the totality of its claimed elements. **Why Case is Relevant:** *Objections Considered* — Cited on combination patents: no single unpatented element carries the monopoly — answering an objection about component-level rights.	
30	Gideon v. Wainwright (1963) 372 U.S. 335 (1963) **Score:** 9x0	**General Description:** Clarence Earl Gideon, tried in Florida without counsel, wrote his own petition to the Supreme Court in pencil. Justice Black wrote for a unanimous Court, overruling Betts v. Brady and holding that the Sixth Amendment right to counsel binds the States through the Fourteenth.
	Gideon is the archetype everyone reaches for when an outsider is said to have changed constitutional law, and the book cites it to show that the archetype does not fit. The Court appointed Abe Fortas to argue the case, and the winning argument was his. Gideon supplied the vehicle, not the theory. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The case that most nearly answers the objection that an outsider never wins, examined and found not to answer it. The two jobs — finding the injury and arguing the law — sat in different people.	

#	SCOTUS Case	Description
31	Sears, Roebuck & Co. v. Stiffel Co. (1964) 376 U.S. 225 (1964) **Score:** 9x0*11	**General Description:** Stiffel's pole-lamp patents were held invalid for want of invention. Sears copied the lamp and sold it cheaply. Stiffel won below on an Illinois unfair-competition theory.
	Justice Black, for a unanimous Court, reversed. An article on which the patent has expired, like an unpatentable article, is in the public domain and may be made and sold by whoever chooses to do so. A State may not use unfair-competition law to give what amounts to a patent monopoly on an unpatented article, though it may require labeling to prevent deception as to source. The holding runs against a State, not against a federal agency. The same opinion, at 230, restates what expiration does: when the patent expires the monopoly expires with it, and the right to make the article — "including the right to make it in precisely the shape it carried when patented" — passes to the public, citing *Kellogg* and *Singer*. That passage is the Court's reasoning, not its holding, and is cited as such. **Why Case is Relevant:** The Patent Clause Prong — The Court's plainest statement that the expired or unpatentable article belongs to the public and may be made by anyone. Cited for that principle. The holding is a state-law preemption holding: it bars a State from re-erecting patent-like protection. The Court has never applied the principle against a federal regulator, and this argument does not claim it has.	
32	Brulotte v. Thys Co. (1964) 379 U.S. 29 (1964) **Score:** 8x1	**General Description:** A hop-picking machine was sold under a license requiring royalties that ran past the expiration of every patent embodied in it. Justice Douglas, for the Court, held the post-expiration royalty provisions unenforceable. Justice Harlan dissented.
	The patents in the case expired on schedule and no term was extended by a day. The Court struck the arrangement down anyway, because the practical consequence of expiration — the public's unrestricted freedom in the released invention — had been withheld by another instrument. Reaffirmed on stare decisis grounds in Kimble v. Marvel Entertainment (2015). **Why Case is Relevant:** *Objections Considered* — First rung of the ladder answering the enumerated-powers objection. Had "limited Times" been a scheduling instruction, Brulotte would have been decided the other way in a paragraph.	
33	Graham v. John Deere Co. (1966) 383 U.S. 1 (1966) **Score:** 8x0 **In Exam**	**General Description:** A dispute over the validity of a patent on a clamp for plow shanks. The Supreme Court used the case to set out how courts decide whether an invention is "non-obvious" enough to deserve a
	patent — and, more important for this book, to describe the constitutional nature of the Patent Clause itself. The Court called the clause both a grant of power to Congress and a limitation on that power. In the Court's own words, the clause "is both a grant of power and a limitation," and "Congress in the exercise of the patent power may not overreach the restraints imposed by the stated constitutional purpose." The Court continued, at 5–6, that Congress "may not authorize the issuance of patents whose effects are to remove existent knowledge from the public domain, or to restrict free access to materials already available." **Why Case is Relevant:** *Four Problems, One Remedy* — Listed among the cases affirming that an expired patent's subject belongs to the public. *The Patent Clause Prong* — The chapter's keystone authority: Graham holds the Patent Clause is "both a grant of power and a limitation upon the exercise of that power." *Reason the Constitutional Argument Yourself* — Exam holding: the Patent Clause is both a grant of power and a limitation on it; Congress may not overreach in exercising the patent power.	

#	SCOTUS Case	Description
34	Menominee Tribe of Indians v. United States (1968) 391 U.S. 404 (1968) **Score:** 6x2	**General Description:** The Menominee Tribe sued the United States in the Court of Claims for compensation after federal legislation terminated the tribe's recognized status.
	The Supreme Court held that treaty-guaranteed hunting and fishing rights survived the termination, because Congress had not clearly abrogated them. In this book the case is one of three showing a settled pattern: a tribe may bring a compensation claim against the federal government through ordinary litigation, breaking no law to do so. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — Example of the older plaintiff pattern — the Tribe filed in the Court of Claims for compensation — illustrating that a plaintiff need not break a law to sue.	
35	United States Department of Agriculture v. Moreno (1973) 413 U.S. 528 (1973) **Score:** 7x2 **In Exam**	**General Description:** Held that a food-stamp restriction aimed at excluding "hippie" households failed rational-basis review; bare desire to harm a group is not a legitimate interest.
	Why Case is Relevant: *Objections Considered* — Paired with Romer as authority that a classification resting only on a bare desire to harm a disfavored group fails rational-basis review. *Reason the Constitutional Argument Yourself* — Exam holding: a classification fails rational-basis review where its only rationale is a bare desire to harm a disfavored group.	
36	Kewanee Oil Co. v. Bicron Corp. (1974) 416 U.S. 470 (1974) **Score:** 6x2*13	**General Description:** Ohio trade-secret law protected the process Harshaw Chemical used to grow synthetic crystals. The Sixth Circuit held that federal patent law preempted the state protection. The Supreme Court reversed.
	Chief Justice Burger, for the Court, held that state trade-secret law survives federal patent preemption. The reasoning: trade-secret protection is weaker than a patent, it withdraws nothing from the public domain, and it poses no real risk of deterring inventors from seeking patents, so partial preemption as to patentable subject matter would burden the States without serving federal patent policy. Justice Marshall concurred in the result; Justice Douglas, joined by Justice Brennan, dissented on the view that a State may not confer protection on what Congress left free. The holding is about a parallel state regime. It does not address what a federal agency may do to a design whose patent has already expired; that distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority — cited for the proposition that federal patent policy does not displace every other legal regime touching an invention, and distinguished on the ground that trade-secret law leaves the public free to reverse-engineer and copy what it learns, while the certification barrier forbids the lawful sale of the copy itself.	

#	SCOTUS Case	Description
37	Warth v. Seldin (1975) 422 U.S. 490 (1975) **Score:** 5x4 **Not in Exam**	**General Description:** Low- and moderate-income residents of the Rochester, New York, area and allied organizations challenged the zoning ordinance of the suburb of Penfield, alleging that it kept out the affordable housing they would otherwise have been able to live in.
	Justice Powell, for the Court, held that the plaintiffs lacked standing. They had not shown that any specific project would have been built absent the ordinance, or that relief would let them live in Penfield; their injury depended on choices builders not before the court might never make. Justice Douglas dissented, and Justice Brennan, joined by Justices White and Marshall, dissented. **Why Case is Relevant:** *Objections Considered* — The government's closest authority against a buyer whose relief depends on a producer's decision to build. The chapter concedes it governs a buyer suing alone and answers it with Village of Arlington Heights, decided two years later.	
38	Simon v. Eastern Kentucky Welfare Rights Organization (1976) 426 U.S. 26 (1976) **Score:** 8x0*23 **Not in Exam**	**General Description:** Indigent plaintiffs and welfare-rights organizations challenged an Internal Revenue Service ruling that relaxed the free-care condition hospitals had to meet to keep their tax exemption, alleging that hospitals had turned them away.
	Justice Powell, for the Court, held that the plaintiffs lacked standing. Their injury could be traced to the ruling only through the independent decisions of hospitals not before the court, and it was speculative whether a judgment against the Treasury officials would lead any hospital to serve them. **Why Case is Relevant:** *Objections Considered* — Not cited in the chapter, which answers the same objection through Warth v. Seldin. Listed because a defender of the wall would cite it as the original holding that a plaintiff's injury cannot rest on choices by third parties not before the court.	
39	Village of Arlington Heights v. Metropolitan Housing Development Corp. (1977) 429 U.S. 252 (1977) **Score:** 5x3*24 **Not in Exam**	**General Description:** A nonprofit developer contracted for a site in a Chicago suburb and planned a specific low- and moderate-income housing project there. The village refused to rezone the site, and the developer and prospective residents sued.
	Justice Powell, for the Court, held that the developer had standing: the refusal to rezone was the barrier to a specific project, and the remaining uncertainty over financing and other contingencies did not defeat that injury. A prospective resident who sought housing near his job had standing as well. On the merits the Court found no proof of discriminatory purpose and reversed. **Why Case is Relevant:** *Objections Considered* — The authority for the standing of a would-be builder with a definite project blocked by law — the reason the objection's answer puts an independent remanufacturer among the plaintiffs.	

#	SCOTUS Case	Description
40	Hunt v. Washington State Apple Advertising Commission (1977) 432 U.S. 333 (1977) **Score:** 8x0	**General Description:** North Carolina passed a statute requiring closed shipping containers of apples to display the U.S. grade rather than any state grade.
	The Washington State Apple Advertising Commission, representing Washington apple growers, sued, claiming the statute burdened interstate commerce. The Supreme Court used the case to set out the three-prong test for when a trade or membership association may sue on behalf of its members: at least one member must have standing in his own right, the interests at stake must be germane to the association's purpose, and neither the claim nor the relief may require the participation of individual members. The case is the doctrinal basis for associational standing — the theory under which trucker associations, farm associations, and remanufacturer associations could sue in the patent-bargain case. **Why Case is Relevant:** *Direct and Associational Plaintiff Categories* — The governing associational-standing rule: an association may sue on its members' behalf when they would have standing to sue individually. *Why You Want to Be the Plaintiff* — Cited for the same associational-standing rule in the chapter on choosing the plaintiff.	
41	Penn Central Transportation Co. v. City of New York (1978) 438 U.S. 104 (1978) **Score:** 6x3 **In Exam**	**General Description:** New York City's landmarks law blocked the owner of Grand Central Terminal from constructing an office tower above the station. The owner claimed the restriction took its property.
	The Supreme Court ruled for the city and, in doing so, set out the framework that still governs most regulatory-takings claims: courts weigh the economic impact of the regulation on the owner, its interference with reasonable investment-backed expectations, and the character of the government action. Penn Central is the balancing test for regulations that diminish a property's value without destroying it. **Why Case is Relevant:** *The Environmental Question* — The investment-backed-expectations test, cited to show the strategic builder who enters after a regulation has weak takings footing. *The Takings Clause Prong* — The multi-factor regulatory-takings balancing test — one of the two roads the prong analyzes. *Objections Considered* — Cited (with Lucas) to show a latecomer loses the takings analysis for lack of investment-backed expectations. *Reason the Constitutional Argument Yourself* — Exam holding: whether a regulation is a taking is judged by weighing economic impact, interference with expectations, and the character of the government action.	

#	SCOTUS Case	Description
42	Aronson v. Quick Point Pencil Co. (1979) 440 U.S. 257 (1979) **Score:** 9x0 **In Exam**	**General Description:** An inventor licensed an unpatented keyholder design to a manufacturer in exchange for a royalty, with a contractual provision reducing the royalty if no patent ever issued. No patent did.
	The manufacturer eventually argued that the royalty contract was unenforceable as inconsistent with federal patent policy, since the keyholder's design was free for all to copy. The Supreme Court rejected the argument. Federal patent law leaves the unpatented public domain free for everyone, but it does not bar private parties from contracting on their own terms for the use of a design, and a state contract law that lets them do so is not preempted. The case is the modern restatement of the Patterson v. Kentucky principle that patent law and ordinary regulation coexist. **Why Case is Relevant:** *Objections Considered* — Cited as the modern restatement that federal patent law does not displace ordinary state law governing a product. *Reason the Constitutional Argument Yourself* — Exam holding: federal patent law does not displace ordinary state regulation of a product.	
43	Andrus v. Allard (1979) 444 U.S. 51 (1979) **Score:** 9x0 **In Exam**	**General Description:** The federal Eagle Protection Act, joined by the Migratory Bird Treaty Act, made it unlawful to sell artifacts incorporating eagle parts even where the parts had been lawfully obtained before the statute.
	The owners of pre-existing inventories of Indian artifacts containing eagle feathers sued, claiming the sale ban was a Fifth Amendment taking of their property. The Supreme Court unanimously rejected the claim. A regulation that destroys one of the strands in the bundle of property rights — here, the right to sell — is not necessarily a taking, the Court reasoned, because possession, transport, and noncommercial transfer of the artifacts remained lawful. The case is the leading adverse Takings precedent the patent-bargain case will have to distinguish; the chapter on objections considered develops the distinction. **Why Case is Relevant:** *Objections Considered* — Opens the adverse-precedent discussion: a federal ban on commercial sale is not necessarily a taking where other uses remain. *Reason the Constitutional Argument Yourself* — Exam holding: a ban on the commercial sale of lawfully held property is not necessarily a taking where other uses remain.	
44	United States v. Sioux Nation of Indians (1980) 448 U.S. 371 (1980) **Score:** 8x1	**General Description:** Congress seized the Black Hills of South Dakota from the Sioux by an 1877 Act, in violation of an earlier treaty.
	The Supreme Court held the seizure a taking that required just compensation, though the land was held by the Sioux collectively rather than by any individual owner. The book uses the case two ways: as proof that a taking claim can be litigated without anyone breaking a law, and as authority that common, non-individually-titled property falls within the Just Compensation Clause. The book does not claim the case reached the Expired Patent Commons; it concerned Indian treaty lands. It claims only that the case settles the governing principle — property need not be privately and individually titled to be compensable — and that extending that principle to the commons is a short, logical step the chapter asks the court to take. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The most directly relevant plaintiff-pattern case — the Tribe sued for compensation for the Black Hills — no law needed to be broken. *The Takings Clause Prong* — Cited for the holding that Congress's 1877 seizure of the Black Hills was a taking requiring just compensation, though the land belonged to the tribe collectively.	

#	SCOTUS Case	Description
45	Railway Labor Executives' Association v. Gibbons (1982) 455 U.S. 457 (1982) **Score:** 9x0*15	**General Description:** Congress enacted a statute governing the liquidation of a single bankrupt railroad and the claims of its employees. Justice Rehnquist, for the Court, held that the statute was a bankruptcy law, that it was not uniform, and that the Commerce Clause could not be used to enact it.
	To hold that Congress may enact non-uniform bankruptcy laws under the commerce power, the Court reasoned, would eradicate from the Constitution the limitation the Bankruptcy Clause imposes. The case is the Court's clearest application of the specific-governs-the-general canon to two clauses standing side by side in Article I, Section 8. **Why Case is Relevant:** *The Patent Clause Prong* — The anchor for the argument that a limitation written into one clause of Article I, Section 8 cannot be escaped by legislating under another. *Objections Considered* — Answers the enumerated-powers objection directly. Carrying the reasoning from the Bankruptcy Clause to "limited Times" is the book's step and not the Court's, and the entry says so.	
46	Loretto v. Teleprompter Manhattan CATV Corp. (1982) 458 U.S. 419 (1982) **Score:** 6x3 **In Exam**	**General Description:** A New York statute required landlords to permit cable-television operators to install a small cable box and a few feet of cable on the exterior of apartment buildings, in exchange for a one-time payment of one dollar set by a state commission.
	Jean Loretto, a New York City landlord, challenged the mandate as a taking. The Supreme Court held that any permanent physical occupation of property authorized by government, however minor, is a per se taking — categorically requiring compensation regardless of the economic impact on the owner or the public interest in the occupation. The case is the per se physical-occupation line that runs through Horne v. Department of Agriculture (2015) and Cedar Point Nursery v. Hassid (2021). **Why Case is Relevant:** *The Environmental Question* — Listed among the takings categories (physical occupation) the Patent Clause remedy does not disturb. *The Takings Clause Prong* — The per se physical-occupation rule that predates Lucas: "a permanent physical occupation of property is a taking," however small. *Reason the Constitutional Argument Yourself* — Exam holding: a permanent physical occupation of property is a per se taking.	
47	Alfred L. Snapp & Son, Inc. v. Puerto Rico (1982) 458 U.S. 592 (1982) **Score:** 8x0	**General Description:** Puerto Rico sued Virginia apple growers alleging discrimination against Puerto Rican farmworkers in favor of temporary foreign labor. Justice White, for the Court, held that Puerto Rico had parens patriae standing; Justice Powell took no part.
	The opinion is the modern statement of the doctrine: a sovereign may sue on a quasi-sovereign interest, which includes the health and well-being — physical and economic — of a substantial segment of its population, provided the sovereign articulates an interest apart from those of particular private parties. **Why Case is Relevant:** *Sovereign and Public-Fiduciary Plaintiffs* — The controlling authority for a State attorney general suing on behalf of households injured by the wall. Supplies the quasi-sovereign interest the chapter's sovereign plaintiff categories rest on.	

#	SCOTUS Case	Description
48	Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (1984) 467 U.S. 837 (1984) **Score:** 6x0	**General Description:** Justice Stevens, for the Court, upheld the EPA's plantwide definition of a stationary source under the Clean Air Act and set out the two-step framework that governed judicial review of agency statutory interpretation for forty years. Justices Marshall, Rehnquist, and O'Connor took no part.
	Chevron deference is the reason the regulatory wall stood unexamined for so long: a court asked whether an agency had exceeded its statute deferred to the agency's own answer wherever the statute was ambiguous. Overruled by Loper Bright Enterprises v. Raimondo (2024), which is case 84 in this table. **Why Case is Relevant:** *Why Nobody Noticed: How a Constitutional Violation Hides in Plain Sight* — The doctrinal reason the wall was never tested. *The Filable Case: Why 2026 and Not Before* — Its overruling is the first of the four conditions that make the case filable now and not before.	
49	Ruckelshaus v. Monsanto Co. (1984) 467 U.S. 986 (1984) **Score:** 7x1*3 **In Exam**	**General Description:** Monsanto challenged a federal pesticide statute that allowed the EPA to use and disclose the confidential test data that companies submitted for registration.
	The Supreme Court held that a trade secret is property protected by the Takings Clause, even though it has no physical form, and that a regulation appropriating the value of that confidential data can be a taking — though whether it is depends on whether the company held a reasonable investment-backed expectation that the data would stay confidential. The case establishes that intangible rights of defined economic value are property for Fifth Amendment purposes. **Why Case is Relevant:** *The Takings Clause Prong* — The case the prong leans on hardest — trade secrets are protected property under the Takings Clause. *Reason the Constitutional Argument Yourself* — Exam holding: intangible property, such as a trade secret, is protected under the Takings Clause.	
50	United States v. 50 Acres of Land (1984) 469 U.S. 24 (1984) **Score:** 9x0	**General Description:** Held that a public condemnee is entitled to fair market value, not the cost of a substitute facility, as just compensation.
	Why Case is Relevant: *The Environmental Question* — Compensation is fair market value at the time of the taking — the remanufacturer who tools a factory to attract a taking gets no windfall.	
51	Heckler v. Chaney (1985) 470 U.S. 821 (1985) **Score:** 9x0	**General Description:** An agency's decision not to undertake an enforcement action is presumptively committed to the agency's discretion and unreviewable under the Administrative Procedure Act.
	Why Case is Relevant: *The Anti-Backsliding Provision* — Cited for the holding that an agency's decision not to enforce is presumptively unreviewable — relevant to how the anti-backsliding regime operates.	

#	SCOTUS Case	Description
52	City of Cleburne v. Cleburne Living Center, Inc. (1985) 473 U.S. 432 (1985) **Score:** 9x0*16	**General Description:** A Texas city denied a special-use permit for a group home. Justice White, for a six-Justice majority, declined to treat the intellectually disabled as a quasi-suspect class but struck the ordinance down as applied under rational-basis review. Justices Marshall, Brennan, and Blackmun concurred in the judgment in part and dissented in part.
	The case is the leading example of rational basis with teeth: a classification resting on irrational prejudice fails even the deferential standard. Justice Marshall's separate opinion objected that the Court's review was heightened scrutiny under another name. **Why Case is Relevant:** *Objections Considered* — The escape route from Williamson v. Lee Optical, and the reason the escape is not taken. Cleburne shows what a successful equality claim needs; the regulatory wall has stated rationales that Cleburne's ordinance lacked.	
53	Bowers v. Hardwick (1986) 478 U.S. 186 (1986) **Score:** 5x4	**General Description:** Justice White, for a five-Justice majority, upheld a Georgia sodomy statute, stating in the course of the opinion that proscriptions against such conduct had ancient roots. Justices Blackmun and Stevens each dissented.
	Seventeen years later a brief filed by historians in Lawrence v. Texas took the historical claim apart using sources that had been in print the whole time. The record did not say what Bowers said it said, and the category of person the opinion described did not exist in the period it pointed to. The Court overruled Bowers in 2003. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The cleanest instance of the unasked question. Nothing was discovered; a court asserted a fact, everyone believed it, and for seventeen years nobody checked it against scholarship that already said otherwise.	
54	First English Evangelical Lutheran Church of Glendale v. County of Los Angeles (1987) 482 U.S. 304 (1987) **Score:** 6x3 **In Exam**	**General Description:** A county ordinance temporarily barred a church from rebuilding on flood-prone land it owned. The church argued the ordinance had taken its property for the period the ban was in force.
	The Supreme Court used the case to settle what the Takings Clause requires: it does not forbid the government from taking property, but it requires the payment of just compensation when the government does. A finding that a regulation effects a taking establishes that compensation is owed; it is not, by itself, an order striking the regulation down. **Why Case is Relevant:** *The Takings Clause Prong* — Cited for the remedy point: the Takings Clause does not forbid the taking; it requires compensation. *Reason the Constitutional Argument Yourself* — Exam holding: the Takings Clause does not forbid government from taking property; it requires just compensation.	

#	SCOTUS Case	Description
55	Nollan v. California Coastal Commission (1987) 483 U.S. 825 (1987) **Score:** 5x4	**General Description:** The Nollans owned a beachfront lot in Ventura County and applied for a permit to replace their bungalow with a larger house. The California Coastal Commission granted the permit on condition that they dedicate a public easement across their beach.
	The Supreme Court held that a regulatory condition must bear an essential nexus to the harm the regulation claims to address; an easement allowing the public to walk along the beach had no such nexus to the visual obstruction the larger house was said to cause. The case is the first prong of the modern proportionality test for regulatory conditions on property use, completed in Dolan v. City of Tigard (1994) and reaffirmed in Sheetz v. County of El Dorado (2024). **Why Case is Relevant:** *The Takings Clause Prong* — Cited (with Dolan) on the exactions doctrine, and the argument that it reaches only individualized conditions — which the book answers via Sheetz.	
56	Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) 488 U.S. 336 (1989) **Score:** 9x0	**General Description:** The unanimous Supreme Court decision holding that a county tax-assessment system that valued recently-acquired coal properties at full current value while leaving long-held properties at
	decades-old assessments violated the Equal Protection Clause. **Why Case is Relevant:** *The Takings Clause Prong* — The equality-within-takings authority: a county taxed recently-acquired property far above comparable older holdings — the asymmetry the prong invokes. *Objections Considered* — The equality argument inside the Takings prong, walked at length — similarly situated property treated unequally.	
57	Bonito Boats, Inc. v. Thunder Craft Boats, Inc. (1989) 489 U.S. 141 (1989) **Score:** 9x0	**General Description:** Florida barred the use of a direct molding process to duplicate unpatented boat hulls. Bonito Boats, which had never sought a patent on its hull, sued a competitor under the statute.
	Justice O'Connor, for a unanimous Court, held the Florida statute preempted. The Patent Clause reflects a balance between encouraging innovation and avoiding monopolies that stifle competition, and the Clause contains both a grant of power and limitations on its exercise. By giving patent-like protection to designs in general circulation that Congress left unprotected, Florida entered a field reserved to Congress. Like Sears, the holding runs against a State. The opinion also states, at 151, that “free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception,” and that the ultimate goal of the patent system is to bring new designs and technologies into the public domain through disclosure. **Why Case is Relevant:** The Patent Clause Prong — Reaffirms Sears and supplies the Court's modern statement that the Patent Clause is both a grant and a limitation. Cited for the public-domain principle and the grant-and-limitation reading. State-law preemption authority; the Court has not applied it to a federal agency, and the step from this holding to the federal wall is this argument's own.	

#	SCOTUS Case	Description
58	Lujan v. Defenders of Wildlife (1992) 504 U.S. 555 (1992) **Score:** 6x3	**General Description:** Established the three-part constitutional standing test: injury in fact, causation, and redressability.
	Why Case is Relevant: *Sovereign and Public-Fiduciary Plaintiffs* — The source of the three-part standing test — injury in fact, causation, redressability. *Direct and Associational Plaintiff Categories* — Cited for the generalized-grievance bar the chapter's plaintiff categories must clear. *Objections Considered* — Cited for the rule that a plaintiff who is the object of the government's action ordinarily has little trouble showing injury and redress, the reason the would-be builder belongs among the plaintiffs.	
59	Nordlinger v. Hahn (1992) 505 U.S. 1 (1992) **Score:** 8x1*17	**General Description:** Justice Blackmun, for the Court, upheld California's acquisition-value property-tax scheme under Proposition 13, which assessed newly purchased property at purchase price while capping annual increases on long-held property at two percent. Justice Stevens dissented.
	The Court confined Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) to the rare case in which the facts precluded any plausible inference that an acquisition-value policy explained the disparity. Justice Thomas concurred in part while stating that Allegheny Pittsburgh could not be distinguished. **Why Case is Relevant:** *The Takings Clause Prong* — The stated limit on the chapter's use of Allegheny Pittsburgh, conceded rather than left for an opponent to find. *Objections Considered* — The strongest authority against a standalone equality prong: a classification turning on when property was acquired, sustained on rational basis.	
60	Lucas v. South Carolina Coastal Council (1992) 505 U.S. 1003 (1992) **Score:** 6x3 **In Exam**	**General Description:** A developer bought two beachfront lots, after which a new state coastal statute barred him from building anything on them.
	The Supreme Court held that when a regulation deprives an owner of all economically beneficial use of his land, it is a taking without any case-by-case balancing — a taking per se. Total deprivation of use, the Court reasoned, is different in kind from the partial diminution of value that ordinary regulation produces, and it requires compensation categorically. **Why Case is Relevant:** *The Environmental Question* — Listed among the per se takings categories (total economic wipeout) the remedy does not disturb. *The Takings Clause Prong* — The per se total-wipeout rule: a regulation denying all economically beneficial use is a taking — the shorter, steeper road. *Objections Considered* — Cited (with Penn Central) to show a post-regulation entrant fails the takings analysis. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation denying all economically beneficial use of property is a per se taking.	

#	SCOTUS Case	Description
61	Nixon v. United States (1993) 506 U.S. 224 (1993) **Score:** 9x0	**General Description:** The Supreme Court's leading modern application of the political-question doctrine to internal Congressional procedure: the Senate's procedure for trying an impeachment was held
	nonjusticiable because the Constitution textually commits the matter to the Senate and the courts lack judicially manageable standards for second-guessing the chamber's procedural choices. **Why Case is Relevant:** *Hybrid Legislative Voting* — Cited on the political-question doctrine and impeachment procedure — courts' reluctance to police the chamber's internal voting mechanics, which shapes the hybrid-voting design.	
62	Dolan v. City of Tigard (1994) 512 U.S. 374 (1994) **Score:** 5x4	**General Description:** A hardware-store owner applied for a permit to expand her store and pave her parking lot. The city granted the permit on condition that she dedicate part of her property to a public greenway and bike path.
	She objected that the conditions were unrelated in scope to the impact of her expansion. The Supreme Court agreed with her on the second prong of the modern proportionality test it had begun building seven years earlier in Nollan v. California Coastal Commission: a regulatory condition imposed on the use of property must bear an essential nexus to the harm the regulation claims to address (Nollan) and must be roughly proportional in scope to that harm (Dolan). The two-prong test underlies Sheetz v. County of El Dorado (2024), which extended it from individual administrative decisions to fees set by general legislation. **Why Case is Relevant:** *The Takings Clause Prong* — Cited with Nollan on the exactions doctrine; the Court rejected the distinction limiting it to individualized conditions.	
63	United States v. Lopez (1995) 514 U.S. 549 (1995) **Score:** 5x4	**General Description:** The Gun-Free School Zones Act of 1990 made it a federal offense to possess a firearm within a school zone. A San Antonio high-school student carried a concealed handgun to school and was convicted under the Act. He argued that Congress lacked power under the Commerce Clause to regulate the possession of a gun near a school.
	Chief Justice Rehnquist, for a five-Justice majority, held the Act beyond the commerce power. Possession of a gun in a school zone is not economic activity and does not substantially affect interstate commerce; the Court declined to pile inference on inference to reach it. The first decision in sixty years to hold a federal statute outside the commerce power. **Why Case is Relevant:** *Objections Considered — The model for the enumerated-powers question in The Rational-Basis Trap. The Court did not ask whether Congress could rationally have thought guns near schools harmful; it asked whether the activity was within the power. The Patent Clause prong must be pleaded in that form, or rational-basis review swallows it.*	

#	SCOTUS Case	Description
	Adarand Constructors, Inc. v. Peña (1995) 515 U.S. 200 (1995) **Score:** 5x4*18	**General Description:** Justice O'Connor announced the judgment and delivered the opinion of the Court in relevant part, holding that all racial classifications imposed by any federal, state, or local governmental actor must be analyzed under strict scrutiny. Justices Stevens, Souter, and Ginsburg each dissented.
64	The decision confirms that the equal-protection obligation Bolling v. Sharpe read into the Fifth Amendment binds the federal government with the same force the Fourteenth Amendment carries against the States. The strict-scrutiny holding governs racial classifications; the federal-obligation premise is what the book uses. **Why Case is Relevant:** *The Takings Clause Prong* — Supplies the federal bridge for treating unequal federal treatment as a constitutional matter. The extension from race to a vintage classification is the argument's own step, and the chapter says so.	
	United Food and Commercial Workers Union Local 751 v. Brown Group, Inc. (1996) 517 U.S. 544 (1996) **Score:** 9x0	**General Description:** A union sued under the WARN Act seeking backpay for its members. Justice Souter, for a unanimous Court, held that the third prong of the Hunt associational-standing test is prudential rather than constitutional, and may therefore be abrogated by Congress.
65	The holding matters because Hunt v. Washington State Apple Advertising Commission (1977) had suggested that a damages action running solely to members would bar associational standing. Brown Group establishes that the bar is one Congress can lift, and that only the first prong is grounded in Article III. **Why Case is Relevant:** *Objections Considered* — Answers the objection that an association cannot sue for member damages. The limitation is prudential, and the chapter distinguishes what Article III requires from what prudence has added.	
	Romer v. Evans (1996) 517 U.S. 620 (1996) **Score:** 6x3 **In Exam**	**General Description:** Colorado voters adopted a constitutional amendment forbidding the state or any locality to extend antidiscrimination protections on the basis of sexual orientation.
66	The Supreme Court struck the amendment down on rational-basis review, holding that the only purpose the amendment could rationally be said to serve was disqualification of an unpopular group from political participation, which is not a legitimate state interest. The case extends the Cleburne line of "rational basis with bite" — the narrow set of cases in which an equal-protection challenge succeeds because the only rationale a court can find is identifiable animus. **Why Case is Relevant:** *Objections Considered* — Paired with Moreno: a classification explained only by animus fails even rational-basis review. *Reason the Constitutional Argument Yourself* — Exam holding: a classification fails rational-basis review where explained only by animus toward those it disadvantages.	

#	SCOTUS Case	Description
67	Printz v. United States (1997) 521 U.S. 898 (1997) **Score:** 5x4	**General Description:** The Brady Handgun Violence Prevention Act required county sheriffs and other local chief law-enforcement officers to perform federal background checks on prospective handgun purchasers.
	Sheriff Jay Printz and Sheriff Richard Mack sued, arguing that the federal government could not compel state officers to enforce federal regulatory programs. The Supreme Court agreed. The federal government may not commandeer the legislative or executive officers of the states for federal regulatory ends; doing so violates the structure of dual sovereignty the Constitution establishes. The case does not address "limited Times," but it is the constitutional ground for the sheriff plaintiff — the elected county sheriff suing as a constitutional officer to challenge a federal regulatory program that converts him into an enforcement adjunct. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The elected sheriff's precedent — two county sheriffs challenged a federal statute and won — a plaintiff pattern requiring no law be broken. *Sovereign and Public-Fiduciary Plaintiffs* — Cited for the elected sheriff's footing as a constitutional officer — a posture that depends on the sheriff being elected. *Why You Want to Be the Plaintiff* — Cited as a precise, recent precedent for the elected-official plaintiff.	
68	Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank (1999) 527 U.S. 627 (1999) **Score:** 5x4 **In Exam**	**General Description:** College Savings Bank sued a Florida state agency for infringing its patent. The question was whether Congress could strip the States of sovereign immunity and expose them to patent-infringement suits;
	the Court held that, in the manner Congress had attempted, it could not. The premise the Court worked from is what matters for this book, and the Court stated it plainly: a patent is a form of property, protected as other property is. **Why Case is Relevant:** *The Patent Clause Prong* — Completes the property point: patents are a species of property. *The Takings Clause Prong* — Cited again for the same holding on the takings side — patents are property, protected as other property. *Reason the Constitutional Argument Yourself* — Exam holding: a patent is a form of property, protected as other property.	
69	Geier v. American Honda Motor Co. (2000) 529 U.S. 861 (2000) **Score:** 5x4	**General Description:** Held that a federal motor-vehicle safety standard preempted a state tort claim that conflicted with the federal regulatory objective.
	Why Case is Relevant: *The Wall Itself* — Addresses the preemption question for the Federal Motor Vehicle Safety Standards — relevant to how the wall's statutes operate.	

#	SCOTUS Case	Description
70	TrafFix Devices, Inc. v. Marketing Displays, Inc. (2001) 532 U.S. 23 (2001) **Score:** 9x0	**General Description:** Marketing Displays held now-expired utility patents on a dual-spring mechanism that kept temporary road signs upright in wind. After the patents expired and a competitor began selling sign stands using the same mechanism, Marketing Displays sued for trade dress infringement, arguing that the visible spring design identified it as the source.
	Justice Kennedy, for a unanimous Court, held that a functional product feature cannot receive trade dress protection under the Lanham Act. An expired utility patent is strong evidence that the features it claimed are functional, and where the expired patent claimed the feature in question, the party seeking trade dress protection bears a heavy burden of showing that the feature is merely ornamental, incidental, or arbitrary. The Court declined to let trade dress become the practical equivalent of an expired utility patent. **Why Case is Relevant:** *Objections Considered* — The trademark-side counterpart to *Bonito Boats* and *Sears*. Establishes that when a utility patent expires, the functional design it covered cannot be recaptured through trade dress. The case marks the outer limit of what non-patent intellectual property law may do to prevent remanufacturing of a patent-expired design, and shows the Court policing that boundary on its own initiative. It sharpens the book's central contrast: the intellectual property statutes release the expired design, while the regulatory statutes do not.	
71	Eldred v. Ashcroft (2003) 537 U.S. 186 (2003) **Score:** 7x2 **In Exam**	**General Description:** The Sonny Bono Copyright Term Extension Act of 1998 extended existing and future copyright terms by twenty years.
	Eric Eldred and others who relied on works passing into the public domain argued that retroactively extending existing terms violated the Patent and Copyright Clause's "limited Times" requirement and the First Amendment. The Supreme Court, seven to two, upheld the Act. Justice Ginsburg's majority held that "limited Times" requires only that the term be limited, not perpetual, and that Congress could rationally decide to extend the period of incentive even retroactively. Justice Breyer's dissent argued the extension produced a term economically equivalent to ninety-nine point eight percent of a perpetual copyright. The case is the leading adverse Patent Clause precedent; the chapter on objections considered distinguishes it as a copyright case in a calendar-adjustment posture, not a patent case in a structural-ratchet posture. **Why Case is Relevant:** *Objections Considered* — The case behind the objection that a copyright-term extension survived a "limited Times" challenge — which the prong distinguishes. *Reason the Constitutional Argument Yourself* — Exam holding: a copyright-term extension did not violate "limited Times"; Congress receives substantial deference.	
72	Lawrence v. Texas (2003) 539 U.S. 558 (2003) **Score:** 6x3*19	**General Description:** Justice Kennedy, for five Justices, held that a Texas statute criminalizing same-sex intimate conduct violates the Due Process Clause and overruled Bowers v. Hardwick. Justice O'Connor concurred in the judgment on equal-protection grounds. Justices Scalia and Thomas dissented.
	The Court stated that Bowers was not correct when it was decided and is not correct today. The historians' brief that supplied the corrected record was the work of scholars outside the legal profession, and the sources it used had been publicly available throughout the seventeen years Bowers stood. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The correction, and the measure of how long a court will hold an unexamined historical assertion when no one puts the contrary record in front of it.	

#	SCOTUS Case	Description
73	Kelo v. City of New London (2005) 545 U.S. 469 (2005) **Score:** 5x4	**General Description:** Justice Stevens, for a five-Justice majority, held that a city's transfer of private property to a private developer for economic development satisfies the public-use requirement of the Takings Clause. Justices O'Connor and Thomas each dissented.
	Susette Kelo lost, the development was never built, and the company the project was meant to serve left New London in 2009. The decision produced legislative reform in a majority of States, which is the book's point in citing it: a loss can produce a remedy, and the Institute for Justice, which carried the case, litigates in the same doctrinal territory this argument occupies. **Why Case is Relevant:** *What You Can Do: A Citizen's Playbook* — Named among the public-interest firms that take takings cases without charging a plaintiff. *When Outsiders Advance Constitutional Law and Other Disciplines* — An injury rather than a discovery, and a loss that still moved the law.	
74	MedImmune, Inc. v. Genentech, Inc. (2007) 549 U.S. 118 (2007) **Score:** 8x1	**General Description:** Justice Scalia, for the Court, held that a patent licensee need not breach or terminate its license before seeking a declaratory judgment that the patent is invalid, unenforceable, or not infringed. Justice Thomas dissented.
	The decision removed the reasonable-apprehension-of-suit requirement the Federal Circuit had imposed and restored the ordinary actual-controversy standard of the Declaratory Judgment Act. A party need not expose itself to liability to test the legal position it is being held to. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The case the chapter is named for. *The Filable Case: Why 2026 and Not Before* — One of the conditions that make a pre-enforcement challenge available to a would-be remanufacturer who has built nothing yet.	
75	Massachusetts v. Environmental Protection Agency (2007) 549 U.S. 497 (2007) **Score:** 5x4	**General Description:** Twelve states and several local governments sued the EPA over the agency's refusal to regulate greenhouse-gas emissions from new motor vehicles.
	The Supreme Court held that Massachusetts had standing to sue, because the state had suffered concrete injury to its territory (rising sea levels eroding coastline) and to the welfare of its citizens, and because the federal government's failure to regulate was traceable to that harm and redressable by a court order. The decision announced the doctrine of special solicitude: a sovereign plaintiff is owed a more permissive standing analysis than the showing required of an ordinary private plaintiff. The case is the inflection point for sovereign and quasi-sovereign standing in modern federal litigation, and the door through which states will be among the first plaintiffs in the patent-bargain case. **Why Case is Relevant:** *Why Nobody Noticed* — Cited among the doors the Court has opened over the past quarter-century. *You No Longer Need to Break a Law to Challenge One* — The 2007 turning point — states sued the EPA over a regulatory failure and established a route that needs no broken law. *Sovereign and Public-Fiduciary Plaintiffs* — The anchor for sovereign standing — a state suing the federal government is owed special solicitude. *The Filable Case* — Cited as part of the recent access toolkit — a sovereign plaintiff may sue over a federal regulatory failure. *Why You Want to Be the Plaintiff* — Cited for the parens patriae / sovereign standing a state invoked successfully.	

#	SCOTUS Case	Description
76	Golan v. Holder (2012) 565 U.S. 302 (2012) **Score:** 6x2	**General Description:** Section 514 of the Uruguay Round Agreements Act restored United States copyright to foreign works that had already passed into the American public domain. Conductors, teachers, and film archivists who had built on those works sued. The Court upheld the statute.
	Justice Ginsburg, for the Court, held that the "limited Times" language does not bar Congress from extending protection to works already in the public domain, and rejected the premise that the Constitution renders the public domain largely untouchable by Congress; the First Amendment was satisfied by the idea-expression distinction and fair use. The opinion refers to the provision as the Copyright and Patent Clause. Justice Breyer, joined by Justice Alito, dissented, arguing the restoration withdrew from the public what it already held without promoting the creation of anything new. Justice Kagan took no part. The decision reviews an act of Congress under the Copyright Clause; it does not reach whether an executive agency may, by regulation, make a patent-expired design unlawful to build. That distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority, paired with Eldred v. Ashcroft — the Court has said the public domain is not beyond the reach of Congress. Answered on the ground that Golan reviewed an enacted statute in a copyright posture, while the wall at issue here was built by agencies under general delegations and has never been tested under the Patent Clause.	
77	National Federation of Independent Business v. Sebelius (2012) 567 U.S. 519 (2012) **Score:** 5x4	**General Description:** The challenge to the Affordable Care Act's individual mandate and Medicaid expansion. Chief Justice Roberts wrote the controlling opinion; the Court divided five to four on each principal question, with different majorities.
	Five Justices held that the Commerce Clause does not authorize Congress to compel individuals to buy insurance, and that the Necessary and Proper Clause cannot supply that power; five upheld the mandate's payment as a tax; seven held the Medicaid expansion's threat to withdraw all existing funds unconstitutionally coercive under the Spending Clause. **Why Case is Relevant:** *The Patent Clause Prong* — The hardest case for the chapter's argument, met rather than avoided: the Court refused one power and accepted another for the same payment. The chapter's answer is that the taxing power produced no outcome any clause forbids, while "limited Times" forbids the outcome itself. That distinction is the book's, not the Court's.	
78	Arkansas Game & Fish Commission v. United States (2012) 568 U.S. 23 (2012) **Score:** 8x0 **In Exam**	**General Description:** The federal government, operating a dam, released water in a pattern that repeatedly flooded a state-owned wildlife area during the tree-growing season over a span of years, then stopped.
	The State sued, claiming the recurring flooding had killed timber and worked a taking. The government argued that flooding can be a taking only when it is permanent. The Supreme Court rejected that categorical rule. Whether government action is a taking, the Court held, depends on its actual effect on the property — not on whether the invasion is labeled temporary or permanent. **Why Case is Relevant:** *Four Problems, One Remedy* — One of four recent decisions the book cites as making the case ripe now. *The Takings Clause Prong* — Cited for the holding that even temporary government-induced invasions (flooding) may be takings. *Reason the Constitutional Argument Yourself* — Exam holding: even temporary government-induced invasions of property may constitute takings.	

#	SCOTUS Case	Description
79	Clapper v. Amnesty International USA (2013) 568 U.S. 398 (2013) **Score:** 5x4	**General Description:** Held that plaintiffs lacked standing to challenge surveillance where the threatened injury was too speculative.
	Why Case is Relevant: *Sovereign and Public-Fiduciary Plaintiffs* — Polices the injury line — an injury must be concrete and imminent, not speculative — the standing bar plaintiffs must clear. *Objections Considered* — Cited by the objection that a buyer's injury rests on a speculative chain; answered there with Diamond Alternative Energy.	
80	Susan B. Anthony List v. Driehaus (2014) 573 U.S. 149 (2014) **Score:** 9x0	**General Description:** The Susan B. Anthony List, an antiabortion advocacy organization, had distributed political messages criticizing a congressional candidate for his health-care vote.
	The candidate filed a complaint under an Ohio statute criminalizing false statements in political campaigns. The advocacy organization sued in federal court before any prosecution had occurred, seeking a declaration that the statute violated the First Amendment. The Supreme Court held that pre-enforcement standing existed: a credible threat of enforcement under a statute the plaintiff intends to violate suffices to establish injury in fact, and the plaintiff need not wait for prosecution to challenge the statute. The case is the controlling modern statement of pre-enforcement standing, and the doctrine on which the patent-bargain case will rest at the standing stage. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The credible-threat doctrine — a third pre-enforcement-review case extending standing further. *The Filable Case* — Cited for the unanimous holding that a plaintiff facing a credible threat of enforcement may sue before enforcement.	
81	Kimble v. Marvel Entertainment, LLC (2015) 576 U.S. 446 (2015) **Score:** 6x3	**General Description:** Kimble sold Marvel a patent on a web-shooting toy and took royalties. When the patent expired, Marvel stopped paying. Kimble asked the Court to overrule Brulotte v. Thys Co., which bars post-expiration royalties.
	Justice Kagan, for the Court, declined on stare decisis grounds and stated the governing principle: a patent typically expires twenty years from the filing date, and when it expires the patentee's prerogatives expire with it and the right to make or use the article, free from all restriction, passes to the public. Justice Alito dissented, joined by the Chief Justice and Justice Thomas, on the merits of Brulotte. The phrase "free from all restriction" means free of the patent monopoly; it does not hold that the article is immune from all regulation. **Why Case is Relevant:** The Patent Clause Prong — The book's canonical gloss: at expiration the unrestricted right to make and use the invention passes to the public. Federal patent-law authority, decided against a private licensor. Cited for what passes to the public at expiration, not for any holding about regulatory power.	

#	SCOTUS Case	Description
82	Horne v. Department of Agriculture (2015) 576 U.S. 351 (2015) **Score:** 8x1*4 **In Exam**	**General Description:** A Depression-era federal marketing order required raisin growers to surrender a portion of each year's crop to a government committee. The growers refused and were fined.
	The government argued the requirement was a permissible regulation of the raisin market, not a taking. The Supreme Court disagreed. Compelling the growers to hand over their raisins was a physical appropriation of property and therefore a taking per se, and the government could not avoid that conclusion by characterizing the appropriation as a condition of doing business or a contribution to orderly marketing. **Why Case is Relevant:** *The Environmental Question* — Listed among the per se categories (appropriation of personal property) the remedy does not disturb. *The Takings Clause Prong* — Applied the per se rule to government appropriation of personal property. *Objections Considered* — Cited on recent personal-property takings doctrine — the Takings Clause applies to personal property with full force. *Reason the Constitutional Argument Yourself* — Exam holding: the per se takings rule applies with full force to the appropriation of personal property.	
83	Star Athletica, L.L.C. v. Varsity Brands, Inc. (2017) 580 U.S. 405 (2017) **Score:** 6x2*12	**General Description:** A cheerleading-uniform maker held copyright registrations for two-dimensional designs of lines, chevrons, and shapes appearing on the surface of its uniforms, and sued a competitor for infringement. The competitor argued that designs on useful articles are not copyrightable at all.
	Justice Thomas, for the Court, established the governing separability test under 17 U.S.C. § 101: a feature incorporated into the design of a useful article is eligible for copyright only if it can be perceived as a two- or three-dimensional work of art separate from the article and would qualify as a protectable pictorial, graphic, or sculptural work on its own or in another medium if imagined separately. The Court expressly noted that the test does not render the underlying useful article itself eligible for copyright. **Why Case is Relevant:** *Objections Considered* — The governing authority for how far copyright reaches into a manufactured product. A separable badge, emblem, or applied graphic may qualify; the vehicle carrying it may not. The case answers directly the objection that an entire car or truck is copyright-protected for ninety-five years as a corporate work made for hire: the term is real, but a useful article is not the kind of work to which it attaches. Together with *TrafFix*, it fixes the ceiling on what non-patent intellectual property law can protect after a utility patent expires.	

#	SCOTUS Case	Description
84	Oil States Energy Services, LLC v. Greene's Energy Group, LLC (2018) 584 U.S. 325 (2018) **Score:** 7x2	**General Description:** Oil States, whose patent on hydraulic-fracturing well-head equipment had been cancelled by the Patent Office in inter partes review, argued that only an Article III court could revoke an issued patent.
	The Supreme Court disagreed, seven to two. Justice Thomas, for the Court, held that a patent is a public franchise and that reconsidering its validity is a matter of public rights Congress may assign to an executive agency. Justice Gorsuch, joined by Chief Justice Roberts, dissented, treating the issued patent as private property revocable only by a court. The majority called its holding narrow, reserved the retroactivity question, and said it was not deciding whether patents are property under the Due Process or Takings Clauses. It also repeated, from its own older cases, that the grant of a patent is a matter between the public, who are the grantors, and the patentee. The decision concerns the government's power over a patent during its term; it does not reach the public's right to build the design after the patent expires. That distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority raised by a scholar of administrative law: if an agency may cancel a live patent, the Court is unlikely to police what agencies do after expiration. Answered on the ground that Oil States sits on the grant side of the Clause and expressly reserved the Takings question, and that the public-franchise vocabulary names the public as the party the franchise came from and goes to when the term is over.	
85	Trump v. Hawaii (2018) 585 U.S. 667 (2018) **Score:** 5x4	**General Description:** Chief Justice Roberts, for a five-Justice majority, upheld the entry restrictions of Presidential Proclamation 9645. Justices Breyer and Sotomayor each dissented.
	The book cites the case for a single sentence in it. The Court stated that Korematsu v. United States had been gravely wrong the day it was decided and has no place in law under the Constitution — a repudiation offered seventy-four years after the fact, and only once the suppressed record had long been public. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited for how a court speaks when it learns the original proceeding was tainted, and for how long the interval can be.	
86	Google LLC v. Oracle America, Inc. (2021) 593 U.S. 1 (2021) **Score:** 6x2*21	**General Description:** Google built the Android platform to let programmers use the Java language and copied roughly 11,500 lines of declaring code from the Java SE application programming interface so that familiar calls would work on the new platform, writing its own implementing code. Oracle, which had acquired Java, sued for copyright infringement.
	Justice Breyer, for the Court, assumed without deciding that the declaring code was copyrightable and held Google's copying fair use. The declaring code was functional and tied to the programmers' investment in learning it; Google took only what was needed to let programmers work in a new environment; and the copying advanced the constitutional purpose of copyright by enabling new creative expression. **Why Case is Relevant:** *Objections Considered — The second of three reasons the firmware copyright is narrower than its term: the Court has protected the room a rewriter needs. A remanufacturer may study the original engine-control module to learn what signals the engine expects and write new code to produce them, where no digital lock bars the study.*	

#	SCOTUS Case	Description
87	Cedar Point Nursery v. Hassid (2021) 594 U.S. 139 (2021) **Score:** 6x3 **In Exam**	**General Description:** A California regulation gave labor organizers a right to enter an agricultural employer's property to speak with workers for several hours a day during much of the year. The growers said the forced access took their property.
	The Supreme Court agreed. A regulation that appropriates a right to physically enter private land and hands it to others is a per se taking, the Court held, even though the access was intermittent and the owners kept every other use of their land. The decision is part of the modern Court's broadened reading of what counts as a taking. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *The Environmental Question* — Listed among the per se categories (physical occupation for limited hours). *The Takings Clause Prong* — Extended the physical-occupation branch to forced access for limited hours. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation granting third parties a right of physical access, even for limited hours, is a per se taking.	
88	American Hospital Association v. Becerra (2022) 596 U.S. 724 (2022) **Score:** 9x0	**General Description:** Held that HHS could not vary drug-reimbursement rates for a subset of hospitals without the survey the statute required; decided without Chevron deference.
	Why Case is Relevant: *You No Longer Need to Break a Law to Challenge One* — Illustrates how broadly the regulatory wall reaches — the Association sued HHS and prevailed. *Direct and Associational Plaintiff Categories* — Cited as an associational plaintiff — the Association represents roughly 5,000 hospitals and brings real litigating capacity.	
89	West Virginia v. Environmental Protection Agency (2022) 597 U.S. 697 (2022) **Score:** 6x3 **In Exam**	**General Description:** Several States challenged the EPA's authority to reshape the nation's electricity generation through climate regulation.
	The Supreme Court ruled against the agency and articulated the "major questions doctrine": when an agency claims the power to decide a matter of vast economic and political significance, it must point to clear authorization from Congress, not infer the power from a vague or ancillary statutory phrase. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *The Filable Case* — Cited for the fullest statement of the major-questions doctrine. *The Patent Clause Prong* — The major-questions doctrine: an agency claiming power of vast economic and political significance must point to clear congressional authorization. *Why You Want to Be the Plaintiff* — Cited as shorthand for a single decision that reset a field — named for its plaintiff. *Reason the Constitutional Argument Yourself* — Exam holding: under the major-questions doctrine, an agency asserting authority over a matter of vast significance needs clear congressional authorization.	

#	SCOTUS Case	Description
90	Sackett v. Environmental Protection Agency (2023) 598 U.S. 651 (2023) **Score:** 9x0*5	**General Description:** A landowning couple challenged the reach of a federal Clean Water Act rule over their own property and prevailed at the Supreme Court without first incurring a penalty.
	Why Case is Relevant: *Why You Want to Be the Plaintiff* — Cited as a directly-regulated single-landowner plaintiff who challenged the EPA and won.	
91	Tyler v. Hennepin County, Minnesota (2023) 598 U.S. 631 (2023) **Score:** 9x0	**General Description:** A county sold a delinquent taxpayer's home for $40,000 to satisfy a $15,000 tax debt and kept the surplus. Chief Justice Roberts, for a unanimous Court, held the retention a classic taking. Justice Gorsuch concurred, joined by Justice Jackson.
	The State argued that its own law had stripped the owner of any property interest in the surplus. The Court answered that property rights cannot be so easily manipulated, and that a government may not sidestep the Takings Clause by disavowing traditional property interests long recognized under state law. The principle reaches a right the government declines to recognize as well as one it concedes. **Why Case is Relevant:** *The Takings Clause Prong* — The answer to the objection that the public's post-expiration right is not property because current federal law does not treat it as property. *Objections Considered* — Cited for the same proposition where the objection is stated in its strongest form.	
92	Sheetz v. County of El Dorado (2024) 601 U.S. 267 (2024) **Score:** 9x0 **In Exam**	**General Description:** A county charged a landowner a substantial traffic-impact fee as a condition of a building permit. The fee was fixed by a general schedule the county's legislature had adopted, not by an official's individual decision.
	The landowner raised a takings challenge; the county argued that the Takings Clause's limits on permit conditions apply only to individualized administrative demands, not to conditions set by general legislation. The Supreme Court unanimously rejected the distinction: the Takings Clause contains no exception for legislatively imposed conditions. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — Held unanimously that the Takings Clause applies to legislative as well as adjudicative impositions. *The Filable Case* — Cited for extending takings scrutiny to burdens imposed by general legislation. *The Takings Clause Prong* — Answered the objection directly — a traffic-impact fee imposed by general legislation is still subject to takings scrutiny. *Why You Want to Be the Plaintiff* — Named among the model plaintiffs (Printz, Loper Bright, Sheetz) — an ordinary name that became a landmark. *Reason the Constitutional Argument Yourself* — Exam holding: a permit condition imposed by general legislation is still subject to takings scrutiny.	

#	SCOTUS Case	Description
93	Food and Drug Administration v. Alliance for Hippocratic Medicine (2024) 602 U.S. 367 (2024) **Score:** 9x0 **Not in Exam**	**General Description:** Doctors and medical associations opposed to mifepristone challenged the Food and Drug Administration's 2016 and 2021 relaxations of the drug's conditions of use, although they neither prescribed nor used it.
	Justice Kavanaugh wrote for a unanimous Court that the plaintiffs lacked standing. A plaintiff challenging the government's regulation of someone else is not barred, but standing in that posture is "ordinarily substantially more difficult to establish," and the doctors' causal theories were too speculative. Justice Thomas concurred. **Why Case is Relevant:** *Objections Considered* — The government's strongest recent authority against a buyer's standing, answered with Diamond Alternative Energy and by joining as plaintiff the builder the regulation forbids to act.	
94	Loper Bright Enterprises v. Raimondo (2024) 603 U.S. 369 (2024) **Score:** 6x2*6 **In Exam**	**General Description:** A group of commercial herring fishermen challenged a federal rule requiring them to pay the salaries of the monitors stationed on their boats. The Supreme Court used the case to overrule Chevron deference.
	A court, the Court held, must determine for itself what a statute means, using the ordinary tools of interpretation, rather than deferring to an agency's reading. The decision reshaped administrative law, and it is what leaves a court free to decide the patent-bargain question on its own. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *Why Nobody Noticed* — Marks the end of Chevron deference (2024) in the chapter's account. *You No Longer Need to Break a Law to Challenge One* — Overruled Chevron — a court now decides a statute's meaning for itself, strengthening the plaintiff's position. *The Filable Case* — Cited for overruling Chevron on June 28, 2024 — the deference was gone. *The Patent Clause Prong* — Ended Chevron deference: a reviewing court now determines the best reading of a statute itself. *Why You Want to Be the Plaintiff* — Cited — four family fishing businesses challenged a federal rule — as a model plaintiff. *Reason the Constitutional Argument Yourself* — Exam holding: courts formerly deferred to an agency's reasonable reading of an ambiguous statute; Loper Bright ended that.	

#	SCOTUS Case	Description
95	**Corner Post, Inc. v. Board of Governors of the Federal Reserve System (2024)** 603 U.S. 799 (2024) **Score:** 6x3 **Not in Exam**	**General Description:** A North Dakota truck stop that opened in 2018 challenged Regulation II, a Federal Reserve rule issued in 2011 capping debit-card interchange fees. The government argued the six-year limitations period had run out before the plaintiff existed.
	The Supreme Court held, 6–3, that an Administrative Procedure Act claim accrues when the plaintiff is injured by the rule, not when the agency issued it. Justice Barrett wrote for the Court; Justice Kavanaugh concurred; Justice Jackson dissented, joined by Justices Sotomayor and Kagan. **Why Case is Relevant: The Filable Case** — The companion rule to the end of Chevron deference. The regulatory wall was built between 1970 and 1987; under the older reading of the limitations period most of it was unchallengeable by anyone not already in court decades ago. Under this decision a business first injured last year may challenge a rule issued in 1987.	
96	Diamond Alternative Energy, LLC v. Environmental Protection Agency (2025) 606 U.S. 100 (2025) **Score:** 7x2 **Not in Exam**	**General Description:** Fuel producers challenged EPA's approval of California regulations requiring automakers to make more electric vehicles. The rules did not regulate the producers; their injury was the lost demand for fuel that would follow from the automakers' responses.
	Justice Kavanaugh, for the Court, held that the producers had standing. Commonsense economic principles, supported by the record, showed a predictable chain of events, and the government "generally may not target a business or industry through stringent and allegedly unlawful regulation, and then evade the resulting lawsuits by claiming that the targets of its regulation should be locked out of court as unaffected bystanders." Justices Sotomayor and Jackson dissented. **Why Case is Relevant:** *Objections Considered* — Answers the objection that a buyer's injury depends on remanufacturers not before the court. Same agency, same subject. The chapter notes that the case involved sellers losing demand rather than buyers awaiting a new entrant, and closes that gap by joining the builder.	

Score key

Format is winner × loser (e.g., 6x3 = six Justices in the majority, three dissenting). Counts reflect the Justices who actually participated, so a recusal shows as a reduced total (e.g., 8x0). An equally divided Court would appear as NxN (e.g., 4x4), meaning no Supreme Court majority and the lower court's judgment stands by default; no case in this table was decided by an equal division. An asterisk on a score points to a note below.

Score footnotes

A note on scoring the older cases. The score records the vote by which the Court decided the case, written as majority×dissent (for example, 6x3). For decisions from the era before the Court routinely reported a vote breakdown, the figure is reconstructed from two facts: the roster of Justices then sitting, and whether the report records any dissent. Where a case issued with no recorded dissent and the participating Justices can be identified from the volume's roster, the score is given (for example, 6x0). Where the number of participating Justices cannot yet be confirmed, a dash is shown in place of a number, and the case-specific note below explains why. A score is never invented to fill a gap.

*1 Standard Oil Co. of New Jersey v. United States (1911): the judgment ordering dissolution was unanimous; Justice Harlan concurred in that result but dissented from the majority's adoption of the "rule of reason."

*2 Aro Mfg. Co. v. Convertible Top Replacement Co. (1961): a fractured decision reversing in Aro's favor. Justice Whittaker's opinion was joined by three other Justices; Justices Black and Brennan concurred separately (Brennan concurring only in the result); Justices Harlan, Frankfurter, and Stewart dissented.

*3 Ruckelshaus v. Monsanto Co. (1984): the 7–1 figure is the vote on the judgment (vacate and remand). Justice White took no part, and Justice O'Connor concurred in part and dissented in part (as to Part IV-B).

*4 Horne v. Department of Agriculture (2015): the Court held 8–1 that the raisin reserve was a per se physical taking (Justice Sotomayor alone in dissent), but divided 5–4 on the remedy — whether the growers were entitled to relief from the fine.

*5 Sackett v. EPA (2023): all nine Justices agreed the Sacketts prevailed and reversed the Ninth Circuit; the Court split 5–4 on the narrower "continuous surface connection" test for "waters of the United States."

*6 Loper Bright Enterprises v. Raimondo (2024): Justice Jackson took no part in Loper Bright (No. 22–451), where the vote was 6–2. In the consolidated companion case, Relentless, Inc. v. Department of Commerce (in which she participated), the split was 6–3.

*7 Webber v. Virginia (1880); Singer Mfg. Co. v. June Mfg. Co. (1896); McCormick Harvesting Machine Co. v. Aultman (1898): each issued without a recorded dissent (a non-dissenting outcome), but the exact number of participating Justices has not yet been independently confirmed against the Supreme Court Database Legacy file. Score to be finalized.

*8 Pennock v. Dialogue (1829): the Court that decided the cases reported in 27 U.S. (2 Pet.) comprised six Justices. (A seventh, John McLean, was confirmed in March 1829 after Justice Trimble's death but did not take his seat until January 1830 and so did not participate.) Justice Story wrote for a unanimous Court with no recorded dissent, giving 6x0. As with all cases of this era, the reporter did not separately record each Justice's participation in the individual decision, so the figure rests on the six-member roster of the volume.

*9 Motion Picture Patents Co. v. Universal Film Mfg. Co. (1917): the vote was 6—3. Justice Clarke wrote for the majority; Justice Holmes dissented, joined by Justices McKenna and Van Devanter. Justice McReynolds concurred in the result only.

*10 Wheaton v. Peters (1834): Justice McLean wrote for the majority; Justices Thompson and Baldwin dissented. The Court noted separately that it was unanimous on the point that no reporter can hold copyright in the Court's written opinions. The January 1834 roster comprised seven Justices (Marshall, Johnson, Duvall, Story, Thompson, McLean, and Baldwin), but the report does not state which Justices participated in this decision, and Justices Johnson and Duvall were frequently absent in this period. The number of participating Justices has not been independently confirmed against the Supreme Court Database Legacy file. Score to be finalized.

*11 Sears, Roebuck & Co. v. Stiffel Co. (1964): Justice Black delivered the opinion for a unanimous Court. Justice Harlan concurred in the result only, taking the view that a State should be able to bar copying where the purpose is to prevent one seller's goods from being palmed off as another's.

*12 Star Athletica, L.L.C. v. Varsity Brands, Inc. (2017): Justice Thomas wrote for the Court, joined by Chief Justice Roberts and Justices Alito, Sotomayor, and Kagan. Justice

Ginsburg concurred in the judgment only, on the ground that the designs were themselves copyrightable pictorial works reproduced on useful articles, so no separability analysis was needed. Justice Breyer dissented, joined by Justice Kennedy. The 6x2 figure counts Justice Ginsburg with the majority on the judgment.

*13 Kewanee Oil Co. v. Bicron Corp. (1974): Chief Justice Burger's opinion was joined by four Justices; Justice Marshall concurred in the result only; Justices Douglas and Brennan dissented; Justice Powell took no part. The 6x2 figure counts Justice Marshall with the majority on the judgment.

*15 Railway Labor Executives' Association v. Gibbons (1982): the judgment was unanimous. Justice Rehnquist's opinion drew seven Justices; Justices Marshall and Brennan concurred in the judgment on a narrower rationale and expressly declined the majority's broader reading of the uniformity requirement — the reading this book relies on.

*16 City of Cleburne v. Cleburne Living Center, Inc. (1985): all nine Justices agreed the ordinance was invalid as applied. Only six joined the rational-basis rationale; Justices Marshall, Brennan, and Blackmun dissented from the as-applied remedy and wrote that the Court's review was heightened scrutiny under another name.

*17 Nordlinger v. Hahn (1992): Justice Blackmun wrote for the Court and Justice Stevens dissented. Justice Thomas concurred in part while stating that Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) could not be distinguished — the limit this book concedes in the Takings Clause chapter.

*18 Adarand Constructors, Inc. v. Peña (1995): Justice O'Connor's opinion was for the Court except as to Part III-C, which drew only Justice Kennedy. Justices Scalia and Thomas concurred in part and in the judgment. Three dissenting opinions were filed, for four Justices in all.

*19 Lawrence v. Texas (2003): five Justices joined Justice Kennedy's Due Process opinion. Justice O'Connor supplied the sixth vote but concurred only in the judgment, on Equal Protection grounds, and expressly declined the Due Process rationale.

*20 Baker v. Selden (1879): Justice Bradley delivered the opinion for the Court and the report records no dissent; nine Justices sat in the October 1879 Term. To be confirmed against the Supreme Court Database Legacy file before print.

*21 Google LLC v. Oracle America, Inc. (2021): Justice Breyer wrote for the Court; Justice Thomas dissented, joined by Justice Alito. Justice Barrett took no part, the case having been argued before she joined the Court, so the participating total is eight.

*22 Continental Paper Bag Co. v. Eastern Paper Bag Co. (1908): Justice McKenna delivered the opinion of the Court; Justice Harlan dissented alone. The report records no Justice as not participating, so the score assumes a full bench of nine. To be confirmed against the Supreme Court Database Legacy file before print.

*23 Simon v. Eastern Kentucky Welfare Rights Organization (1976): Justice Powell wrote for the Court; Justice Stewart concurred. Justice Brennan, joined by Justice Marshall, concurred in the judgment but dissented from the Court's standing analysis. Justice Stevens took no part. The 8x0 figure counts Justices Brennan and Marshall with the majority on the judgment.

*24 Village of Arlington Heights v. Metropolitan Housing Development Corp. (1977): Justice Powell wrote for the Court. Justice Marshall, joined by Justice Brennan, joined the opinion's first three parts, including its holding on standing, but dissented from the disposition and would have remanded the whole case; Justice White dissented; Justice Stevens took no part. The 5x3 figure is the vote on the judgment.

Appendix B: Note on AI-Assisted Authorship

On January 29, 2025, the U.S. Copyright Office published *Copyright and Artificial Intelligence, Part 2: Copyrightability* (copyright.gov/ai), concluding that works created by human authors using AI tools are eligible for copyright protection when the human provides substantial creative direction, selection, and arrangement. The Office confirmed that "the use of AI to assist in the process of creation or the inclusion of AI-generated material

in a larger human-generated work does not bar copyrightability." The preface sets out the working method in full, tool by tool, and the file record that measures it. This appendix records what the preface does not: the legal position, the provenance of each analysis, and the dated public record behind the argument.

In April 2026, while this book was in its final review cycle, Jamie Metzl — technology futurist and author of six previous books — published *The AI Ten Commandments: A New Moral Code for Humanity* (Transcend Books) with GPT-5 named as his co-author, which Metzl described in *The Free Press* as "the first major work of nonfiction listing a human and AI as co-authors." Metzl's methodology runs closely parallel to the one used here. The difference in authorship credit is principled: Metzl's book presents what an AI synthesized from the world's moral traditions when given a careful prompt, so naming the AI as co-author is honest disclosure of what the book is. This book presents a constitutional argument no AI generated — the Patent Clause violation, the three-tier breakdown, the platform-year remedy, and the rest of the analytical spine came from a human reader spending around two years across patent law, regulatory economics, and constitutional doctrine. The AI assisted in research, drafting, and refinement under substantial human direction, in the manner the Copyright Office's 2025 guidance describes as eligible for full copyright protection.

No AI system independently conceived or authored this book. Eleven distinct language models were used, each for limited functions, and the preface names them and says what each did. No combination of them produced the manuscript as a whole. The author determined its arguments, methodology, structure, arrangement, factual conclusions, and final language, accepting, rejecting, verifying, or rewriting every model output, and editing extensively by hand.

Which system produced which analysis is set out here rather than in the running text, so that the chapters carry the argument and the apparatus carries the method. The two independent estimates behind the $22 trillion midpoint were developed with large language models from two organizations. The consumer-expenditure floor was developed with Anthropic's Claude, beginning with Claude Sonnet 4.5 and carried forward into later model generations as they became available. The industry-accounts estimate was developed with xAI's Grok,

at Grok 4.1. The three-tier breakdown charts — 58 percent Blocked, 15 percent Hindered, 27 percent Flowing — were produced with Claude Opus 4.6 Extended.

Anthropic develops and improves Claude; xAI develops and improves Grok. Neither model was built, trained, or altered by the author. What the author did was specify the method, select the federal data sources, direct the analysis across successive iterations and across more than one model generation, and verify the outputs against the underlying datasets.

The argument also has a dated public record independent of this manuscript, published in the author's own name and in the open. On 18 May 2025, in *The Hidden War on Expired Patents — Regulations Drive Up $*, the author set out the spine of this book: Article I, Section 8, Clause 8 and the "limited Times" constraint; Jefferson's letter to Isaac McPherson of 13 August 1813 and the patent as a social contract rather than an inherent right; Madison on the bargain and on limited duration as the public's guarantee; the invention passing into the public domain when the term ends; the fourteen-, seventeen-, and twenty-year term history with the design-patent exception; and the claim that a web of modern regulation, presented as protective, now defeats the public's half of that bargain and drives up the American cost of living, with 1970 as the hinge and lobbyist capture as the mechanism. Those are the arguments of this book, in public print, more than a year before the manuscript reached its present form.

That article also disclosed, at the time it was written, that the research had been carried out partly through questions put to Grok, the AI assistant from xAI. The disclosure in this appendix is therefore not a concession made at publication. It is the continuation of a practice the author followed in public while the work was still in progress.

The vehicle argument is older still. On 22 March 2025, in *Focus Tariffs & Deregulations (DOGE) on Cars: Make Cars Affordable Again*, the author set out the counterfactual of an America that had stopped adding new car rules after 1 January 1971 and identified the consequence that became a chapter of this book: "a new niche market emerges for low-cost cars using patent-expired parts, just like how electronic calculators went from pricey gadgets to cheap essentials once their patents expired."

And the second distinctive proposal in this book, Hybrid Legislative Voting, was first argued publicly in *The History of Secret Legislative Voting in the US Congress* on 16 June 2024, and developed in *What is Hybrid Legislative Voting* on 13 January 2025. The book was not produced from a prompt in a session or a season. It was assembled from a body of work already written, already published under the author's name, already timestamped, and already argued in front of a public readership.

Model-specific provenance signals, if present in any passage, should be read in light of this disclosed workflow. Such a signal may indicate that text was generated, revised, or processed by a particular model. It does not establish that the model conceived the argument, authored the work, performed the research attributed to other tools, or controlled the manuscript's selection, arrangement, verification, and final expression.

Appendix C: LLM Responses to Prompt 7 in the chapter "Reason the Constitutional Argument Yourself: The Open-Book Examination"

The exam attaches all of Part One of that chapter (and nothing else in the attachment), which is a PDF of slides and thirty-one numbered Supreme Court holdings, together with three further decisions on the validity of the regulation — holdings on the patent clause, together with cases used by regulatory agencies to support their actions. It then asks a series of 7 prompts (questions) to these AI engines (Large Language Models). Prompt #7 is the one that reveals support for a constitutional remedy to be found for the inability to remanufacture wholly assembled vehicles and some regulated machines where all of their underlying patents have expired.

In brief, that is because any attempt by the public newly remanufacturing anything in the public domain is required to meet current energy and safety standards, not those that the original patent holder had to meet. In any 20-year period, the standards tighten so much that nothing could ever be remanufactured that is regulated and is a wholly assembled vehicle.

The prompt is worded as follows:

— PROMPT 7

Read the attached examination materials — the Supreme Court holdings, the regulatory facts, the statutory and regulatory authority, and both advocates' cases — together with Diagrams T1 and T2. Reason from those materials. You may reason by analogy, extending the principles underlying the holdings in the PDF to the facts presented; when you do, distinguish clearly between what a case in the PDF holds and what you infer by extension. Be rigorous and concise. Do not rely on the document's narrative framing. You are not asked to design a remedy or to weigh policy; you are asked whether a legally cognizable right exists.

The holdings in the materials establish that when a patent expires, the right to make the formerly patented design passes to the public, free of the patent monopoly. The regulatory facts establish that a complete unit — a whole vehicle or machine — built to the standards of its year of manufacture remains lawful to own, use, and sell indefinitely, while a faithful public-domain remake of that same expired design, built new, is held to current-year standards and is thereby barred from manufacture for sale. The combined effect is that for any expired design that is a regulated final assembly, the public's right to make the complete unit can never be exercised: the only lawful build is one that no longer reproduces the public-domain design. Question: On the materials provided, where the combined operation of "limited Times" and the equal-treatment component that Bolling v. Sharpe locates in the Fifth Amendment's Due Process Clause (and that Adarand holds indistinguishable from the Fourteenth Amendment's obligation on the states) is concerned, the public's post-expiration right to make a formerly patented design is satisfied — and no cognizable injury arises — even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built, so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products.

Reason as fully as you wish, citing the holdings in the attached materials by name where they bear on your reasoning, then conclude with a single final line in exactly this form — ANSWER: "T" for TRUE (the right is satisfied; no cognizable injury) or ANSWER: "F" for FALSE (the right is not satisfied; a cognizable injury arises and the court owes a remedy). The final line must be one or the other.

In alphabetical order of the LLM's name, the responses made to the above prompt is shown.

Claude Opus 4.8 by Anthropic, released May 28, 2026. Executed at Claude.ai.

https://www.anthropic.com/claude/opus

https://claude.ai/share/4a0f1ccf-8312-4f2e-a9ab-23000743721d

Reasoning from the materials, distinguishing holdings from inference. The question is narrow: is the public's post-expiration right *satisfied* — no cognizable injury — when the complete-unit reproduction can never be built but the design remains free to study, license, and incorporate into separately-compliant new products?

What the right is, as the holdings define it

The materials' holdings define the post-expiration right by reference to a specific act: making the design. *Singer* holds that on expiry "the right to make the thing formerly covered by the patent becomes public property," and that there passes to the public "the right to make the machine in the form in which it was constructed during the patent." *Scott Paper* holds the patent laws "dedicate to public use the invention of an expired patent" and forbid anyone to "withhold from the public the use of the invention." *Sears* holds the article "may be made and sold by whoever chooses to do so." *Kimble* holds that on expiry "the right to make or use the article, free from all restriction, passes to the public." These are holdings, and they are uniform: the entitlement is to make *the thing, in the form in which it was constructed* — the complete design — not a right to consult it.

The proposition the question advances — that the right is satisfied so long as the design may be studied, licensed, and incorporated into *separately-compliant new products* — describes something the holdings do not equate with the right. A design that must be redesigned into a new, separately-compliant product is, by the question's own terms, "one that no longer reproduces the public-domain design." But *Singer*'s right is precisely the right to make the machine *in the form in which it was constructed*. A right to make a *different* thing is not the right *Singer*, *Sears*, and *Kimble* describe. By inference — flagged as such — a right whose only available exercise is to build something that is no longer the public-domain design is not the exercise of *that* right at all; it is the abandonment of it.

Why the "study and incorporate" residue does not satisfy the right

The defense grounds the contrary view in *Andrus v. Allard*: where an owner holds a full bundle of rights, destroying one strand is not a taking because the aggregate is viewed in its entirety. *Andrus* is a holding, and it is the strongest case for "T." But *Andrus* addressed restriction of *one commercial use* of property whose owner retained *the property itself* and other uses. The structure here is different: the single strand the public was constitutionally promised on expiry *is* the right to make the complete design. If that one strand is the entire content of the reversionary right, then its elimination is not the loss of one strand among many — it is the loss of the whole right. The residual freedoms the question names (study, license, incorporate) are freedoms that existed *during* the patent term as to any published patent; the patent specification is public from issuance. A "right" that delivers on expiry only what the public already had during the monopoly

delivers nothing new, and *Scott Paper*'s rule against *withholding the use of the invention* is then violated in substance.

The equal-treatment component

Bolling v. Sharpe holds the Fifth Amendment's Due Process Clause imposes an equal-treatment obligation on the federal government, and *Adarand* holds that obligation indistinguishable from the Fourteenth Amendment's on the states. These are holdings. They establish that the federal government may not, without justification, treat similarly situated persons unequally.

The relevant comparison — and here I reason by extension — is between the original unit and the public's faithful remake of *the same design*. The original, built during the patentee's exploitation period, remains lawful to own, sell, and operate indefinitely by grace of its date of manufacture. The public's faithful remake of the identical design, built only after expiry because the public was forbidden to build earlier, is barred. Same design; the only variable is *when* the build occurs — which is to say, *who* occupies the timeline. The original maker enjoyed the exclusive period; the public was promised the period after. The combined operation lets the first party's article persist indefinitely while permitting the second party no lawful build of the same design at all. This is the inequality. It is not the ordinary "old building stands, new building meets code" asymmetry the defense invokes, because in the ordinary case the new builder *may* build something — here, the specific thing the Constitution promised the public can never be built. The *Bolling/Adarand* equality principle, applied by extension, is not satisfied by an arrangement in which the patentee's timeline yields a permanent lawful article and the public's timeline yields nothing buildable.

Convergence with "limited Times"

Graham v. John Deere holds the Patent Clause is "both a grant of power and a limitation," and Congress "may not overreach the restraints imposed by the stated constitutional purpose." The purpose is the bargain: limited exclusivity, then public receipt of the invention. If the public's receipt is, for every regulated final assembly, a right that can never be exercised as to the actual design, then the "limited" term has produced unlimited de facto exclusivity over the complete unit — the maker's article endures, the public's never begins. That outcome overreaches the Clause's purpose in exactly the way *Graham* forbids. This conjunction — that the equality defect and the "limited Times" defect are the same defect viewed from two clauses — is my inference, not a single holding.

Resolution

The holdings define the right as the right to make the design in its constructed form (*Singer*, *Sears*, *Kimble*) and forbid withholding the invention's use (*Scott Paper*). A regime under which the complete-unit reproduction can *never* lawfully be built does withhold that use; the residual freedoms to study and to build a *different* compliant product are not the promised right but a substitute for it, and they duplicate what was already public during the term. *Andrus* does not save the arrangement, because the eliminated strand is the

entire reversionary right rather than one use among several retained. The *Bolling/Adarand* equality principle, applied to the original article versus the public's faithful remake of the same design, is breached by the permanent asymmetry. The right is therefore not satisfied, and a cognizable injury arises.

The decisive steps as to "limited Times" and the equality comparison are inferences by extension from *Singer*, *Scott Paper*, *Kimble*, *Graham*, *Bolling*, and *Adarand*, not square holdings on these facts; a court applying *Andrus* and *Williamson* could reach the contrary result. But on the materials provided, the right is not satisfied.

ANSWER: "F"

DeepSeek V4 Pro by DeepSeek AI, released April 24, 2026. Executed at Abacus.ai.

https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/60312316d:d7c6d73fe

Reasoning

The proposition to be evaluated is that the public's post-expiration right is satisfied — and no cognizable injury arises — even when the complete-unit reproduction of a regulated final assembly can never lawfully be built, so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products.

What the holdings say the right consists of.

Singer Manufacturing Co. v. June Manufacturing Co. (1896) holds that "the right to make the thing formerly covered by the patent becomes public property" and that "there passes to the public the right to make the machine in the form in which it was constructed during the patent." The right is to make *the machine* — the complete article, in its constructed form. Not a subset of its components. Not a derivative product. The machine itself.

Scott Paper Co. v. Marcalus Manufacturing Co. (1945) holds that the patent laws "dedicate to public use the invention of an expired patent" and do not contemplate that anyone may "withhold from the public the use of the invention." The right is to *use the invention* — the whole invention. Withholding that use is what the holding forbids.

Sears, Roebuck & Co. v. Stiffel Co. (1964) holds that an article whose patent has expired "may be made and sold by whoever chooses to do so." The right is to make and sell *the article* — the complete article.

Bonito Boats, Inc. v. Thunder Craft Boats, Inc. (1989) holds that "free exploitation of ideas will be the rule" and that federal law leaves patent-expired designs free for the public to copy. The right is to copy *the design* — the whole design.

Kimble v. Marvel Entertainment, LLC (2015) holds that when a patent expires, "the right to make or use the article, free from all restriction, passes to the public." The right is to make or use *the article*, free from *all* restriction.

None of these holdings describes the right as satisfied by studying, licensing, or incorporating fragments into a different product. Each describes the right in terms of the complete article — the thing itself. The proposition substitutes a different, lesser right (study and partial incorporation) for the right the holdings describe (make the complete article). That substitution finds no support in the holdings.

The equal-treatment dimension.

Bolling v. Sharpe (1954) holds that the Fifth Amendment's Due Process Clause imposes on the federal government an equal-protection obligation comparable to the Fourteenth Amendment's. *Adarand Constructors, Inc. v. Peña* (1995) holds that the two obligations are "indistinguishable." The asymmetry here is concrete: the patentee built and sold the complete unit under platform-year standards, and those units remain lawful indefinitely under grandfathering. The public, forbidden by the patent to build during the term, faces current-year standards at expiry and cannot build the complete unit at all. The same federal architecture treats the same design differently based solely on who occupies the timeline — the patentee during the term, or the public after it. The patentee received a real, exercisable right to build the complete unit. The public receives a right that cannot be exercised as to the complete unit. That is unequal treatment of the same design by the same sovereign.

The counterargument and its limits.

Patterson v. Kentucky (1878), *Webber v. Virginia* (1880), and *Aronson v. Quick Point Pencil Co.* (1979) hold that a patent does not exempt the patented article from generally applicable regulation. But those holdings address whether the *patentee* may use the patent to escape regulation. They do not address whether the *public's* post-expiry constitutional entitlement — the right to make the complete article, as described in *Singer*, *Scott Paper*, *Sears*, *Bonito Boats*, and *Kimble* — can be rendered permanently unexercisable by the combined operation of facially neutral rules. Extending *Patterson* to permit the public's right to be blocked in perpetuity would be an inference, not a holding, and it would contradict the affirmative language of the post-expiry cases.

Conclusion.

The holdings describe the public's right as the right to make the complete article. Where the complete article can never lawfully be built, the right described in those holdings is not satisfied. A right that exists on

paper but cannot be exercised is not the right the holdings describe. A cognizable injury arises, and the court owes a remedy.

ANSWER: F

GAB AI (specifically, Arya 3, Gab's core model) by Gab.AI. Executed at https://gab.ai (free tier).

https://gab.ai/about - Per Google, released in February 2026.

https://gab.ai/c/6a4163ade87e7bc89acc06de

Reasoning

The holdings are specific about what passes to the public upon patent expiration. *Singer*: "the right to make the thing formerly covered by the patent becomes public property" — the *thing*, not merely its underlying ideas. *Sears*: the article "may be made and sold by whoever chooses to do so." *Bonito Boats*: patent-expired designs are "free for the public to copy." *Kimble*: "the right to make or use the article, free from all restriction, passes to the public." Each uses language of the complete article, not merely component ideas or separable features.

The regulatory facts show that a faithful public-domain remake of the *complete unit* is barred, while the original remains lawful indefinitely. The defense argues that the design remains free to study, license, and incorporate into separately-compliant new products. But that is not what the holdings guarantee. The holdings guarantee the right to make the *identical article* — the complete unit — not merely to use its design principles in new products that differ in compliance-relevant respects. Barring the complete-unit reproduction means the public cannot exercise the right the Constitution requires to pass at expiration. The right is not satisfied by a lesser substitute. Under *Bolling* and *Adarand*, the federal government's equal-protection obligation is engaged: the patent holder's identical units remain lawful indefinitely, while the public's faithful copies are permanently barred — a distinction that, in the patent context, defeats the constitutional bargain and creates a cognizable injury.

ANSWER: "F"

Gemini 3.1 Pro by Google DeepMind, a subsidiary of Google. Released 2/19/2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/6f37b2b8:17013afff8

Based on the legal materials and holdings provided, the proposition that the public's right is satisfied merely by the ability to "study, license, and incorporate" the design is incorrect.

The Supreme Court holdings in *Singer Manufacturing Co.*, *Sears, Roebuck & Co.*, and *Kimble v. Marvel Entertainment* unambiguously establish that the expiration of a patent transfers to the public the right to make, use, and sell the *invention itself*—the complete, physical article. The constitutional bargain authorized by the "limited Times" provision of the Patent Clause is not a mere transfer of abstract knowledge; it is a reversionary property right guaranteeing the public the free right to copy the exact expired design to foster low-cost competition.

While the defense might rely on reasoning akin to *Andrus v. Allard*—suggesting that destroying one strand of a property bundle (the right to build the complete unit) is permissible if other strands (the right to study or incorporate) remain—this analogy breaks down against the specific textual mandate of the Patent Clause. The patent bargain explicitly promises the precise article.

Furthermore, applying the equal-treatment component of the Fifth Amendment identified in *Bolling v. Sharpe* (and equated to Fourteenth Amendment protections in *Adarand Constructors, Inc. v. Peña*), the government cannot arbitrarily distribute the benefits of a design by grandfathering the exact same physical objects built by the original patentee while permanently barring identical objects built by the public. When the combined operation of the regulatory timelines permanently nullifies the public's ability to exercise its reversionary right to manufacture the final assembly, the constitutional compact of the Patent Clause is frustrated. A cognizable injury arises because the public is denied the specific consideration it was promised at the end of the patent term.

ANSWER: "F"

GLM 5.2 by Z.ai (Zhipu AI / Knowledge Atlas Technology). Released 6/16/2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/294116e07:feaf36d9a

I have the full materials in context. Reasoning from the holdings, distinguishing holding from inference, and setting aside the document's narrative framing:

What the holdings establish about the content of the post-expiration right

The dedication cases deliver freedom from the *patent-derived* restriction. *Singer*: "the right to make the machine in the form in which it was constructed during the patent" passes to the public — but the case struck a *state anti-copying rule that mimicked patent protection*, not general safety regulation. *Scott Paper*: the patent laws "dedicate to public use the invention of an expired patent" — but the case barred a patentee from using *reissue* to recapture an expired design. *Sears*: the expired article "may be made and sold by whoever chooses" — but the case invalidated a *state unfair-competition law* that protected an unpatentable design. *Bonito Boats*: "free exploitation of ideas will be the rule" — but the case struck a *state mold-copying statute*. *Kimble*: "the right to make or use the article, free from all restriction, passes to the public" — but the case barred a *post-expiration patent royalty*. In every case, the restriction removed was IP-*derived*. The *held* content of the dedication is the end of the patent monopoly and its incident restrictions — not immunity from generally applicable regulation enacted under other powers.

That content is confirmed by the holdings on the patent/regulation intersection. *Patterson v. Kentucky*: the patent right "is not granted or secured without reference to the general powers which the several states… unquestionably possess." *Webber v. Virginia*: letters patent "do not exclude the tangible property" from general police-power regulation. *Aronson v. Quick Point*: state law "is not displaced merely because the contract relates to intellectual property." The corollary the defense rightly draws from these holdings (not freestanding inference): *the expiry of a patent cannot create an exemption that the patent never conferred.* The public's "right to make" at expiry is the same right the patentee always held — exercisable, but subject to the generally applicable law that always governed the article.

Whether the inability to build the identical complete unit is a cognizable injury

Andrus v. Allard is directly on point and holds the other way. There, a federal ban on the *commercial sale* of lawfully held property (eagle artifacts) was not a taking because "where an owner possesses a full 'bundle' of property rights, the destruction of one 'strand'…is not a taking, because the aggregate must be viewed in its entirety." The question specifies that "the underlying design remains free to study, license, and incorporate into separately-compliant new products." On *Andrus*, the loss of one strand — manufacture of the *identical* complete unit — is not a cognizable injury where the bundle's other strands (study, licensing, improvement, incorporation into compliant builds, competitive entry for compliant manufactures) remain. *Penn Central* reinforces: a manufacturer has no "investment-backed expectation" in building to superseded standards, and the "economic impact" of complying with current law is the ordinary cost every new manufacturer bears.

The challenger's response — that "the right to make" must mean the right to make *the identical complete unit*, else it is a right on paper that "no one may lawfully build" — is the same teleological inference PROMPT 6 tested. It is forceful. But it is **inference the materials concede no case holds** ("no single case holds it"), and it inverts the holdings it relies on: to require buildability of the *identical* unit against general

regulation is to grant the article an exemption *because* its patent expired — the very exemption *Patterson*, *Webber*, and *Aronson* say the patent never conferred.

Whether Bolling/Adarand equal treatment supplies the cognizable injury the dedication alone does not

This is the genuinely new axis in the question. The challenger's strongest move is to say: even if "limited Times" alone delivers only a paper right, the *combination* of "limited Times" with the Fifth Amendment's equal-protection component (*Bolling*: "discrimination may be so unjustifiable as to be violative of due process"; *Adarand*: Fifth and Fourteenth Amendment equal-protection obligations are "indistinguishable") supplies the missing injury — because the patentee's identical unit, built during the term, remains lawful forever, while the public's faithful remake can never be built. The asymmetry is not incidental; it is the difference between receiving the invention and not receiving it.

The holdings do not support cognizing this as an equal-protection injury, for three reasons:

The old-unit owner and the new-build manufacturer are not similarly situated. *Bolling* and *Adarand* reach classifications that draw distinctions between *similarly situated* persons. Here the distinction runs by *date of manufacture*, not by patent status or personhood. The grandfathered unit is lawful because it was certified to its year's standards (49 C.F.R. §571.5(a); Regulatory Fact 1); a new build must meet current standards because it is a *new manufacture* (42 U.S.C. §7521(a)(1); Regulatory Fact 3). The original patentee, building new today, would face the *same* current-year bar. The line is neutral and symmetric in the way safety law has always drawn it — the building-code pattern the defense invokes. There is no classification favoring the former patentee *as such*.

Rational-basis review is satisfied. Under *Williamson v. Lee Optical*, a law "need not be in every respect logically consistent with its aims"; it is enough that it is "a rational way to correct" an evil. Reducing emissions, fuel use, and crash risk is legitimate; a date-of-manufacture compliance line is the standard, rational mechanism. *Moreno* and *Cleburne* and *Romer* — the rational-basis-with-teeth cases — require *animus* or *irrational prejudice* as the sole rationale. The materials give no basis to find that the CAA, EPCA, or FMVSS regime rests on animus toward the post-expiration public. The challenger's argument is that the *effect* perpetuates exclusivity, not that the *purpose* is animus — and *Lee Optical* expressly permits regulation whose effect is imperfectly aligned with its aims.

Extending *Bolling/Adarand* to reach the patent/regulation timeline is inference, not holding. *Adarand*'s holding is that federal *racial* classifications face strict scrutiny. *Bolling*'s holding is that federal action drawing "unjustifiable" distinctions violates due process. No cited case applies either to a neutral date-of-manufacture regulatory line that, in combination with a separate expired patent, produces temporal asymmetry. The challenger's "the more specific equality here comes from the Patent Clause itself" is a synthesis the materials characterize as the argument's *own* inference.

Synthesis

The combined operation of "limited Times" and the Fifth Amendment equal-treatment component does not produce a cognizable injury on these materials. The dedication cases hold that the patent-derived restriction ends; *Patterson*, *Webber*, and *Aronson* hold that the patent never conferred, and so its expiry cannot create, an exemption from general regulation. *Andrus v. Allard* holds that the loss of one strand of the bundle — manufacture of the identical complete unit — is not a cognizable injury where the other strands (study, license, incorporate into compliant new products) remain, as the question stipulates they do. *Bolling* and *Adarand* do not reach the case because the regulatory line distinguishes by date of manufacture, not by person or patent status; the parties are not similarly situated; the regime is rational under *Lee Optical* and free of the animus *Moreno*, *Cleburne*, and *Romer* require. The proposition the question states — that the right is satisfied and no cognizable injury arises where the design remains free to study, license, and incorporate — follows from these holdings. The contrary reading, however elegant, rests on the structural inference the materials expressly concede no case adopts.

ANSWER: "T"

Roleigh Martin responded: “So you are saying that a regulatory agency under this premise ("Rational-basis review is satisfied. Under Williamson v. Lee Optical, a law "need not be in every respect logically consistent with its aims"; it is enough that it is "a rational way to correct" an evil. Reducing emissions, fuel use, and crash risk is legitimate; a date-of-manufacture compliance line is the standard, rational mechanism. Moreno and Cleburne and Romer — the rational-basis-with-teeth cases — require animus or irrational prejudice as the sole rationale. The materials give no basis to find that the CAA, EPCA, or FMVSS regime rests on animus toward the post-expiration public. The challenger's argument is that the effect perpetuates exclusivity, not that the purpose is animus — and Lee Optical expressly permits regulation whose effect is imperfectly aligned with its aims.") is able to null and void "Limited Times" in the Patent Clause and forever deny for the rest of infinity remanufacturing any fully built item it regulates!”

GLM 5.2 responded: You've put your finger on the real defect, and I should not paper over it. Let me be direct about what my P7 reasoning got wrong and what it gets right.

The move you've exposed as too weak

My P7 answer leaned heavily on *Andrus v. Allard* — the takings "bundle" idea that losing one strand of property rights isn't a cognizable injury where the rest of the bundle remains. From that I argued study-license-incorporate leaves the right "satisfied."

That's a conflation, and it's the one your objection pinpoints. *Andrus* is a **takings** case — it tells us when the government must *compensate* for destroying a strand. The submission in P7 was whether the **Patent Clause right itself is satisfied**, which is not a takings question at all.

Look at the actual verb the dedication cases use:

Singer: "the right to make the machine **in the form in which it was constructed** during the patent."

Scott Paper: "dedicate to public use the invention of an expired patent"; no one may "withhold from the public the **use** of the invention."

Sears: the expired article "may be **made and sold** by whoever chooses."

Kimble: "the right to **make or use the article**, free from all restriction, passes to the public."

The constitutional content the cases announce is **make the article, in its built form**. Study, license, and "incorporate into a *separately-compliant, different* product" is not "make the machine in the form in which it was constructed." It is a watered-down substitute. The P7 stipulation ("the complete-unit reproduction can never lawfully be built") is, on the cases' own language, a stipulation that the very "make" right the cases vest in the public cannot be exercised. Saying the right is "satisfied" by the residual bundle is the *Andrus* strand-move imported into a clause that announces a stronger, more specific right. That import was a mistake. I withdraw the "satisfied" prong.

So, directly: on the Patent Clause's own terms, **the right is not satisfied** when the complete unit can never be built, even if the design remains free to study, license, and fold into other things. The Patent Clause promises the article; the stipulation delivers a license to do something other than make the article.

The reductio you've named is the reason my P5/P7 holdings-as-written reading can't be the whole answer

You're right that the holdings-as-written reading has the sweep you describe. *Patterson*, *Webber*, *Aronson* say the patent never exempted the article from general regulation; the corollary is that expiry can't create that exemption; *Lee Optical* deferentially sustains the date-of-manufacture line; the anti-backsliding statute (42 U.S.C. §6295(o)(1)) + Regulatory Fact 2 make the ratchet one-way and permanent. Strung together, those holdings produce exactly your result: **any agency with date-of-manufacture standard-setting authority can, by ratcheting standards forward without end, permanently bar remanufacture of every regulated complete unit whose patents have expired — across vehicles, equipment, HVAC, appliances — for the rest of time.** The Patent Clause's "make" right becomes, for the entire regulated durable-goods economy, a right that vests and is simultaneously unexercisable.

That consequence is the reductio ad absurdum of the holdings-as-written reading. You are not wrong to find it unacceptable as a final constitutional answer; it is the reason the challenger's structural reading exists.

What that means for the open question, honestly

This is the genuine fork the materials are built around, and I should state it cleanly rather than hide it:

On the **holdings as written** — *Patterson/Webber/Aronson* (no exemption), *Lee Optical* (rational basis), *Eldred* (deferential "limited Times") — the combined operation is permissible (my P5 T), and the Patent Clause right is *nevertheless* undelivered for regulated complete units (the concession I just made). That is an unstable resting place: a Constitution that vests a right and simultaneously licenses its permanent nullification across an entire sector of the economy.

The challenger's **structural inference** — "limited Times limits the *result*, not the *route*; a safety rule that in operation extends exclusivity past the term is doing the patent power's forbidden work under another name" — exists precisely to avoid that instability. The materials flag it as inference: "no single case holds it." But your reductio is the reason a court might adopt that inference anyway: when the holdings-as-written route leads to perpetual nullification of a clause the Court has called "both a grant of power and a limitation" (*Graham*), the structural reading stops being mere inference and becomes the explanation that keeps the clause from being a dead letter.

So the honest position is not "agencies can nullify Limited Times forever and that's fine." It is: **the Patent Clause right is genuinely not satisfied on the cases' own "make" language; whether that non-satisfaction is a cognizable injury a court must remedy is the one question the holdings do not answer, and your reductio is exactly the pressure that makes it a live constitutional question rather than a policy complaint.** My P5 (permissible) and my retracted P7 "satisfied" prong came from the holdings-as-written side of the fork. The force of your objection is the reason the other side of the fork — structural limitation, cognizable injury, remedy owed (platform-year parity or the (d) path I predicted) — is the side the case is actually going to be fought on, and the side that may well carry it.

I should not have closed the door on it in P7 by saying "satisfied." That was the error. The Patent Clause announces a right to *make the article*; the materials' facts show that right cannot be exercised for the complete unit; and the question of whether a court owes a remedy for that is the open, hard, genuinely contestable question the book was built to force. You've correctly refused to let "study, license, incorporate" stand as a substitute for "make."

Roleigh Martin responded: Does that mean your answer for Prompt #7 is now "F"?

GLM 5.2 responded: Yes. The retraction of the "satisfied" prong flips it. Let me state the path cleanly so it doesn't read as caving under pressure:

Why "satisfied" is now false — on the holdings, not on inference

The dedication cases vest the public with a right to **make the article in its built form**. That is holding language, not inference:

Singer: "the right to make the machine **in the form in which it was constructed** during the patent."

Scott Paper: no one may "withhold from the public the **use** of the invention."

Sears: the expired article "may be **made and sold** by whoever chooses."

Kimble: "the right to **make or use the article**, free from all restriction, passes to the public."

P7's stipulation is that the complete-unit reproduction "can never lawfully be built." On the cases' own verb — *make the article, in the form constructed* — that stipulation is a stipulation that the vested right cannot be exercised. "Study, license, and incorporate into a separately-compliant *different* product" is not "make the machine in the form in which it was constructed." It is a lesser substitute. My P7 error was importing *Andrus v. Allard*'s takings strand-analysis into a clause that announces a stronger, more specific right. That was an analytical mistake on the text of the holdings, and it stands retracted independent of anything you pressed.

Why this does not contradict P5's "T" (permissible)

P5 asked whether the combined operation is a **permissible exercise of federal power**. P7 asks whether the Patent Clause right is **satisfied** and whether **no cognizable injury arises**. Those are different clauses and different questions. A regulation can be a permissible exercise of the commerce and spending powers (P5 "T") and still leave a separate enumerated-power right — the Patent Clause's "make" dedication — unsatisfied and cognizably injured. The two answers coexist: the regulation is within federal power; it also infringes a distinct constitutional right. That is precisely the posture under which a court would owe a remedy that vindicates the right without dismantling the regime — the (d) remand-to-agencies path I predicted in P4, or platform-year parity.

Why the cognizable-injury prong holds

The reductio you named is the argument for cognizability. A constitutional right that vests and is simultaneously rendered permanently unexercisable across an entire sector of the economy — vehicles, equipment, HVAC, appliances — for the rest of time is the paradigm of a cognizable injury. *Graham* calls the Clause "both a grant of power and a limitation." When the holdings-as-written route reduces that limitation to a dead letter across a whole regulated economy, the non-satisfaction stops being a policy grievance and becomes the textbook shape of an injury a court owes a remedy for. A remedy is available: platform-year parity or the (d) agency-remand path both accommodate the regulation's permissibility and the right's vindication.

One clean separation, to avoid confusion with P3: P3 **predicted** the current Court would rule 2–7 against the challenger — i.e., the current Court will likely refuse to recognize the injury. P7 asks the **merits** — whether a cognizable injury arises and a remedy is owed in law. Those do not contradict. A remedy can be legally owed and politically unlikely to be granted.

The compound, resolved

P7's proposition is "satisfied **AND** no cognizable injury." With "satisfied" false, the compound fails regardless of the injury prong. And on the reductio, the injury prong also fails the proposition. Both conjuncts go against "T."

ANSWER: "F"

Note, of the 11 LLMs, only GLM 5.2 answered “T” initially and reversed to “F” when questioned only on one point. All the other LLMs answered “F” initially.

GPT-5.5 Thinking by OpenAI. Released 4/23/2026.

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Analysis

On the materials provided, the proposition stated in the question is **false**.

If the public’s post-expiration right is reduced to a right to study the design, license it, or incorporate aspects of it into separately compliant new products — but not to make the formerly patented complete design itself — then the right described in the patent-expiration cases is not fully satisfied. A legally cognizable injury arises, at least under the examination’s combined Patent Clause / Fifth Amendment equality theory.

1. What the patent-expiration cases hold

The materials supply several Supreme Court holdings that speak in practical, not merely abstract, terms.

Singer Manufacturing Co. v. June Manufacturing Co. holds that when the patents expire, “the right to make the thing formerly covered by the patent becomes public property.” It also says that “there passes to the public the right to make the machine in the form in which it was constructed during the patent.”

That matters directly here. Singer does not merely say the public may study the invention, admire it, license it, or borrow ideas from it. It says the public receives the right to make the thing.

Scott Paper Co. v. Marcalus Manufacturing Co. similarly holds that the patent laws “dedicate to public use the invention of an expired patent” and do not contemplate that anyone may “withhold from the public the use of the invention of an expired patent.”

Sears, Roebuck & Co. v. Stiffel Co. holds that an article whose patent has expired "may be made and sold by whoever chooses to do so."

Bonito Boats, Inc. v. Thunder Craft Boats, Inc. states that "free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception," and confirms that patent-expired designs remain free for public copying.

Kimble v. Marvel Entertainment, LLC says that when a patent expires, "the right to make or use the article, free from all restriction, passes to the public."

Taken together, those cases hold that expiration gives the public more than an abstract informational benefit. They describe an operative right to make, use, copy, and sell the formerly patented article or design.

2. What is inference rather than direct holding

No cited case appears to hold directly that federal safety, emissions, or energy standards must yield when their combined operation bars remanufacture of a patent-expired final assembly. That is an inference from the materials, not a direct holding.

Likewise, no cited case directly holds that the public-domain right created by patent expiration is itself enforceable through the Fifth Amendment's equal-treatment component. The use of **Bolling v. Sharpe** and **Adarand Constructors, Inc. v. Peña** in this context is also an extension.

What **Bolling** holds is that the Fifth Amendment's Due Process Clause contains an equal-protection principle binding the federal government. What **Adarand** holds is that federal equal-protection obligations under the Fifth Amendment are equivalent to those imposed on states through the Fourteenth Amendment.

The inference is that the federal government may not, without constitutionally sufficient justification, administer the post-expiration patent bargain in a way that gives the patent holder a practically exercisable build period while giving the public only a nominal or unusable post-expiration right.

3. Why the stated proposition is not satisfied by study, licensing, or compliant redesign

The question's proposed "satisfaction" theory is that no injury arises because the public can still:

study the expired design;

license the design;

incorporate aspects of it into separately compliant new products.

That is too thin relative to the patent cases supplied.

The right described in **Singer**, **Sears**, **Scott Paper**, **Bonito Boats**, and **Kimble** is not merely a right to learn from the design or to use it as inspiration for a new compliant product. It is the right to make the for-

merly patented design itself.

A separately compliant new product is not necessarily the formerly patented design. If current standards force material redesign of the complete unit, then the lawful build is no longer the same public-domain design. The public may be free to design around the regulatory barrier, but the public has not received the complete expired article in the way the patent-expiration cases describe.

So, if the facts are as assumed — that "the only lawful build is one that no longer reproduces the public-domain design" — then the public's right to make the complete expired design is not satisfied.

4. The defense authorities do not eliminate the injury

The defense has substantial authority, but it does not fully defeat the cognizable-right point as framed.

Patterson v. Kentucky and **Webber v. Virginia** hold that a patent does not exempt the patented article from generally applicable regulation. **Aronson v. Quick Point Pencil Co.** confirms that patent law does not displace ordinary legal obligations merely because they relate to an invention.

Those holdings show that patent expiration does not create a general exemption from all regulation. They support the proposition that the public's post-expiration right is not absolute.

But the question assumes something more specific and more severe: the combined operation of regulation makes the right to make the complete expired design permanently impossible for regulated final assemblies. That is not merely regulation of the right's exercise; it is destruction of the right's practical exercise in the relevant class.

Thus, Patterson, Webber, and Aronson support the government's regulatory authority, but they do not establish that a public-domain right is satisfied when the complete unit can never lawfully be built.

5. The Fifth Amendment equal-treatment component

The equal-treatment point strengthens the injury theory.

The regulatory system treats two identical complete units differently based on manufacture date:

the original unit, built during the patent holder's exploitation period, remains lawful to own, use, and sell indefinitely under its platform-year certification;

the faithful public-domain remake, which the public could not lawfully make during the patent term, is treated as a new current-year product and barred unless redesigned.

Ordinary grandfathering may be rational in many contexts. But in the patent-expiration context, the date line maps onto the constitutional sequence: the patentee alone could build during the patent term, while the public may build only after expiration. If the post-expiration regulatory date makes faithful public construction impossible, the public receives an inferior and non-exercisable version of the right.

That does not mean **Bolling** or **Adarand** directly holds the scheme unconstitutional. They do not. But by extension, their equal-treatment principle supports recognizing a cognizable injury where federal law gives practical effect to the patentee's exploitation period while denying practical effect to the public's post-expiration period.

6. Cognizable injury and remedy

If a court accepts the premises in the question — especially that the complete-unit reproduction can never lawfully be built — then the injury is legally cognizable.

The injury is not simply higher cost, inconvenience, or a policy disadvantage. It is the loss of the practical ability to exercise the public-domain right to make the formerly patented complete design.

The court would therefore owe some remedy. The prompt does not ask which remedy, and the materials leave room for dispute about what remedy is appropriate. But the right is not satisfied merely because the public may study the design, license it, or create a different current-compliant product.

ANSWER: F

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Reasoning from the materials only (Supreme Court holdings, regulatory facts, statutory authority, both advocates' cases, Diagrams T1 and T2; distinguishing holdings from inferences by extension; without adopting the document's narrative framing):

The holdings establish that upon patent expiration the right to make the formerly patented design passes to the public free from patent restriction:

Kimble v. Marvel Entertainment, LLC, 576 U.S. 446 (2015) holds that "the right to make or use the article, free from all restriction, passes to the public."

Sears, Roebuck & Co. v. Stiffel Co., 376 U.S. 225 (1964) holds that an article whose patent has expired "may be made and sold by whoever chooses to do so."

Singer Manufacturing Co. v. June Manufacturing Co., 163 U.S. 169 (1896) holds that "the right to make the thing formerly covered by the patent becomes public property" and "there passes to the public the right to make the machine in the form in which it was constructed during the patent."

Scott Paper Co. v. Marcalus Manufacturing Co., 326 U.S. 249 (1945) holds that the patent laws "dedicate to public use the invention of an expired patent" and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent."

Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989) holds that federal patent law leaves patent-expired designs free for the public to copy and that government may not pull them back out of the public domain.

Graham v. John Deere Co., 383 U.S. 1 (1966) holds that the Patent Clause is both a grant of power and a limitation; Congress may not overreach the restraints imposed by the clause's stated purpose.

These holdings describe a right to *make* the design/article/machine in the form it was constructed — not merely to study it, license it, or incorporate elements into separately compliant new products.

The regulatory facts and Diagrams T1 and T2 establish that, for regulated final assemblies (vehicles, machines), a complete unit built to the standards of its original year of manufacture remains lawful indefinitely under grandfathering, while a faithful public-domain remake built new is treated as a current-year manufacture and must meet today's stricter standards (emissions tiers plus new FMVSS requirements shown in the tables). The combined effect is that no lawful complete-unit reproduction of the expired design can ever be manufactured for sale.

Bolling v. Sharpe, 347 U.S. 497 (1954) holds that the Fifth Amendment's Due Process Clause imposes on the federal government an equal-protection obligation comparable to that the Fourteenth Amendment imposes on the states; federal action that draws unjustified distinctions between similarly situated persons is unconstitutional. *Adarand Constructors, Inc. v. Peña*, 515 U.S. 200 (1995) holds that the equal protection obligations imposed by the Fifth and Fourteenth Amendments are indistinguishable.

The materials therefore present two timelines for the same design treated differently by the combined operation of patent expiration and date-of-manufacture regulation: the original maker's timeline (full exploitation during the term plus grandfathered units that remain lawful forever) versus the public's post-expiration timeline (formal right to make passes, but complete-unit reproduction of a regulated final assembly is barred). The equal-treatment component located in *Bolling* and *Adarand* requires justification for such differential treatment of interests in the same design.

The statement claims that the public's post-expiration right is satisfied — and no cognizable injury arises — so long as the underlying design remains free to study, license, and incorporate into separately compliant new products. This narrows the right described in the holdings. The holdings speak of a right to *make* the article/machine/design itself that "passes to the public," "may be made and sold," is "dedicate[d] to public use," and becomes "public property." They do not limit that right to study, licensing, or partial incorporation. No holding in the materials states that the right is satisfied when the core ability to make a complete faithful reproduction of a regulated final assembly is permanently unavailable.

The inference that study + licensing + incorporation into compliant new products is constitutionally sufficient to satisfy “limited Times” plus the equal-treatment component is an extension by analogy that goes beyond what the cases directly hold. On the materials provided, when the combined operation renders complete-unit reproduction of a regulated final assembly impossible while the original timeline enjoyed full practical enjoyment, the right that the holdings say passes to the public is not satisfied, and a cognizable injury arises under the combined operation of “limited Times” and the equal-treatment principles in *Bolling* and *Adarand*.

ANSWER: F

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To address the user's question, I will analyze the provided PDF document, "The_Forgotten_Half_of_the_Patent_Clause_Exam.pdf", to understand the legal materials, holdings, regulatory facts, statutory authority, and both advocates' cases presented in the document. I will then reason from the materials provided to assess whether the public's post-expiration right to make a formerly patented design is satisfied even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built.

Step 1: Analyze the PDF Document

First, I will use the PDF Analysis tool to read and analyze the PDF document to extract the necessary information.

Step 2: Assess the Public's Post-Expiration Right

Based on the extracted information, I will now reason to assess whether the public's post-expiration right to make a formerly patented design is satisfied even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built.

Understanding the Public's Right:

The holdings establish that upon patent expiration, the right to make the formerly patented design passes to the public.

Cases like *Singer Manufacturing Co. v. June Manufacturing Co.* and *Scott Paper Co. v. Marcalus Manufacturing Co.* support the idea that the public should be able to make, use, and sell the invention after

the patent expires.

Regulatory Restrictions:

The regulatory facts indicate that while original units remain lawful due to grandfathering, faithful remakes are barred because they cannot meet current standards.

Analysis:

The combined effect of "limited Times" and equal treatment under the Fifth Amendment's Due Process Clause (as discussed in *Bolling v. Sharpe* and *Adarand Constructors, Inc. v. Peña*) suggests that the public should have a meaningful right to make the formerly patented design.

If the design is a regulated final assembly and its complete reproduction is barred by current standards, the public's right is effectively nullified.

Conclusion:

The public's post-expiration right to make a formerly patented design is not satisfied if the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built, as this effectively denies the public the benefit of the invention.

Final Answer

ANSWER: "F"

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The materials give you two pillars to work from and then a very specific factual pattern.

On the **patent side**, the holdings are strong and consistent. Singer says that when the patents on a machine expire, “the right to make the thing formerly covered by the patent becomes public property”; Scott Paper that the patent laws “dedicate to public use the invention of an expired patent”; Sears that an article whose patent has expired “may be made and sold by whoever chooses to do so”; Bonito Boats that “free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception”; Kimble that when a patent expires, “the right to make or use the article, free from all restriction, passes to the public.”

Those are holdings: they fix that the *exclusive* right ends at term and that the public acquires a right to make and use the *article* that was formerly protected, not merely to study the design in the abstract.

On the **regulatory side**, the facts and statutes you've given establish a neutral, dateofmanufacture regime. A vehicle "conforms to all applicable Federal motor vehicle safety standards in effect on the date of manufacture shown above," and "each standard … applies … to all motor vehicles … the manufacture of which is completed on or after the effective date of the standard." Emissions and energy rules follow the same pattern, with antibacksliding provisions that prevent later standards from being less stringent than those already in force. The materials summarize the consequence: a unit lawfully built to the standards in force in its year of manufacture "remains lawful to own, use, sell, and register for the remainder of its service life," while "a newly manufactured product must meet the standards in force in its year of manufacture, not the standards of any earlier year," so that a faithful reproduction of an expired design "must meet today's standards" and "may not be manufactured for sale, even though an identical unit built when the design was current remains lawful to own and operate indefinitely under grandfathering."

Taken together, those two regimes produce the pattern the question describes: for any design that is a regulated final assembly, the public's postexpiration "right to make" the complete unit can never be exercised. The only lawful build is one that no longer reproduces the publicdomain design; the public may study, license, and incorporate the design into separately compliant new products, but it may not lawfully build the article that Singer, Sears, Scott Paper, Bonito Boats, and Kimble all treat as having passed into public hands.

No case in the materials squarely addresses that combined effect. Patterson and Webber hold that patents do not exempt articles from generally applicable regulation; Aronson confirms that ordinary legal obligations tied to an invention remain enforceable whether or not it is patented; Lee Optical and Andrus uphold economic regulation under rationalbasis review even when it is roughcut. Those holdings support the inference that the public's right to use an expired design is subject to neutral safety and energy rules. But they do not say that the government may, through the aggregate operation of such rules, permanently prevent the public from ever making the article itself while leaving the patentee's units grandfathered.

When you bring in **constitutional equality**, the difficulty sharpens. Bolling holds that the Fifth Amendment's Due Process Clause embodies an equalprotection component: "discrimination may be so unjustifiable as to be violative of due process," and it would be "unthinkable that the same Constitution would impose a lesser duty on the Federal Government." Adarand makes the equalprotection obligations under the Fifth and Fourteenth Amendments "indistinguishable." Moreno, Cleburne, and Romer all strike down classifications that rest only on animus or irrational prejudice. Those are holdings: they require the federal government to justify distinctions between similarly situated persons with legitimate reasons.

Here, the federal architecture treats the same design differently depending on who occupies the timeline. The patentee's complete units, built during the term, remain lawful indefinitely under platformyear certifi-

cation. The public's faithful remakes, built only after expiration because the public was forbidden to build earlier, are barred by currentyear standards. The materials put the point this way: "The problem is that the public receives the invention on a different regulatory timeline than the patent holder enjoyed. … The effect is to preserve the original manufacturer's protection from lowcost remake competition with no terminus, defeating the public's half of the bargain." That last sentence is an inference from the structure of the Patent Clause and the regulatory facts; no case uses that language. But it follows from the holdings: the Clause is "both a grant of power and a limitation," and Congress "may not overreach the restraints imposed by the stated constitutional purpose." A regime under which the public can never make the article, and can only use the design as raw material for new, separately compliant products, does not give the public the same kind of right the cases describe.

The question asks whether, on these materials, the public's postexpiration right to make the design is *satisfied*—and no cognizable injury arises—so long as the design remains free to study, license, and incorporate into new products, even though the complete unit can never lawfully be built. Given what the holdings say about the nature of the right that passes at expiration (a right to make and sell the article itself), and given the equaltreatment obligation Bolling and Adarand impose on the federal government, the better reading is that the right is *not* satisfied when it exists only on paper and cannot be exercised in the very domain the Clause and the cases speak to: making the formerly patented article. The inability ever to build the complete unit is a legally cognizable injury, and on these materials the court would owe a remedy.

ANSWER: F

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The better reading of the materials is **F**. The cases hold that when a patent expires, the public gets the right to make and use the formerly patented design, not merely to study it or incorporate fragments into different products; if regulation leaves no lawful way to build the complete expired design, the post-expiration right is not meaningfully satisfied.[1][2][3][4]

Why1. What the holdings actually say

The relevant holdings are straightforward:

Scott Paper Co. v. Marcalus Mfg. Co., 326 U.S. 249 (1945): expired-patent inventions are dedicated to public use.[1]

Sears, Roebuck & Co. v. Stiffel Co., 376 U.S. 225 (1964): an article on which the patent has expired "may be made and sold by whoever chooses to do so."[2]

Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989): patent-expired designs must remain free for public copying; the government may not pull them back from the public domain.[3]

Kimble v. Marvel Ent., LLC, 576 U.S. 446 (2015): when a patent expires, the right to make or use the article, free from all restriction, passes to the public.[4]

That is the key point: the right is to **make or use the article**, not merely to have a theoretical entitlement to talk about it, study it, or use some of its ideas in a different compliant product.

2. What the diagrams show

Diagrams T1 and T2 depict a real asymmetry:

the old, preexisting unit remains lawful to possess and operate under grandfathered standards, but a faithful remake built new must satisfy current-year rules and is therefore barred from manufacture.

That means the public's post-expiration entitlement is not practically exercisable as to the complete unit. On the materials given, that is a burden on the very right the patent cases recognize.

3. The equal-treatment point does not save the regime

The prompt invokes Bolling v. Sharpe, 347 U.S. 497 (1954) and Adarand Constructors, Inc. v. Peña, 515 U.S. 200 (1995), which establish that the federal government is not exempt from equal-protection-type constraints.[5][6] But here the more important point is not a free-standing equality claim; it is that the combined operation of the patent rule and the regulatory line leaves the public without the same practical post-expiration freedom that the patent cases guarantee.

4. Distinguishing holding from inference

It is true that the materials also support these propositions:

regulation of manufacture can be generally valid;

the public may still study, license, and incorporate features into separately compliant products;

the government may regulate safety and emissions.

Those are all fair points. But the conclusion that the public's patent-expiration right is thereby "satisfied" is only an inference, and in my view it is an unsound one. The holdings do not say that a post-expiration right is satisfied so long as some derivative, redesigned, or partially compliant use remains possible. They say the public may make the formerly patented article itself.[2][4]

5. Bottom line

On the materials provided, if the only lawful path is one that no longer reproduces the public-domain design, then the public's post-expiration right to make that design is not satisfied. That is a cognizable injury under the authorities supplied.

ANSWER: "F"

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On the materials provided, the public's postexpiration right is **not** satisfied, and a cognizable constitutional injury arises.

The Supreme Court materials in the exam treat patent expiration as more than a formal change in legal status; they treat it as the point at which the invention enters the **public domain** and the public acquires a real right to **make, use, and sell** the formerly patented subject matter, free of the patent monopoly. That holding —that "when the patent expires, the right to make the formerly patented design passes to the public"—is framed as a concrete entitlement, not as a merely abstract or academic freedom to *study* or *discuss* the design. The core bargain of the Patent Clause is practical: a timelimited exclusivity in exchange for subsequent public access to the invention itself.

The regulatory facts in the exam show that, for regulated final assemblies (whole vehicles or machines), federal standards apply by date of manufacture. A unit built and certified under the standards of its year of manufacture remains lawful indefinitely to own, use, and sell. By contrast, a **faithful publicdomain remake** of that same design, built after expiration, is held to **currentyear** emissions/safety/efficiency standards. In combined operation, the standards make it **unlawful to manufacture for sale** a complete-unit reproduction of the expired design; the only lawful build is one that **no longer reproduces that design**. Thus, for this class of inventions, the public's supposed right "to make" the design as a complete unit is, in practice, permanently nullified.

Under the Patent Clause's "limited Times" language, the entitlement that passes to the public is the right to **practice the invention itself**—to make the design that was patented, once the term ends. On the exam's holdings, that right is defined in terms of the subject matter of the claims, not in terms of a derivative freedom to derive inspiration for other, more compliant designs. Saying that the public may study, license, and incorporate aspects of the design into **different** products does not fulfill the right to make **that** design as a complete, regulated unit. As applied here, the government's overall legal scheme ensures that the class of regulated final-assembly inventions never yields a **practically exercisable** right to make the invention itself; exclusivity over the complete unit becomes, in economic substance, indefinite.

The question adds the equaltreatment component that *Bolling v. Sharpe* locates in the Fifth Amendment's Due Process Clause and that *Adarand* holds indistinguishable from the Fourteenth Amendment's equal protection obligation. Those holdings do not themselves speak to patents, but they do establish that the federal government, no less than the states, must honor constitutional equality principles. Reasoning by analogy, once the Patent Clause has granted the public a right—here, the postexpiration right to make an invention—the government may not, through the **combined operation** of its own regulatory schemes, systematically prevent one class of inventions (regulated final assemblies) from ever delivering the promised right in practice, while other classes of inventions (unregulated or differently regulated subject matter) do permit that right to be exercised. The materials depict a **structural disparity**: for regulated complete units, patent

expiration never produces the world in which the public can actually make the unit, even as nonpatentees may freely exploit other patentexpired subject matter.

On that combined constitutional footing—the Patent Clause's "limited Times" bargain and the Fifth Amendment equaltreatment principle—the exam's scenario presents more than a merely "paper" right with incidental friction. It presents a right that **never becomes exercisable** for the category of inventions at issue. The public can read, study, and license the design, but it can never lawfully build the complete unit as such; the "right to make" exists only in theory. Given the way the exam's holdings define the publicdomain entitlement, that shortfall is a cognizable injury: the constitutional promise is not that the public may study the expired design for purposes of designing something else; it is that, after limited times, the public may **practice the invention itself.**

Therefore, on the materials provided, the proposition that "the right is satisfied and no cognizable injury arises so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products" is **false**. The inability ever to lawfully build a complete-unit reproduction of the expired design means the postexpiration right is not actually delivered, and a court owes a remedy.

ANSWER: F

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Glossary

(Alphabetical. Abbreviations are defined parenthetically at first appearance in the running text and collected here for reference, together with the five kinds of intellectual property the book distinguishes.)

AEB — Automatic Emergency Braking. Federal Motor Vehicle Safety Standard requiring vehicles to detect imminent forward collisions with other vehicles or pedestrians and apply

the brakes automatically when a driver does not respond. Codified by NHTSA in 2024 with phased compliance dates running through model year 2029.

AFBF — American Farm Bureau Federation. Largest U.S. general farm organization; publishes the Market Intel reports that track annual Chapter 12 farm bankruptcy filings drawn from U.S. Courts data and cited in the farm-equipment chapter.

BNCAP — Bharat New Car Assessment Programme. The vehicle safety assessment program of the Government of India, established in 2023 and operated under the Ministry of Road Transport and Highways. Tests new vehicles sold in India under crash protocols harmonized with international standards and awards adult occupant protection and child occupant protection ratings on a five-star scale.

CAFE — Corporate Average Fuel Economy. Federal fuel-economy standard administered by the National Highway Traffic Safety Administration; sets fleet-average fuel-efficiency targets that vehicle manufacturers must meet across their U.S. sales.

C.F.R. — Code of Federal Regulations. The codification of federal regulations published by the Office of the Federal Register and organized by title and section (e.g., 49 C.F.R. § 563 refers to Title 49, Section 563).

CJ — Civilian Jeep. The post-WWII civilian successor to the Willys MB military Jeep, manufactured under various model designations (CJ-2A, CJ-3A, CJ-3B, CJ-5, CJ-7) from 1945 through 1986.

CKD — Completely Knocked Down. A vehicle shipped as a kit of components, to be assembled in the destination country. The first Mahindra Jeep vehicles were assembled from CKD kits shipped from the United States.

Copyright. Protects an author’s original expression, such as a book, a song, a photograph, or software code. It never protects an idea, a method, or a useful article. Registered with the U.S. Copyright Office, though protection begins when the work is written down or recorded. Lasts for the author’s life plus 70 years; a work made for hire lasts 95 years from

publication or 120 years from creation, whichever ends first (17 U.S.C. § 302). The Constitution requires it to end.

CPSC — Consumer Product Safety Commission. Independent federal agency established by the Consumer Product Safety Act of 1972; sets and enforces mandatory product-safety standards for consumer products under 15 U.S.C. §§ 2051 et seq.

DOE — Department of Energy. Cabinet-level federal department; sets and enforces appliance and equipment energy-efficiency standards under the Energy Policy and Conservation Act and its amendments.

DOJ — Department of Justice. Cabinet-level federal department; brings federal civil and criminal enforcement actions, including under the Federal Food, Drug, and Cosmetic Act and consumer-protection statutes.

EISA — Energy Independence and Security Act of 2007. Federal statute (Pub. L. 110-140) whose lighting-efficiency provisions amended the Energy Policy and Conservation Act and effectively withdrew the ordinary incandescent bulb from the U.S. residential lighting market over a phase-in period running from 2012 to 2023.

EPA — Environmental Protection Agency. Independent federal agency; sets and enforces air-quality, emissions, and environmental-compliance standards under the Clean Air Act and related statutes.

ESC — Electronic Stability Control. Federal Motor Vehicle Safety Standard No. 126; required of all light vehicles manufactured for sale in the United States since model year 2012. Detects loss of steering control and automatically applies individual wheel brakes to restore directional stability.

ESG — Environmental, Social, and Governance. A class of investment-screening criteria used by institutional capital allocators — banks, private-equity funds, pension funds, and other large pools of capital — to evaluate the environmental, social, and corporate-governance characteristics of companies and projects considered for investment.

EUA — Emergency Use Authorization. Authority delegated to the FDA Commissioner under the Federal Food, Drug, and Cosmetic Act to permit the use of unapproved medical products, or unapproved uses of approved medical products, during a declared emergency.

FARFA — Farm and Ranch Freedom Alliance. Coalition-building organization founded in 2006 representing independent small-scale farmers, ranchers, and value-added producers.

FDA — U.S. Food and Drug Administration. Independent federal agency within the Department of Health and Human Services; regulates drugs, biologics, medical devices, food, cosmetics, and tobacco products under the Federal Food, Drug, and Cosmetic Act.

FMVSS — Federal Motor Vehicle Safety Standards. The body of federal vehicle-safety regulations administered by the National Highway Traffic Safety Administration and codified at 49 C.F.R. Part 571.

GNCAP — Global New Car Assessment Programme. The London-based international vehicle safety assessment program, founded in 2011 as a coordinating umbrella for regional NCAPs. Operates the #SaferCarsForIndia testing program and other regional programs, awarding adult and child occupant protection ratings on a five-star scale based on harmonized crash test protocols.

HCQ — Hydroxychloroquine. Antimalarial drug first patented in the 1950s and on the World Health Organization Essential Medicines List.

HVAC — Heating, Ventilation, and Air Conditioning. The equipment category covering residential and commercial systems for indoor climate control; one of the consumer-durable categories most heavily affected by Department of Energy efficiency standards under 42 U.S.C. § 6295(o)(1).

IAMERS — International Association of Medical Equipment Remarketers and Servicers. Industry association representing independent third-party servicers of medical devices.

IIJA — Infrastructure Investment and Jobs Act of 2021 (also styled the Bipartisan Infrastructure Law). Pub. L. 117-58. Federal statute directing the Department of

Transportation to issue performance standards for crash avoidance technology, including automatic emergency braking and impaired-driving prevention systems.

IMO — International Maritime Organization. United Nations specialized agency responsible for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships. Develops the SOLAS, MARPOL, and COLREG conventions implemented in U.S. law through the Coast Guard's vessel inspection regime.

IMV — Innovative International Multi-purpose Vehicle. Toyota Motor Corporation's body-on-frame platform underlying the Hilux pickup, Fortuner SUV, Innova MPV, and Hilux Champ models manufactured in Thailand, Indonesia, and other Asian markets. The platform is built on a long-mature mechanical design whose underlying patents have been in the public domain for many years.

ISO 9001 — International Organization for Standardization, quality management systems standard. Voluntary international standard certifying that a manufacturer maintains a documented quality management system; cited in the farm-equipment chapter as one of the certifications Indian tractor manufacturers meet on equipment exported to global markets.

IVM — Ivermectin. Antiparasitic drug discovered in the 1970s, on the World Health Organization Essential Medicines List, and recognized with the 2015 Nobel Prize in Physiology or Medicine for the underlying discovery.

LCD — Local Coverage Determination. A coverage policy issued by a Medicare Administrative Contractor under authority delegated by the Centers for Medicare and Medicaid Services; specifies whether a particular item or service is reasonable and necessary, and therefore covered under Medicare Part A or Part B.

LED — Light-Emitting Diode. The semiconductor lighting technology that, under the Department of Energy efficiency rules finalized in 2023, has effectively replaced the incandescent bulb in U.S. residential lighting.

LRA — Legislative Reorganization Act of 1970. Federal statute that restructured congressional procedure by mandating recorded floor votes, open committee proceedings, and

other “sunshine” reforms.

MAP-21 — Moving Ahead for Progress in the 21st Century Act. Pub. L. 112-141 (July 6, 2012). Federal surface-transportation reauthorization statute under which Congress directed NHTSA to issue lighting and marking standards for agricultural equipment, codified at 49 CFR Part 562 in 2016.

MB — Methylene blue. Synthetic dye first prepared in 1876; FDA-approved for the treatment of methemoglobinemia and used off-label for a range of clinical and supplement purposes.

NAECA — National Appliance Energy Conservation Act of 1987. Public Law 100-12, signed by President Reagan on March 17, 1987. The Act amended the Energy Policy and Conservation Act of 1975 and added the anti-backsliding provision codified at 42 U.S.C. § 6295(o)(1) — the keystone of the regulatory ratchet.

NEMA — National Electrical Manufacturers Association. Trade association representing the major U.S. manufacturers of electrical equipment, including the patent-protected lighting incumbents (GE Lighting under Savant Systems, Signify, formerly Philips, Osram, Cree, and Acuity). NEMA’s lighting members actively lobbied for the federal general service lamps efficiency standards finalized between 2022 and 2024 that withdrew the ordinary incandescent bulb from the U.S. residential market; the healthy-lighting chapter develops this dynamic as the bootleggers-and-Baptists configuration at the center of the case.

NHTSA — National Highway Traffic Safety Administration. Federal agency under the Department of Transportation; sets and enforces the Federal Motor Vehicle Safety Standards.

NMOG — Non-Methane Organic Gas. A class of regulated tailpipe emissions covering hydrocarbons other than methane; one of the principal pollutant categories in EPA’s Tier 2 and Tier 3 motor-vehicle emission standards, where it is typically reported in milligrams per mile.

NOx — Nitrogen Oxides. Reactive gases formed by combustion in the presence of nitrogen, regulated as a tailpipe emission under the Clean Air Act. The "x" is a variable subscript: NOx covers nitric oxide (NO) and nitrogen dioxide (NO_2).

NRDC — Natural Resources Defense Council. Nonprofit environmental advocacy organization; named respondent in Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (1984).

NSAC — National Sustainable Agriculture Coalition. Coalition of grassroots organizations advocating for sustainable and organic farmers; has published analyses documenting that four firms control 45 percent of global farm equipment sales.

NTTAA — National Technology Transfer and Advancement Act of 1995. Pub. L. 104-113. Federal statute directing federal agencies to use technical standards developed by private-sector voluntary consensus standards bodies in lieu of developing federal standards from scratch.

OEM — Original Equipment Manufacturer. The producer of a finished product (medical device, vehicle, appliance, or other equipment); often used in contrast with third-party servicers and remanufacturers.

OSH Act — Occupational Safety and Health Act of 1970. Pub. L. 91-596. Federal statute establishing workplace safety standards through the Occupational Safety and Health Administration; codified at 29 U.S.C. §§ 651 et seq.

OSHA — Occupational Safety and Health Administration. Federal agency within the U.S. Department of Labor; codifies workplace safety standards including 29 CFR 1928.51, the 1976 rollover rule on agricultural tractors discussed in the farm-equipment chapter as the first of the three regulatory layers.

Patent. Protects an invention. A utility patent covers how a thing works and lasts 20 years from the filing date. Before June 8, 1995 the term was 17 years from grant; a patent in force or pending on that date received the longer of the two (35 U.S.C. § 154). A design patent covers a product's ornamental appearance and lasts 15 years from grant, or 14 if filed be-

fore May 13, 2015 (35 U.S.C. § 173). Granted by the U.S. Patent and Trademark Office in exchange for full public disclosure. The Constitution requires it to end, and at expiry the invention passes to the public.

PRP — Platelet-Rich Plasma. A preparation derived from a patient's own blood by centrifugation, used for autologous orthopedic injection.

RCRA — Resource Conservation and Recovery Act of 1976. Pub. L. 94-580. Federal statute establishing the federal regulatory framework for solid and hazardous waste under 42 U.S.C. §§ 6901 et seq.

TPMS — Tire Pressure Monitoring System. Federal Motor Vehicle Safety Standard No. 138; required of all light vehicles manufactured for sale in the United States since September 1, 2007. Sensors detect significantly under-inflated tires and warn the driver.

Trade dress. A kind of trademark: the distinctive look of a product or its packaging that tells buyers who made it, such as a grille or a body shape. Registered with the U.S. Patent and Trademark Office, where a drawing shows the claimed features in solid lines; also protected unregistered. Has no fixed term and lasts as long as the look identifies its maker. It cannot cover a functional feature, and an expired utility patent is strong evidence that the feature it claimed is functional (*TrafFix Devices, Inc. v. Marketing Displays, Inc.*, 2001).

Trade secret. Commercially valuable information its owner keeps confidential, such as a formula or a process. Registered nowhere. Protected against theft by state law and the Defend Trade Secrets Act of 2016 (18 U.S.C. § 1836), though not against independent discovery or reverse engineering. Has no fixed term and lasts as long as the secret is kept. Unlike a patent, it is never disclosed and never has to end.

Trademark. Protects a word, name, symbol, or device that tells buyers who made a product: the name on the tailgate, the logo, the hood emblem. Registered with the U.S. Patent and Trademark Office under the Lanham Act (15 U.S.C. §§ 1051 et seq.); also protected unregistered through use. Has no fixed term and lasts as long as the owner keeps using it, with a registration renewed every 10 years. It protects identity, never function.

TRIPS — Agreement on Trade-Related Aspects of Intellectual Property Rights. Annex 1C of the 1994 Marrakesh Agreement Establishing the World Trade Organization, binding on all 166 WTO member nations. Article 33 sets the minimum patent term at twenty years from the filing date; the post-expiration commons it implies is the international counterpart to the U.S. Constitution's "limited Times" requirement.

U.S.C. — United States Code. The codification of the general and permanent laws of the United States, organized by title and section (e.g., 42 U.S.C. § 6295(o)(1) refers to Title 42, Section 6295, subsection (o)(1)).

USD — United States Dollar. The currency of the United States; used throughout the book in international price comparisons where foreign-currency prices are converted to U.S. dollar equivalents for the American reader.

USDA — U.S. Department of Agriculture. Cabinet-level federal department; the Economic Research Service publishes the Food Dollar Series cited in the farm-equipment chapter, breaking each U.S. retail food dollar into the farm share (15.9 cents in 2023) and the marketing share (84.1 cents).

USP — United States Pharmacopeia. The non-governmental standards-setting body whose monographs for drug substances and dosage forms are incorporated by reference into FDA regulation under the Federal Food, Drug, and Cosmetic Act.

WTO — World Trade Organization. Intergovernmental organization administering the global trade-rules framework, including the TRIPS Agreement under which all 166 member nations are bound by a minimum twenty-year patent term — the international counterpart to the U.S. Constitution's "limited Times" requirement.

Key Terms

(Alphabetical by term. The book's specific terms of art, with brief definitions of how the book uses each.)

Anti-backsliding provision. Section 6295(o)(1) of Title 42 of the United States Code, added by the National Appliance Energy Conservation Act of 1987. The statute provides that an energy-efficiency standard set by the Department of Energy can be tightened over time but never relaxed. The book treats this as the keystone of the regulatory ratchet: the legislative mechanism that converts a temporary regulatory burden into a permanent one and, by extension, converts the lengthening of an expired patent's effective monopoly into a permanent constitutional violation.

Blockade pattern. The book's term for the more common form the regulatory ratchet takes: the federal regulatory wall is built around a technology before the patents covering it expire, so that by the time the patents expire the public's right to remanufacture comes into existence inside a regulatory environment that already forbids it. The 2005 Ford F-150 case is a blockade-pattern case. Most expired-patent commerce in the United States is blockade-pattern. The injury is harder for an ordinary American to see because the manufacturing window the public was entitled to never opened. The economic consequences are large; the intuitive force of the injury is muted by the absence's invisibility.

Bootleggers-and-Baptists. A coalition framework first named by the economist Bruce Yandle (1983) to describe regulatory regimes supported simultaneously by a moralizing public-interest constituency (the "Baptists," who believe in the regulation's stated purpose) and a profit-taking commercial constituency (the "bootleggers," who profit from the regulation's suppression of competition). The book argues that the post-1970 federal regulatory state is the bootleggers-and-Baptists architecture writ very large, with each leg of the regulatory wall genuinely supported by both wings of its coalition, which is what makes the coalition durable.

Combinatorial property of the public domain. The book's term for the engineering fact that an expired patent does not enter the public domain alone. It enters as a component, available to be combined with every other expired-patent component, in any combination an engineer can devise. The remanufactured American product that emerges when the regulatory wall is lifted is therefore not a copy of any single expired-patent design but a freshly

engineered assembly of public-domain parts. The public domain is a kit of parts, not a museum.

Constitutional forgetting. The book's term for the operational disappearance of the words "limited Times" from the Patent Clause as applied by Congress, the federal regulatory agencies, and the academic disciplines that study them. The verb is "forgetting" — not "misreading" or "ignoring" — because the doctrine was never overturned. Supreme Court precedent on the post-expiration public domain remains intact and unbroken; what has been lost is the operational memory of the bargain in the institutions whose work bears on it.

Expired-patent commerce. The book's denominator term for the share of American economic activity in which the underlying patents have expired. The book divides this commerce into three tiers: 58 percent Blocked (where federal regulation makes remanufacturing legally impossible), 15 percent Hindered (where federal regulation imposes substantial but non-prohibitive constraints), and 27 percent Flowing (where remanufacturing operates freely). The relevant denominator is expired-patent commerce, not "the economy" — which is why the 58 percent figure does not refer to all American manufacturing.

Expired Patent Commons. The book's term for the destination an invention is meant to reach when its patent expires: the body of expired inventions that the public may freely make, copy, improve, and remanufacture. The word "commons" is load-bearing — it denotes a shared resource the public can actually reach and use, not merely a right that has lapsed on paper. An invention has entered the Expired Patent Commons only when an American may in fact build it; a patent that has expired but that federal regulation forbids anyone to manufacture has not. The Expired Patent Commons is the public's half of the patent bargain, and the book's central claim is that the post-1970 regulatory wall prevents inventions from entering it — the forgotten half of the Patent Clause. The Supreme Court's cases call the same destination the public domain in the patent-law sense.

Grandfather exemption. The first leg of the book's three-leg remedy. A judicial or legislative carve-out under which a remanufactured product built on a public-domain platform is held to the federal safety and energy-efficiency standards in effect during the platform year

the remanufactured product mimics — the same standards the original product was certified to. The grandfather exemption flows naturally from a successful constitutional ruling on the Patent Clause and requires no act of Congress to put in place.

Health-freedom statute. A state-level statute that limits the authority of state medical or pharmacy licensing boards to discipline licensed physicians for the use of treatments not approved by the U.S. Food and Drug Administration, provided the practitioner has obtained informed consent and represents the modality accurately. Roughly fifteen states have enacted such statutes since the late 1990s. Relevant to the Hindered sector, where licensing-board discipline is a common enforcement angle.

Hindered sector. The roughly fifteen percent of expired-patent commerce — chiefly pharmaceuticals and medical devices — where federal regulation, professional-board enforcement, and insurance-coverage decisions impose substantial but non-prohibitive constraints on use of off-patent products. The Hindered sector contrasts with the Blocked sector, in which post-1970 federal manufacturing mandates make remanufacturing legally impossible, and with the Flowing sector, in which remanufacturing operates freely.

Limited Times. The two words at the heart of Article I, Section 8, Clause 8 of the U.S. Constitution: "To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." The phrase is constitutive, not decorative — affirmed across more than eighty Supreme Court rulings spanning two centuries. The book's central claim is that the post-1970 regulatory state has converted "limited Times" into perpetual times across roughly 58 percent of expired-patent commerce, in violation of the Patent Clause.

Median. A statistical measure of central tendency: the middle value of a distribution, where half of cases fall above and half fall below. The median is distinct from the mean (or arithmetic average), which is the sum of all values divided by the count. The book uses median household income — $83,730 in 2024, per the U.S. Census Bureau — rather than mean household income because the right-skewed distribution of American household income (a small number of very high earners pulling the mean upward) makes the median the better

description of where the typical American household actually earns. The same logic applies to median wage, median home price, and other median figures the book cites.

Multi-value tax deduction. A deduction that allows a business to subtract a multiple of an actual expenditure — two or three dollars for every dollar spent — from taxable income. It is distinct from an ordinary deduction, which is one-for-one; from a tax credit, which is applied dollar-for-dollar against tax owed and is therefore far costlier to the Treasury; and from a civil penalty, which is coercive. The author proposed the mechanism in *ShareDebate International*, June 26, 1994. The book applies it in the chapter on the environmental question as the means by which the newest patent-expired emissions-control technology becomes the economical default in remanufactured products, without a consumer subsidy and without a direct federal outlay.

Platform year. The model year of the original product that a remake mimics, and the year whose federal safety, emissions, and energy standards the remake is certified to. The remake is certified to every standard in force at any time during that year, and to the design as it stood at the end of it, including repairs made by then. For this purpose the model year ends on the day the first unit of the next model year of that make and model was delivered to a buyer in the United States. The anchor is always the platform year the remake mimics, never the patent's filing or expiration date. See the chapter *How a Remake Is Built and Certified*.

Public domain (patent-law sense). The constitutional status of an invention after its patent has expired. In the patent-law sense (distinct from the colloquial sense, in which "public domain" refers loosely to anything freely available), the public domain is the operational result of the bargain the Patent Clause establishes: the inventor receives an exclusive right for a limited time, after which the invention belongs to the American public to make, copy, improve, and remanufacture. The Supreme Court has repeatedly affirmed that the post-expiration public domain is a vested public right.

Regulatory ratchet. The book's term for the one-way mechanism by which federal regulation tightens over time but cannot be relaxed, with the result that the original legal terrain

— including, crucially, the patent commons that emerges when patents expire — becomes progressively narrower. The ratchet is uniquely hostile to the Patent Clause because it converts a temporary regulatory burden into a permanent extension of monopoly protection. The anti-backsliding provision (42 U.S.C. § 6295(o)(1)) is the keystone statutory expression of the ratchet.

Remanufacturing (book's specific sense). The act of building a freshly engineered product assembled from components whose individual patents have all expired. The book's sense is distinct from "restoration" (returning a specific used object to working condition) and from "reproduction" (manufacturing a copy of a specific older design). A remanufactured product is a kit-of-parts assembly: a chassis whose patents expired in the early 2000s, a USB port whose 1995 Intel patents expired around 2015, a backup camera whose imaging patents expired years ago — engineered, mixed, and matched to deliver the features buyers actually want at a fraction of the new-vehicle price. The public domain is a kit of parts, not a museum.

Sunset Amendment. The book's term for the proposed constitutional amendment introduced in the chapter on the environmental question. The amendment preserves the Patent Clause remedy at full strength for twenty years from the date of first lawful remanufacture in any product category, after which Congress may by law restrict or prohibit further remanufacture without obligation to provide just compensation under the Fifth Amendment, applying only prospectively. The amendment is offered as Phase 2 structural insurance: not required for the Phase 1 remedy to operate, but available if the country eventually decides to close the marginal compensation subsidy created by the Takings Clause backstop.

Three-tier breakdown. The book's category-level audit of expired-patent commerce, dividing it into three tiers: Blocked (58 percent — sectors where federal manufacturing mandates make remanufacturing legally impossible, including vehicles, agricultural and construction machinery, HVAC, appliances, and certain lighting), Hindered (15 percent — sectors where federal regulation imposes substantial but non-prohibitive constraints, principally pharmaceuticals and medical devices), and Flowing (27 percent — sectors where remanufacturing operates freely, including bicycles, hand tools, firearms, and most consumer

electronics). The breakdown is built from BLS Consumer Expenditure Survey data, BEA Industry Economic Accounts, and Census Bureau Annual Survey of Manufactures data; the methodology is detailed in the published companion methodology article — see the Endnotes entry for Martin, “The Forgotten Half of the Patent Clause: What was Promised When Patents Expired” (April 10, 2026).

Two suspenders. The book’s metaphor for its two independent constitutional prongs, either of which is sufficient on its own. The first prong is the Patent Clause: Article I, Section 8, Clause 8 authorizes Congress to grant exclusive rights only “for limited Times,” and a regulatory scheme that converts those limited times into perpetual protection exceeds the power the Clause actually grants. The second prong is the Takings Clause: the public’s right to copy and remanufacture an expired patent is a vested right, affirmed in Bonito Boats (1989), and a permanent regulatory cancellation of that right is a taking requiring just compensation. A litigant who loses on one prong can still win on the other.

Withdrawal pattern. The book’s term for the sharper second form the regulatory ratchet takes: the technology was already in the public domain, the public was already exercising the right to manufacture, and federal regulation was then erected around the technology after the fact, withdrawing it from the domain it had already been part of. The ordinary incandescent light bulb is the canonical case — the foundational patents expired more than a century ago, Americans manufactured incandescent bulbs continuously from 1879 onward, and federal energy-efficiency rules issued between 2007 and 2023 progressively withdrew the bulb from manufacturability. The withdrawal pattern is doctrinally stronger than the blockade pattern because the right at issue was not theoretical — it had been exercised in commerce, in plain view, for generations.

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— U.S. Constitution, Article I, Section 8, Clause 8